SAIM RESEARCH / TECHNICAL BOOK

Generalized Hierarchical Bayesian Segmentation with Irregular Designs, Multi-Sequence Hierarchies, and Grouped/Latent-Group Designs

Statistical formulation, theory, algorithms, and empirical results

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Exponential-family segment models, irregular designs, multi-sequence hierarchies, and latent groups.

A technical treatment of generalized hierarchical Bayesian segmentation for ordered data. Segment marginal likelihoods from conjugate exponential-family models are combined with exact dynamic programming over contiguous partitions, with extensions for irregular designs, shared and group-specific changepoints, latent groups, and explicitly approximate nonconjugate segment models.

sAIm LABS

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Scope and inferential objectives

This book develops *Generalized Hierarchical Bayesian Segmentation with Irregular Designs, Multi-Sequence Hierarchies, and Grouped/Latent-Group Designs*. Its basic objects are segmentwise integrated likelihoods, proper priors over ordered partitions, posterior distributions over segment counts and changepoints, Bayes regression curves, and joint MAP partitions. The mathematical constructions build on established product-partition models and exact offline Bayesian changepoint inference. Extensions cover physical-coordinate priors, shared-boundary pooling, finite-template clustering, and numerical integration of nonconjugate segment models.

Exact and approximate calculations

Analytic segment integration combined with sum-product dynamic programming yields exact finite posterior quantities under the specified model. Max-sum backtracking yields a joint MAP partition, not an independently chosen collection of marginal boundary modes. Where segment integrals require Laplace approximation, variational approximations, expectation propagation, or quadrature, the global recursion is exact only for the resulting approximate segment scores. No unverified finite error certificate is implied by convergence of a local approximation.

Empirical interpretation

The reported figures and tables are computations under their stated observation laws, data transformations, priors, and sample units. Results from different likelihood families or different observation supports are not directly comparable marginal likelihoods. Agreement with an estimated segmentation is distinct from independent boundary accuracy. The current array-CGH precision model requires a corrected likelihood implementation and a new fit before its numerical result can support a precision-aware sampling claim. More extensive experiments are defined prospectively, with outputs left unfilled until actually executed and checked.

Organization

Chapters 1–6 present the formulation, conjugate block integration and partition recursions. Chapters 7–10 develop irregular designs, shared boundaries, finite templates and local numerical approximations. Chapters 11– 14 cover prediction, algorithms, verification and examples. The concluding chapters state failure conditions and open experimental requirements. The appendices provide annotated literature, numerical implementation principles, and prospective experimental designs.

CHAPTER 1

Generalized hierarchical Bayesian segmentation: problem, outputs, and scope

CHAPTER PATH

The research problem, the meaning of generality and hierarchy, the returned posterior quantities, and the conditions for exact computation.

LEARNING OBJECTIVES

- State the generalized hierarchical Bayesian segmentation problem and its principal outputs.
- Explain why the partition recursions are common across observation models once segment marginal likelihoods are available.
- Identify the roles of irregular designs, multiple sequences, known groups, and latent groups.
- Distinguish exact conjugate calculations from explicitly approximate nonconjugate calculations.

Multiple-changepoint analysis partitions an ordered set of observations into contiguous segments within which a statistical parameter is constant or follows a specified segment model. Given responses $y_1, \ldots, y_n$ observed at ordered coordinates $u_1 < \cdots < u_n$, the inferential questions include the number of segments, the changepoint locations, the segment-specific parameters, and uncertainty in all of these quantities. Bayesian formulations place a prior on the partition and segment parameters and therefore yield posterior probabilities for competing segmentations rather than only a single penalized estimate [3, 4, 14, 19].

The central statistical structure is a product-partition model. Conditional on a partition, observations in different segments are independent after segment parameters are introduced. When a segment parameter can be integrated out, each candidate segment $(i, j]$ contributes a segment marginal likelihood. A prior that factorizes over segments or transitions then permits exact finite summation over all admissible ordered partitions by dynamic programming. This yields marginal likelihoods, posterior distributions for the segment count, ordered-changepoint marginals, changepoint-event probabilities, and posterior moments of segment functionals [3, 14, 19].

BayesBreak develops this calculation as a generalized hierarchical Bayesian segmentation method. The term *generalized* refers to the following mathematical property: the recursion over partitions does not depend on the particular observation family once the required segment marginal likelihoods and posterior moments have been computed. For a regular exponential

family with a proper conjugate prior and finite normalizing constants, these segment quantities are available by exact integration from sufficient statistics. Gaussian observations with an unknown segment mean, Poisson counts with exposure, and Binomial observations with known trial counts are examples. A nonconjugate segment model may use numerical integration or another stated approximation, but exact partition summation for an approximate score array is not the same as exact Bayesian inference for the intended observation model.

The term *hierarchical* refers to probability models that couple sequences or groups through shared partition structure or group-level parameters. The book treats: (i) a single ordered sequence; (ii) irregularly spaced observations through design-dependent segment cohesion and boundary hazards; (iii) multiple aligned sequences with common changepoints and conditionally independent segment likelihoods; (iv) known groups of sequences; and (v) a finite latent-group criterion in which sequences are assigned to candidate changepoint templates. The first four are Bayesian models with explicit joint distributions under their stated assumptions. The finite latent-group extension is presented according to its actual optimization criterion and is not relabelled as a normalized mixture likelihood.

Principal contributions.

- **Exponential-family segment integration.** A common notation separates sufficient statistics, exposure or trial descriptors, and optional likelihood powers. Proper conjugate priors give exact segment marginal likelihoods and posterior moments through ratios of normalizing constants.
- **Posterior computation over ordered partitions.** Forward and backward sum-product recursions compute fixed-count marginal likelihoods, $p(k \mid y)$, ordered-changepoint marginals, changepoint-event probabilities, and model-averaged segment functionals. A distinct max-sum recursion with backtracking computes the joint MAP partition.
- **Irregular-design priors.** Segment cohesion and interior-boundary hazard are separated mathematically. This permits priors based on physical segment length or local design spacing without confusing design geometry with exposure in the observation likelihood.
- **Multiple-sequence hierarchies.** For aligned conditionally independent sequences with a common partition, sequence-specific segment marginal likelihoods multiply. Known groups use the same calculation within each group. A finite-candidate consistency result states conditions under which the posterior concentrates on the common partition as the number of informative sequences increases.
- **Grouped and latent-group segmentation.** Known group memberships lead to groupwise shared-changepoint posteriors. For unknown memberships, a finite latent-group objective alternates auxiliary allocation weights with exact responsibility-weighted max-sum updates of the group templates; monotonicity follows from a Jensen minorization argument.
- **Numerical segment integration and prediction.** For nonconjugate observation models, the effect of a uniform segmentwise log marginal-likelihood error is propagated through the partition sums. Posterior prediction is defined by the fitted observation family and coordinate support, and changepoint metrics use one-to-one matching.

1.1 Research question and returned quantities

The research question is whether one generalized Bayesian formulation can combine observation-model-specific segment integration with posterior computation over contiguous partitions while

retaining irregular-design and hierarchical multiple-sequence structure. For a sequence y, the principal returned quantities are

$$\{p(y \mid k),\ p(k \mid y),\ p(t_p = h \mid y, k),\ p(b_h = 1 \mid y),\ \mathbb{E}[m(\theta_s) \mid y],\ \widehat{t}^{\,\mathrm{MAP}}(k)\}\,.$$

These quantities have different meanings. In particular, a marginal mode of each changepoint coordinate need not form the joint MAP partition, and a posterior mean curve is not a selected piecewise-constant partition.

KEY IDEA

The observation model determines the segment marginal likelihoods and posterior moments. The partition model determines how these quantities are summed over ordered segmentations. Generality comes from separating these two mathematical calculations while keeping their probability models explicit.

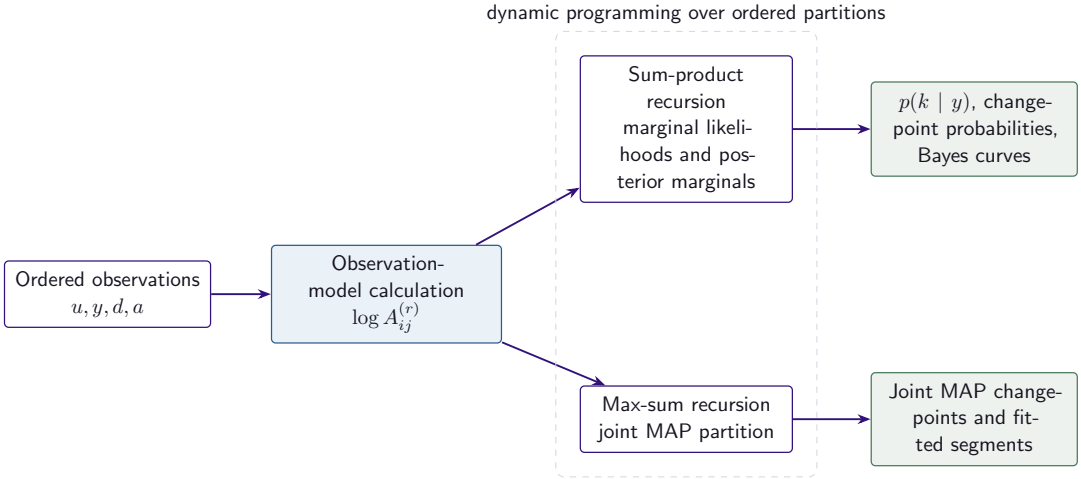


Figure 1.1. Observation-model calculations and inference over ordered partitions. Sum-product and max-sum recursions use the same prior-adjusted segment scores but return different posterior or optimization quantities.

1.2 Conditions and exclusions

The exact results concern offline segmentation on a finite ordered candidate set with a prior and likelihood that factor into segment or transition contributions. The method is not an online run-length filter, a general nonparametric detector, or a proof that arbitrary dependence can be represented without enlarging the state space. Exact Bayesian inference requires exact segment marginal likelihoods. When these quantities are approximated, the resulting partition calculation is exact only for the supplied approximate array; comparison with the intended posterior requires additional numerical error analysis.

CHAPTER 2

Statistical foundations and relation to prior work

CHAPTER PATH

The established literature on exponential families, product partitions, exact Bayesian changepoint recursion, irregular designs, multiple sequences, latent groups, and numerical approximation.

2.1 Exponential families and conjugate analysis

For a regular exponential family, likelihood contributions can be represented through sufficient statistics, a natural parameter, and a log-partition function. Conjugate priors for exponential families preserve the natural form of the posterior and express integrated likelihoods through normalizing-constant ratios [5, 12]. This is established Bayesian theory, not a new property of BayesBreak. Its role here is to provide exact segment marginal likelihoods and posterior moments for every candidate interval when the required normalizers are finite. Exposure in Poisson sampling and trial counts in Binomial sampling enter the likelihood through family-specific sufficient summaries; they are not generic likelihood weights.

2.2 Product-partition models and exact Bayesian changepoint inference

Product-partition models assign a probability to a partition through products of segment cohesion terms and combine this prior with segmentwise integrated likelihoods [3, 4]. Exact Bayesian multiple-changepoint recursions exploit the same factorization to sum over possible previous changepoints and compute posterior distributions without enumerating every partition explicitly [14]. Bayesian piecewise-constant regression further shows how partition averaging yields posterior regression curves, changepoint probabilities, and marginal likelihoods [19].

Accordingly, this book does not claim to introduce product partitions, conjugate segment integration, or dynamic programming for Bayesian changepoint models. BayesBreak uses these established results as the foundation for a broader model specification covering multiple observation families, design-dependent priors, multiple aligned sequences, and group structure. The sum-product recursion for posterior marginals and the max-sum recursion for a joint MAP partition are stated separately because they solve different algebraic problems even when they use the same segment scores.

2.3 Frequentist segmentation and alternative Bayesian computation

Frequentist methods optimize additive segment costs or multiscale criteria. Segment-neighborhood and optimal-partitioning algorithms provide exact minimizers for fixed segment counts or penalties [2, 21]; pruned dynamic programming and PELT reduce practical cost under additional conditions [22, 30]. Total-variation and fused-lasso estimators use convex regularization, including group penalties for multiple sequences [7, 31, 35]. SMUCE and wild binary segmentation provide multiscale or recursive detection procedures with different inferential targets [16, 17]. Circular binary segmentation is widely used for array-CGH analysis [25]. These procedures are relevant comparators but do not, without additional construction, return the same posterior distribution over partitions.

When segment parameters cannot be marginalized or the dependence structure is not amenable to a finite recursion, reversible-jump Markov chain Monte Carlo and other trans-dimensional sampling methods provide a general Bayesian alternative [11, 18]. Bayesian online changepoint detection instead filters a run-length distribution as data arrive and solves a different online problem [1]. BayesBreak is an offline finite-partition method.

2.4 Irregular designs and design-dependent partition priors

Product-partition models have been generalized so that cohesion depends on covariates or other predictors [24, 26]. In changepoint problems, irregular observation coordinates affect the prior plausibility of segment lengths and boundary locations. This book therefore distinguishes a segment-cohesion factor $c_u(i, j)$ from an interior-boundary hazard $h_u(j)$. The distinction prevents two different prior constructions—a prior on segment length and a local hazard of a changepoint—from being described as the same object. It also keeps design spacing separate from exposure or trial information in the observation likelihood.

2.5 Hierarchical and multiple-sequence changepoint models

Hierarchical Bayesian changepoint models have a long history [8]. Methods for simultaneous changepoints in multiple data sequences use hierarchical priors to borrow information about the propensity for changes across sequences [13]. Joint random partition models provide more flexible dependence among partitions across several ordered processes [29]. Other work permits dependence between successive segments or richer within-segment regression structures [15, 32].

The exact multiple-sequence calculation developed here is deliberately narrower: aligned sequences share the same candidate partition, are conditionally independent given that partition, and may have sequence-specific segment parameters. Their segment marginal likelihoods then multiply. This is an exact hierarchical model under those assumptions, not a claim to encompass every form of partial synchrony or dependence among sequence-specific changepoints. Known groups repeat the calculation within each group.

2.6 Latent groups and optimization

Model-based clustering of longitudinal or ordered data with common structural changes is relevant prior work for clustering sequences through changepoint patterns [9]. Finite-mixture identifiability results concern normalized component distributions [34], and the EM algorithm concerns likelihoods or objectives for which the auxiliary-function construction is valid [10].

The latent-group calculation in this book is stated more narrowly. It defines a finite objective

$$\mathcal{F}(\pi, \tau) = \sum_{s} \log\left\{ \sum_{g} \pi_g S_s(\tau_g) \right\},$$

where $S_s(\tau_g)$ is a positive sequence-template score. A Jensen minorizer yields auxiliary allocation weights; exact maximization of the minorizer over template partitions uses a responsibility-weighted max-sum recursion. The proof establishes monotone ascent of this stated criterion. It does not establish that $S_s(\tau_g)$ is a normalized component density, identify a finite mixture, or average over uncertainty in the templates.

2.7 Nonconjugate segment models

Laplace approximation, variational bounds, Pólya–Gamma augmentation, expectation propagation, and numerical quadrature are established tools for nonconjugate Bayesian calculations [20, 23, 27, 36]. BayesBreak uses them, when appropriate, to approximate individual segment marginal likelihoods. The general theorem in Chapter 10 is conditional: if every admissible segment log marginal likelihood is within ε of a reference value, then partition marginal likelihoods and posterior odds obey explicit bounds. The theorem does not supply a method-specific value of ε.

2.8 Position of the present work

The contribution is a generalized hierarchical Bayesian segmentation formulation built from established components whose assumptions are kept explicit. It combines:

1. exact segment integration for a class of conjugate exponential-family observation models;
2. sum-product posterior computation and separate max-sum MAP optimization over ordered partitions;
3. priors for irregular designs that distinguish segment cohesion from boundary hazard;
4. exact common-changepoint pooling for aligned conditionally independent sequences and known groups;
5. a finite latent-group template criterion with a proved monotone alternating algorithm;
6. conditional propagation of numerical segment-score error; and
7. family-specific posterior prediction and clearly defined changepoint metrics.

The novelty claim is attached to this combined generalized method and its derivations, not to the prior existence of dynamic programming, exponential-family conjugacy, product partitions, or EM.

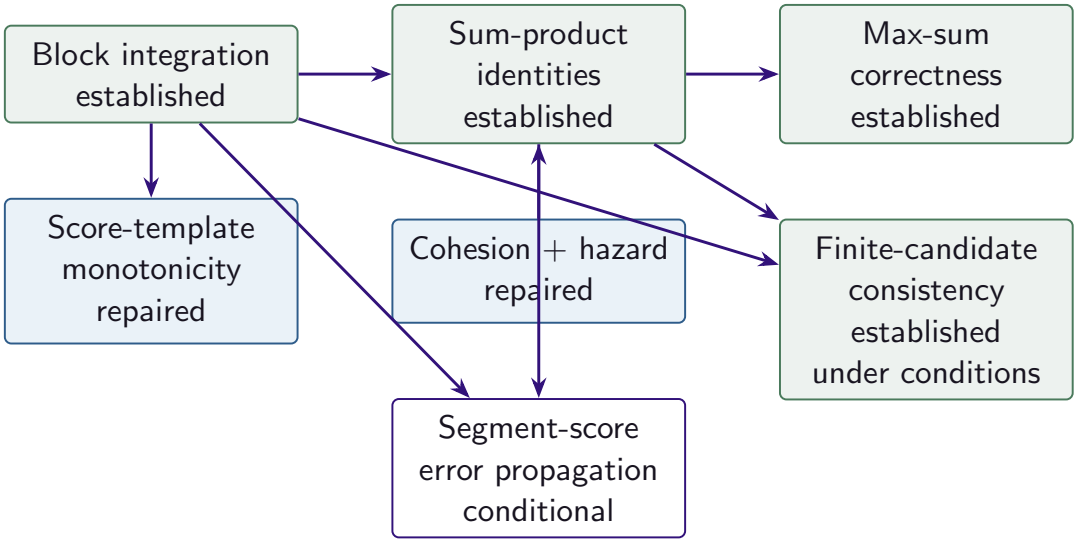


Figure 2.1. Logical dependence of the principal results. Exact partition calculations require finite segment marginal likelihoods and factorized prior contributions. The nonconjugate result additionally requires a uniform segmentwise error bound.

2.9 Recommended reading sequence

A reader new to the subject should begin with exponential-family conjugacy, product-partition models, and exact Bayesian multiple-changepoint recursion; then proceed to Bayesian piecewise-constant regression, design-dependent product partitions, and hierarchical multiple-sequence methods. The literature annotations in Appendix A state the specific role of every reference retained in the book. Numerical implementation additionally requires stable log-sum-exp evaluation and the approximation methods used for nonconjugate segment models [6].

Formal setup, notation, and model specification

CHAPTER PATH

Ordered segmentation, exposure-aware local likelihoods, posterior targets, and the exact factorization required by the partition dynamic program.

3.1 Notation and symbols

This section fixes notation used throughout the book. We use lower-case Roman letters for indices, reserve $t = (t_0, \dots, t_k)$ for boundary vectors, and adopt the half-open block notation $(i, j] := \{i + 1, \dots, j\}$ for $0 \le i < j \le n$.

For irregular designs, physical segment lengths are defined using candidate boundary coordinates $u_0 < u_1 < \cdots < u_n$ rather than by implicitly referencing an undefined observation location x_0. The physical length of block $(i, j]$ is $\Delta_x(i, j) := u_j - u_i$. On a regular index grid set $u_i = i$; for interval data the u_i are the observed interval endpoints. Point observations without meaningful physical interval endpoints use the index-uniform prior or provide an external left endpoint u_0.

Conjugate-prior normalization convention. For exponential-family blocks we use the following (Diaconis–Ylvisaker) convention for the conjugate prior normalizer. For hyperparameters (α, β),

$$\begin{aligned} p(\theta \mid \alpha, \beta) &= \exp\Big\{\eta(\theta)^\top \alpha - \beta A(\theta) - \log Z(\alpha, \beta)\Big\}, \\ Z(\alpha, \beta) &:= \int \exp\Big\{\eta(\theta)^\top \alpha - \beta A(\theta)\Big\}\, d\theta. \end{aligned} \tag{3.1}$$

With this convention, Z is the integral itself (not its reciprocal) and block evidences take the ratio form $Z(\alpha_0 + S_{ij}, \beta_0 + W_{ij})/Z(\alpha_0, \beta_0)$ (Theorem 4.1 and Chapter 4).

Table 3.1. Notation and symbols used throughout the book. This is a notation glossary; no source CSV is associated with it.

Symbol	Meaning
n	Number of observations on the ordered design.
x_i	Design point / location of observation y_i.

u_i	Candidate boundary coordinate used for physical segment lengths, with $u_0 < \cdots < u_n$.
$\Delta_x(i,j)$	Physical length of block $(i,j]$, defined as $u_j - u_i$.
y_i	Observation at location x_i (single-sequence case).
d_i	Family-specific exposure, trial-count, offset, or precision descriptor when required by the observation model. Optional power/reliability weights are denoted a_i; neither role is inferred from irregular spacing.
k; $k_{\max}$	Number of segments; maximum segment count considered by the DP.
$t = (t_0, \ldots, t_k)$	Boundary vector, with $0 = t_0 < t_1 < \cdots < t_k = n$.
$I_q(t)$	Index set of segment q: $\{t_{q-1}+1, \ldots, t_q\}$.
$(i,j]$	Candidate block $\{i+1, \ldots, j\}$.
$S(y,d)$, $E(d)$, $\eta(\theta)$, $A(\theta)$, $h(y,d)$	Family-specific statistic contribution, exposure multiplier, natural-parameter map, log-partition function, and base measure for the chosen exponential-family block model.
$m_\star(\theta) = m(\theta; d_\star)$	Declared segment functional on the reporting scale, evaluated at a fixed reference descriptor $d_\star$ when the observation mean depends on exposure or trial structure. Examples include a Gaussian mean, a Poisson rate at unit exposure, and a Binomial success probability.
(α, β)	Conjugate hyperparameters with normalizer $Z(\alpha,\beta)$ defined in (3.1).
(α_0, β_0)	Prior hyperparameters before observing block data.
S_{ij}	Block statistic sum $\sum_{t=i+1}^{j} a_t S(y_t, d_t)$.
W_{ij}	Block log-partition multiplier $\sum_{t=i+1}^{j} a_t E(d_t)$.
H_{ij}	Powered base-measure sum $\sum_{t=i+1}^{j} a_t \log h(y_t, d_t)$.
$A_{ij}^{(0)}$	Block evidence on $(i,j]$.
$A_{ij}^{(r)}$	Block moment numerator $A_{ij}^{(0)}\, \mathbb{E}[m_\star(\theta)^r \mid y_{i+1:j}]$ (for integer $r \in \{1,2,\ldots\}$; conjugate-case definition). In the non-conjugate Laplace discussion of Section 10, the superscript is a test function rather than a moment order; those objects are denoted $M_{ij}^{[g]}$ to avoid collision.
$\widetilde{A}_{ij}^{(r)}$	Prior-adjusted block quantity $A_{ij}^{(r)} \gamma_x(i,j)$, where γ_x combines cohesion and an interior-boundary hazard when appropriate.
$L_{k,j}$	Prefix evidence for $y_{1:j}$ with exactly k segments.
$R_{k,i}$	Suffix evidence for $y_{i+1:n}$ with exactly k segments.
C_k	Normalizer of $p(t \mid k)$ under a segment-factorized partition prior.
$p(k \mid y)$	Posterior over segment counts.
$p(t_p = h \mid y, k)$	Marginal posterior of the pth boundary given k.
$p(b_i = 1 \mid y, k)$	Boundary-event marginal at interior index i, equal to $\sum_{p=1}^{k-1} p(t_p = i \mid y, k)$. This is the calibration target in Section 13.
$\widehat{t}^{\,\mathrm{MAP}}(k)$	Joint MAP segmentation given k, obtained by max-sum DP and backtracking.

$\widehat{\mu}_q^{(r)}$	Posterior moment of the segment-q mean parameter conditional on a chosen segmentation.
$\widehat{f}_t^{\mathrm{Bayes}}$	Bayes regression curve at index t (posterior mean of the latent piecewise-constant signal).
$s \in \{1, \dots, S\}$	Sequence / subject index in multi-sequence settings.
$g \in \{1, \dots, G\}$	Group index.
r_{sg}	Auxiliary allocation weight for sequence s and latent group g in the finite latent-group criterion.
$\mathcal{M}_g$	Fitted group-specific model exported for prediction.
ℓ_g	Log predictive score of new data under group g.
π_g	Prior group weight used for prediction or in the score-template objective.
ε	Uniform per-block log-evidence error tolerance for non-conjugate block approximations; appears as the hypothesis controlling the posterior-odds stability bound in Section 10.
$\widehat{A}_{ij}^{(0)}$	Approximate (non-conjugate) block evidence; satisfies $\lvert \log \widehat{A}_{ij}^{(0)} - \log A_{ij}^{(0)} \rvert \le \varepsilon$ under the stability assumption.
ϕ_t	Known per-observation precision used in the Beta-block likelihood for $y_t \in (0,1)$.
ω_t	Pólya–Gamma augmentation variable for logistic / negative-binomial mean-field block approximations.
$M_{ij}^{[g]}$	Block test-function numerator $\int g(\theta)\, p(y_{i+1:j} \mid \theta)\, p(\theta)\, d\theta$, used in the Laplace-approximation discussion of Section 10 where the superscript denotes a test function rather than a moment order.
f_{Lap}, f_{JJ}, f_{EP}, f_{PG}, f_{Q}	Local block-evidence approximations: Laplace, Jaakkola–Jordan variational bound, expectation propagation, Pólya–Gamma mean field, and one-dimensional quadrature. Each instantiates $\widehat{A}_{ij}^{(0)}$ and inherits the same DP layer.

Remarks. Two recurring distinctions: *design irregularity* may enter through the partition prior (§7.1); d_i enters the likelihood only when the observation model supplies genuine exposures or known precisions. *Boundary marginals* and the *joint MAP segmentation* are different posterior summaries (§5) and must not be conflated.

BayesBreak addresses *offline* Bayesian segmentation of ordered data. Throughout, the fundamental unknown is a contiguous partition of the index set $\{1, \dots, n\}$ together with a segment parameter attached to each block. This section states the objects of interest in a model-agnostic way; Section 3.1 introduces the detailed notation used later.

Unless stated otherwise, all inferential statements in the book use the following standing assumption.

> **Assumption 3.1** (Offline segmentation support and block independence)**.** The observations are ordered on a fixed set of candidate boundary locations and the segment-count set

$\{1, \dots, k_{\max}\}$ is finite. Structural admissibility of a candidate block is determined before observing the response values by the candidate grid, declared segment-length restrictions, and prior support. Every structurally admissible block has finite nonnegative block evidence; a zero evidence contributes zero to the likelihood-weighted partition sum but does not change the prior support or its normalizer C_k. Structurally excluded blocks have local prior factor zero and are stored with log-score $-\infty$. Segment parameters are conditionally independent across blocks given a segmentation, and the conditional partition prior factorizes over segments as in Assumption 3.6. For approximate non-conjugate blocks, the same statement applies to the supplied approximate score array, while comparison with the exact non-conjugate model requires the additional error hypothesis of Assumption 10.6.

3.1.1 Single-sequence segmentation and Bayes regression

We observe ordered pairs $\{(x_i, y_i)\}_{i=1}^n$ with strictly increasing design points $x_1 < \cdots < x_n$. The design can be irregular. When the observation model includes known exposures, trial counts, or precision weights, we also allow deterministic nonnegative quantities w_i; unless explicitly stated otherwise, point-observation models use $w_i \equiv 1$.

A segmentation into k contiguous blocks is represented by a boundary vector

$$0 = t_0 < t_1 < \cdots < t_{k-1} < t_k = n, \tag{3.2}$$

where block $q \in \{1, \dots, k\}$ is the index set

$$I_q(t) := \{t_{q-1} + 1, \dots, t_q\}. \tag{3.3}$$

Definition 3.2 (Segmentation space and structural support)**.** For fixed n and k, let

$$\mathcal{T}_{k,n} := \{t = (t_0, \dots, t_k) : 0 = t_0 < t_1 < \cdots < t_k = n\}.$$

The *structural support* $\mathcal{T}_{k,n}^+ \subseteq \mathcal{T}_{k,n}$ consists of those boundary vectors for which every block satisfies the declared grid and length restrictions and has positive local prior factor $\gamma_x(t_{q-1}, t_q)$. Conditional on the design and model specification, this set is fixed independently of the observed response values and of $A_{ij}^{(0)}$. A structurally admissible partition may have zero likelihood for the observed data; it then contributes zero to the posterior partition sum, while the prior normalizer C_k continues to sum over the full structural support.

The corresponding segment parameter is denoted $\theta_q \in \Theta$. Conditionally on $(k, t, \theta_{1:k})$, observations factorize across blocks:

$$p(y_{1:n} \mid k, t, \theta_{1:k}) = \prod_{q=1}^{k} \prod_{i \in I_q(t)} p(y_i \mid \theta_q, w_i), \tag{3.4}$$

where $p(y_i \mid \theta_q, w_i)$ is interpreted according to the chosen family. For example, depending on the family, w_i encodes exposure in Poisson models, trial count in Binomial models, or known precision in Gaussian models. Irregular spacing by itself does not force a non-unit w_i; design geometry can instead enter through the partition prior.

For any candidate block $(i, j]$ with $0 \le i < j \le n$, the key primitive is the integrated

single-block evidence

$$\mathcal{L}(i, j; \alpha_0, \beta_0) \coloneqq p(y_{i+1:j} \mid \alpha_0, \beta_0) = \int \prod_{t=i+1}^{j} p(y_t \mid \theta, w_t)\, p(\theta \mid \alpha_0, \beta_0)\, d\theta, \tag{3.5}$$

where $p(\theta \mid \alpha_0, \beta_0)$ is the segment prior with hyperparameters (α_0, β_0) (see Section 3.1). When $\mathcal{L}(i, j)$ is available in closed form, the entire offline posterior can be computed exactly by dynamic programming. When it is only approximated, BayesBreak isolates the approximation at the block level and reuses the same global DP.

Definition 3.3 (Boundary-event variables, Bayes curve, and exported estimator)**.** For a boundary vector $t \in \mathcal{T}_{k,n}$, define the interior boundary indicator $b_i(t) := \sum_{p=1}^{k-1} \mathbf{1}\{t_p = i\}$ for $i = 1, \ldots, n-1$. Because the boundaries are strictly ordered, $b_i(t) \in \{0, 1\}$. For fixed k, the Bayes regression curve at index i is the posterior mean of the observation-scale segment mean covering i after marginalizing over $t \in \mathcal{T}_{k,n}^{+}$ and over segment parameters. The exported segmentation estimator is a deterministic boundary vector, usually the joint MAP vector $\widehat{t}^{\,\mathrm{MAP}}(k)$, paired with segment-level posterior summaries for downstream prediction. These two objects need not coincide because the Bayes curve averages over boundary uncertainty whereas the exported estimator chooses one segmentation.

The inferential targets are:

(i) the marginal evidence at fixed segment count,

$$p(y_{1:n} \mid k) = \sum_{t \in \mathcal{T}_{k,n}^{+}} p(t \mid k) \prod_{q=1}^{k} \mathcal{L}(t_{q-1}, t_q);$$

(ii) the posterior $p(k, t \mid y)$ over segment counts and boundaries and its marginals $p(k \mid y)$, $p(t_p = h \mid y, k)$, and the boundary-event marginal $p(b_i = 1 \mid y, k) = \sum_{p=1}^{k-1} p(t_p = i \mid y, k)$ (the calibration target of §13);

(iii) posterior summaries of the latent piecewise-constant signal, namely $p(\theta_i \mid y)$ and $\mathbb{E}[m_\star(\theta_i) \mid y]$ with $m_\star$ the observation-scale mean parameter (the Bayes regression curve; conditional on the selected $\widehat{k}$ unless stated otherwise);

(iv) point estimators for downstream use. In particular the *joint* MAP segmentation $\widehat{t}^{\,\mathrm{MAP}}(k) \in \arg\max_t p(t \mid y, k)$ differs from the vector of marginal boundary modes $\arg\max_h p(t_p = h \mid y, k)$ and is recovered by max-sum DP. The prior on k and the normalizer $\log C_k$ enter the objective only when also optimizing over k.

The three posterior summaries entering downstream analysis are isolated as labelled objects so that later sections can refer to them without re-deriving notation.

Definition 3.4 (Posterior summaries: marginal boundaries, segment count, and joint MAP)**.** Fix a sequence $y = y_{1:n}$, a maximum segment count $k_{\max}$, and Assumption 3.1. Define:

(a) the *marginal boundary posterior* $\mathrm{MB}_i(k) := p(b_i = 1 \mid y, k)$ for interior indices $i \in \{1, \ldots, n-1\}$ and admissible $k \in \{1, \ldots, k_{\max}\}$;

(b) the *posterior over segment count* $\mathrm{PK}(k) := p(k \mid y) \propto p(k)\, p(y \mid k)$ for $k \in \{1, \ldots, k_{\max}\}$;

(c) the *joint MAP segmentation at fixed* k $\widehat{t}^{\,\mathrm{MAP}}(k) := \arg\max_{t \in \mathcal{T}_{k,n}^{+}} p(t \mid y, k)$, with ties broken by a deterministic rule (Section 5.0.5).

These three objects are returned by distinct DP recursions (sum-product, sum-product with k-aggregation, and max-sum, respectively), and they are not in general consistent with each other: MB_i and $\widehat{t}^{\mathrm{MAP}}(k)$ can place boundary mass at different indices (Remark 5.3), and the mode of PK need not coincide with the cardinality of any single thresholded set of marginal boundaries.

3.1.2 Multi-sequence structure

Many applications provide multiple related sequences observed on a common ordered grid or on a common set of candidate boundary locations. Let sequence $s \in \{1, \dots, S\}$ be denoted by $y_{1:n}^{(s)}$, with optional weights $w_i^{(s)}$ and subject-specific segment parameters $\theta_q^{(s)}$ when boundaries are shared. We use $z_s \in \{1, \dots, G\}$ for group membership, $\mathcal{S}_g := \{s : z_s = g\}$ for known groups, and r_{sg} for latent-group responsibilities. BayesBreak treats three progressively richer settings.

Definition 3.5 (Multi-sequence segmentation classes)**.** A shared-boundary replicate model has one boundary vector t and subject-specific segment parameters $\theta_{1:k}^{(s)}$. A known-group model has observed labels z_s and one boundary vector $t^{(g)}$ per group. A latent-template model has unobserved labels z_s and a finite set of templates $\tau_g = (k_g, t^{(g)})$ optimized under the finite latent-group criterion of Section 9.3. These three classes preserve the block-evidence interface; fully Bayesian averaging over both unknown labels and all group segmentations is a different target and is not claimed for the score-template model.

Shared-boundary replicates. All sequences share a common segmentation $t \in \mathcal{T}_{k,n}^{+}$, but each sequence has its own within-block parameters $\theta_{1:k}^{(s)}$. The common-grid requirement is an assumption on the candidate boundary coordinates; subject-specific missingness or exposures are handled inside the block evidences. Conditional on t, the sequences are independent:

$$p(\{y^{(s)}\}_{s=1}^{S} \mid t, \{\theta^{(s)}\}) = \prod_{s=1}^{S} \prod_{q=1}^{k} \prod_{i \in I_q(t)} p\big(y_i^{(s)} \mid \theta_q^{(s)}, w_i^{(s)}\big). \tag{3.6}$$

Integrating out the subject-specific segment parameters yields pooled block evidences obtained by multiplying subject-level block evidences.

Known groups. Sequences are partitioned into known groups $g \in \{1, \dots, G\}$. Each group has its own segmentation $t^{(g)} \in \mathcal{T}_{k_g,n}^{+}$, while sequences within a group share that segmentation and retain their own segment parameters. Conditional independence is within group and across subjects given $t^{(g)}$.

Latent groups. Group labels are unknown. A fully Bayesian formulation averages over both assignments and group-specific segmentations. BayesBreak adopts a tractable *template-mixture* formulation: each group is associated with a single latent boundary template $\tau_g = (k_g, t^{(g)})$, and coordinate ascent alternates between soft assignments (normalized auxiliary weights for the current templates) and exact responsibility-weighted max-sum DP updates of those templates. The finite latent-group criterion is non-decreasing under the exact alternating updates (Theorem 9.1); this is a finite-template optimization model rather than exact Bayesian averaging over latent-group segmentations.

3.1.3 Computational scope

For fixed $k_{\max}$, the core BayesBreak workflow consists of:

(a) precomputing all block evidences and required moment numerators;
(b) running sum-product DP recursions for evidences and posterior marginals; and
(c) optionally running a max-sum/backtracking recursion for a joint MAP segmentation.

When block evaluation is constant time after prefix-sum preprocessing, this yields $\mathcal{O}(n^2)$ block precomputation and $\mathcal{O}(k_{\max}n^2)$ DP time. These worst-case costs make the exact method most natural for moderate sequence lengths (hundreds to low thousands), where exact posterior quantities are still computationally feasible and often worth the cost.

3.1.4 Prediction as a first-class inferential target

Beyond posterior inference on the training sequence(s), BayesBreak treats *prediction for new data* as an additional inferential target at the same level as evidence, posteriors, and point estimators. Given a fitted model $\mathcal{M}_g$ (one per group) and new observations $(X^{\text{new}}, y^{\text{new}})$, the relevant quantities are (i) group-conditional posterior-predictive likelihoods $p(y^{\text{new}} \mid \mathcal{M}_g)$, (ii) group membership posteriors $p(g \mid y^{\text{new}})$, and (iii) signal predictions $\widehat{f}_g^{\text{MAP}}(x^\star)$ or $\widehat{f}_g^{\text{Bayes}}(x^\star)$ at query points $x^\star$. Section 11.1 develops the prediction layer in detail, covering pointwise, set-valued, and vector-valued new data, and separates exported-segmentation scoring from optional full resegmentation of new data.

3.2 Exponential-family segment likelihood with exposure and power weights

Let d_t collect family-specific known information, such as Poisson exposure, Binomial trial count, or Gaussian precision. Let $a_t \geq 0$ be an optional power or reliability weight that, when used, multiplies the complete log contribution. The canonical local model is

$$p(y_t \mid \theta, d_t)^{a_t} = h(y_t, d_t)^{a_t} \exp\Big\{ a_t S(y_t, d_t)^\top \eta(\theta) - a_t E(d_t) A(\theta) \Big\}. \tag{3.7}$$

Thus

$$S_{ij} = \sum_{t=i+1}^{j} a_t S(y_t, d_t), \qquad W_{ij} = \sum_{t=i+1}^{j} a_t E(d_t), \qquad H_{ij} = \sum_{t=i+1}^{j} a_t \log h(y_t, d_t).$$

This notation is deliberately more general than a scalar "weight": a Binomial trial count changes both the statistic and the log-partition coefficient, while a power weight scales the entire observation contribution.

Family	$S(y, d)$	$E(d)$	Role of d
Gaussian, known precision ω	ωy	ω	precision/exposure in the quadratic family
Poisson, exposure e	y	e	mean is $e\lambda$; base measure contains $e^y/y!$
Binomial, trials m	y	m	log-partition is multiplied by the trial count
Unweighted point observation	$T(y)$	1	d is empty and $a = 1$

Table 3.2. Valid specializations of the exposure-aware segment likelihood. Optional a_t is a separate likelihood-power weight.

3.3 Ordered partitions and factorized priors

Assumption 3.6 (Factorizing ordered-partition prior)**.** For fixed k and candidate-boundary coordinates x, the prior on $t = (0 = t_0 < \cdots < t_k = n)$ has the form

$$p(t \mid k, x) = C_k(x)^{-1} \prod_{q=1}^{k} c_x(t_{q-1}, t_q) \prod_{q=1}^{k-1} h_x(t_q), \tag{3.8}$$

where $c_x(i, j) \geq 0$ is a segment cohesion and $h_x(j) \geq 0$ is an interior-boundary hazard. The terminal endpoint n receives no hazard factor. Define $\gamma_x(i, j) = c_x(i, j)h_x(j)^{\mathbf{1}\{j<n\}}$.

The normalizer is

$$C_k(x) = \sum_{t \in \mathcal{T}_{k,n}^{+}} \prod_{q=1}^{k} c_x(t_{q-1}, t_q) \prod_{q<k} h_x(t_q), \tag{3.9}$$

and the prior-adjusted local arrays are

$$\widetilde{A}_{ij}^{(r)} = A_{ij}^{(r)} \gamma_x(i, j). \tag{3.10}$$

This factorization subsumes an index-uniform prior, a length cohesion, a local boundary-intensity model, or their product without confusing their interpretations.

CHAPTER 4

Integrated block evidence and posterior moments

CHAPTER PATH

One local interface covers exact conjugate families and clearly labeled deterministic quadrature without changing the global partition engine.

Plate model (single segment). For a block $(i, j]$ and conjugate hyperparameters (α_0, β_0), the generative model is

$$\theta \sim p(\theta \mid \alpha_0, \beta_0), \qquad y_t \mid \theta \overset{\text{ind}}{\sim} p(y_t \mid \theta, d_t) \text{ as in (3.7)}, \quad t \in \{i+1, \dots, j\}. \tag{4.1}$$

The block evidence $A_{ij}^{(0)}$ is obtained by marginalizing θ out of (4.1).

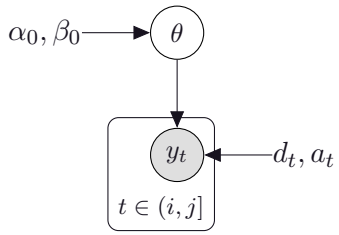


Figure 4.1. Plate model for a single conjugate EF block. Shaded nodes are observed; unshaded are latent; point nodes are fixed hyperparameters and family descriptors or power weights. Marginalizing θ gives $A_{ij}^{(0)}$.

4.1 Conjugate block integration

Assume the segment likelihood in Equation (3.7) and the conjugate prior

$$p(\theta \mid \alpha_0, \beta_0) = \exp\{\eta(\theta)^\top \alpha_0 - \beta_0 A(\theta) - \log Z(\alpha_0, \beta_0)\}.$$

Theorem 4.1 (Exposure-aware conjugate block integration). *For a candidate block $(i, j]$, if $Z(\alpha_0 + S_{ij}, \beta_0 + W_{ij}) < \infty$, then*

$$A_{ij}^{(0)} = \exp\{H_{ij}\} \frac{Z(\alpha_0 + S_{ij}, \beta_0 + W_{ij})}{Z(\alpha_0, \beta_0)}. \tag{4.2}$$

The posterior remains conjugate with hyperparameters $(\alpha_0 + S_{ij}, \beta_0 + W_{ij})$. *For any measurable observation-scale target* $m_\star(\theta)$ *whose* r*th posterior moment exists,*

$$A_{ij}^{(r)} = A_{ij}^{(0)}\, \mathbb{E}[m_\star(\theta)^r \mid y_{i+1:j}, d_{i+1:j}, a_{i+1:j}]. \tag{4.3}$$

Proof. Multiplying the powered likelihood contributions gives

$$\exp\{H_{ij}\} \exp\{\eta(\theta)^\top S_{ij} - W_{ij} A(\theta)\}.$$

After multiplication by the prior, the terms depending on θ are exactly the conjugate kernel with updated hyperparameters $\alpha_0 + S_{ij}$ and $\beta_0 + W_{ij}$. Integrating this kernel yields its normalizer and hence Equation (4.2). Division by that evidence gives the updated posterior. Finally, inserting $m_\star(\theta)^r$ into the same integral factors it as the evidence times the corresponding posterior moment, provided the moment is finite. □

Remark 4.2 (Scope of "exact"). The theorem is exact for the stated powered/exposure-aware model. It does not authorize treating an arbitrary sample weight as exposure, a Binomial trial count as a likelihood power, or an approximate numerical integral as a conjugate closed form.

4.2 Family-specific derivations

We now instantiate (4.2)–(4.3) for three closed-form conjugate families (Gaussian, Poisson, and Binomial), an overdispersed count model with a conjugate distributional parameter, and one low-dimensional quadrature example for $(0, 1)$-valued responses. In each case we give (i) the segment model and prior, (ii) posterior hyperparameters, (iii) the block evidence $A_{ij}^{(0)}$, and (iv) first two posterior moments on the scale used by the Bayes-curve interface. For non-Gaussian families this scale must be stated explicitly: natural-parameter, distribution-parameter, and observation-mean moments are not interchangeable. Implementation routines are provided immediately after each derivation (Algorithms 1–5).

4.2.1 Gaussian with known variance (heteroscedastic via weights)

Proposition 4.3 (Gaussian-block evidence). *Suppose* $y_t \mid \mu \sim \mathcal{N}(\mu, \sigma^2/w_t)$ *with known variance* σ^2 *and positive weights* $w_t > 0$, *and a Gaussian conjugate prior* $\mu \sim \mathcal{N}(\nu, \rho^2)$. *Define the block summaries* $n_{ij} = j - i$, $W_{ij} = \sum_{t=i+1}^{j} w_t$, $S_{ij} = \sum_{t=i+1}^{j} w_t y_t$, $Q_{ij} = \sum_{t=i+1}^{j} w_t (y_t - \nu)^2$, *and* $\kappa_{ij} = \rho^2 W_{ij} + \sigma^2$. *Then*

$$\mu \mid y_{(i,j]} \sim \mathcal{N}\Big(\tfrac{\rho^2 S_{ij} + \sigma^2 \nu}{\kappa_{ij}},\ \tfrac{\sigma^2 \rho^2}{\kappa_{ij}}\Big),$$

and the block evidence admits the closed form

$$A_{ij}^{(0)} = (2\pi\sigma^2)^{-n_{ij}/2} \Big(\prod_{t=i+1}^{j} w_t^{1/2} \Big) \Big(1 + \tfrac{\rho^2}{\sigma^2} W_{ij}\Big)^{-1/2} \exp\Big\{ -\frac{Q_{ij}}{2\sigma^2} + \frac{(S_{ij} - \nu W_{ij})^2}{2(\sigma^2 W_{ij} + \sigma^4/\rho^2)} \Big\}.$$

The corresponding first and second moment numerators are $A_{ij}^{(1)} = A_{ij}^{(0)}\, \mathbb{E}[\mu \mid y_{(i,j]}]$ *and* $A_{ij}^{(2)} = A_{ij}^{(0)} \left(\mathbb{V}[\mu \mid y_{(i,j]}] + \mathbb{E}[\mu \mid y_{(i,j]}]^2\right)$.

Proof. Multiplying the weighted Gaussian likelihood by the Gaussian prior produces a quadratic kernel in μ. Completing the square gives posterior precision $\rho^{-2} + W_{ij}/\sigma^2$, posterior mean ν_B, and the stated normalizing-factor ratio. Multiplying the evidence by the first two moments of $\mathcal{N}(\nu_B, \rho_B^2)$ yields the moment numerators. □

The proof specializes Theorem 4.1 to the Gaussian–Gaussian conjugate pair with weighted base measure.

Implementation. Algorithm 1 gives a numerically stable implementation of the Gaussian block log-evidence and posterior moments using cumulative weighted sums. The routine is written so that the DP layer consumes only $\log \mathcal{L}_{ij}$ and does not depend on Gaussian-specific algebra.

Algorithm 1: GaussianBlock$(y, w, i, j, \nu, \rho^2, \sigma^2)$

Input: Block indices $(i, j]$; observations $y_{i+1:j}$; weights $w_{i+1:j}$

$n \leftarrow j - i$; $\quad W \leftarrow \sum_{t=i+1}^{j} w_t$; $\quad S \leftarrow \sum_{t=i+1}^{j} w_t y_t$; $\quad Q \leftarrow \sum_{t=i+1}^{j} w_t (y_t - \nu)^2$;

$\kappa \leftarrow \rho^2 W + \sigma^2$;

$\log A0 \leftarrow -\frac{1}{2} n \log(2\pi) - \frac{1}{2} \sum_{t=i+1}^{j} \log(\sigma^2 / w_t) - \frac{Q}{2\sigma^2} + \frac{(S - \nu W)^2}{2(\sigma^2 W + \sigma^4/\rho^2)} - \frac{1}{2} \log\left(1 + \frac{\rho^2}{\sigma^2} W\right)$;

$A0 \leftarrow \exp(\log A0)$;

$\mu_{\text{mean}} \leftarrow (\rho^2 S + \sigma^2 \nu)/\kappa$; $\quad \mu_{\text{var}} \leftarrow \sigma^2 \rho^2 / \kappa$;

Output: $A0$, μ_{mean}, μ_{var}

4.2.2 Poisson with exposure

Proposition 4.4 (Poisson-block evidence). *For $y_t \mid \lambda \sim \mathrm{Pois}(\lambda w_t)$ with $w_t > 0$ and a Gamma conjugate prior $\lambda \sim \mathrm{Gamma}(a_0, b_0)$, the block posterior is $\lambda \mid y_{(i,j]} \sim \mathrm{Gamma}(a_0 + C_{ij}, b_0 + W_{ij})$, where $C_{ij} = \sum_{t=i+1}^{j} y_t$ and $W_{ij} = \sum_{t=i+1}^{j} w_t$. The block evidence is*

$$A_{ij}^{(0)} = \exp\{H_{ij}\} \frac{b_0^{a_0} \, \Gamma(a_0 + C_{ij})}{\Gamma(a_0) \, (b_0 + W_{ij})^{a_0 + C_{ij}}}, \qquad H_{ij} = \sum_{t=i+1}^{j} \left[y_t \log w_t - \log(y_t!) \right],$$

and moment numerators take the standard form $A_{ij}^{(r)} = A_{ij}^{(0)} \, \mathbb{E}[\lambda^r \mid y_{(i,j]}]$ for $r \in \{1, 2, \dots\}$.

Proof. The block likelihood is $\{\prod_t e_t^{y_t} / y_t!\} \lambda^{C_{ij}} e^{-W_{ij}\lambda}$. Multiplication by the Gamma(a_0, b_0) density gives a Gamma kernel with shape $a_0 + C_{ij}$ and rate $b_0 + W_{ij}$. Integrating the kernel gives the evidence; its ordinary moments give the stated moment numerators. □

The proof specializes Theorem 4.1 to the Gamma–Poisson conjugate pair.

Implementation. Algorithm 2 summarizes the Poisson-with-exposure block evidence computation (and associated posterior summaries) in a form that matches the BayesBreak block-evidence interface.

Algorithm 2: PoissonBlock(y, w, i, j, a_0, b_0)

Input: Block $(i, j]$; counts $y_{i+1:j}$; exposures $w_{i+1:j}$

$C \leftarrow \sum_{t=i+1}^{j} y_t$; $\quad W \leftarrow \sum_{t=i+1}^{j} w_t$;

$\log A0 \leftarrow \sum_{t=i+1}^{j} y_t \log w_t - \sum_{t=i+1}^{j} \log(y_t!) + a_0 \log b_0 + \log \Gamma(a_0 + C) - \log \Gamma(a_0) - (a_0 + C) \log(b_0 + W)$;

$A0 \leftarrow \exp(\log A0)$;

$\lambda_{\text{mean}} \leftarrow (a_0 + C)/(b_0 + W)$; $\quad \lambda_{\text{var}} \leftarrow (a_0 + C)/(b_0 + W)^2$;

Output: $A0$, λ_{mean}, λ_{var}

4.2.3 Binomial with Beta prior

Proposition 4.5 (Binomial-block evidence). *For $y_t \mid p \sim \mathrm{Binom}(m_t, p)$ with known trial counts m_t and a Beta conjugate prior $p \sim \mathrm{Beta}(a_0, b_0)$, the block posterior is $p \mid y_{(i,j]} \sim \mathrm{Beta}(a_0 + C_{ij}, b_0 + M_{ij} - C_{ij})$, where $C_{ij} = \sum_{t=i+1}^{j} y_t$ and $M_{ij} = \sum_{t=i+1}^{j} m_t$. The block evidence is*

$$A_{ij}^{(0)} = \exp\{H_{ij}\} \frac{B(a_0 + C_{ij},\, b_0 + M_{ij} - C_{ij})}{B(a_0, b_0)}, \qquad H_{ij} = \sum_{t=i+1}^{j} \log \binom{m_t}{y_t},$$

with moment numerators $A_{ij}^{(r)} = A_{ij}^{(0)}\, \mathbb{E}[p^r \mid y_{(i,j]}]$.

Proof. Collecting success and failure powers gives $p^{C_{ij}}(1-p)^{M_{ij}-C_{ij}}$ times the product of binomial coefficients. Multiplication by the Beta prior gives a $\mathrm{Beta}(a_0 + C_{ij}, b_0 + M_{ij} - C_{ij})$ kernel. The ratio of Beta normalizers is the evidence, and Beta moments yield the remaining formulas. □

The proof specializes Theorem 4.1 to the Beta–Binomial conjugate pair.

Implementation. Algorithm 3 provides pseudocode for Binomial/Bernoulli block evidence evaluation under Beta conjugacy, along with posterior mean/variance computations used for segment-wise moments and MAP/Bayes curve summaries.

Algorithm 3: BinomialBlock(y, m, i, j, a_0, b_0)

Input: Block $(i, j]$; successes $y_{i+1:j}$; trials $m_{i+1:j}$

$C \leftarrow \sum_{t=i+1}^{j} y_t$; $\quad M \leftarrow \sum_{t=i+1}^{j} m_t$;

$\log A0 \leftarrow \sum_{t=i+1}^{j} \log \binom{m_t}{y_t} + \log B(a_0 + C, b_0 + M - C) - \log B(a_0, b_0)$;

$A0 \leftarrow \exp(\log A0)$;

$p_{\text{mean}} \leftarrow (a_0 + C)/(a_0 + b_0 + M)$;

$p_{\text{var}} \leftarrow \frac{(a_0+C)(b_0+M-C)}{(a_0+b_0+M)^2(a_0+b_0+M+1)}$;

Output: $A0$, p_{mean}, p_{var}

4.2.4 Negative-Binomial block (overdispersed counts) with fixed dispersion

Proposition 4.6 (Negative-Binomial-block evidence)**.** *Fix dispersion $r > 0$ and assume $y_t \mid p \sim \mathrm{NegBin}(r, p)$ with Beta conjugate prior $p \sim \mathrm{Beta}(a_0, b_0)$. The block posterior is $p \mid y_{(i,j]} \sim \mathrm{Beta}(a_0 + N_{ij},\, b_0 + C_{ij})$, where $C_{ij} = \sum_{t=i+1}^{j} y_t$ and $N_{ij} = \sum_{t=i+1}^{j} r$. The block evidence is*

$$A_{ij}^{(0)} = \exp\{H_{ij}\}\,\frac{B(a_0 + N_{ij},\, b_0 + C_{ij})}{B(a_0, b_0)}, \qquad H_{ij} = \sum_{t=i+1}^{j} \log \binom{y_t + r - 1}{y_t},$$

and the moment numerators $A_{ij}^{(r)}$ on the observation-mean scale are obtained from the Beta posterior moments via the count-mean transformation $m_*(p) = r_*(1-p)/p$.

Proof. Under the displayed parameterization, the block likelihood is the product of combinatorial terms times $p^{N_{ij}}(1-p)^{C_{ij}}$. Multiplication by the Beta prior therefore updates its two shape parameters to $a_0 + N_{ij}$ and $b_0 + C_{ij}$. Integration gives the Beta-normalizer ratio. The observation-scale moments follow by integrating $r_*(1-p)/p$ and its square, which requires the stated posterior-shape conditions. □

Remark 4.7 (Observation-scale vs. parameter-scale moments). The Beta posterior moments of p are useful diagnostics, but the Bayes-curve target on the observation scale is the count mean $m_*(p) = r_*(1-p)/p$ for a future count with dispersion r_*. With $a_B = a_0 + N_{ij}$ and $b_B = b_0 + C_{ij}$, $\mathbb{E}[m_*(p) \mid (i,j]] = r_*\, b_B/(a_B - 1)$ when $a_B > 1$, and $\mathbb{V}[m_*(p) \mid (i,j]] = r_*^2\, b_B(a_B + b_B - 1)/\{(a_B-1)^2(a_B-2)\}$ when $a_B > 2$. Moment numerators $A_{ij}^{(r)}$ therefore multiply $A_{ij}^{(0)}$ by the observation-mean moments, not by raw Beta moments of p, unless the exported quantity is explicitly labelled parameter-scale. When r is unknown, a semi-conjugate EM or one-dimensional quadrature in r can be layered on top.

Implementation. Algorithm 4 summarizes the Negative-Binomial block computation in the same interface as Algorithms 1–3.

Algorithm 4: NegBinBlock(y, r, i, j, a_0, b_0)

Input: Block $(i, j]$; counts $y_{i+1:j}$; dispersion r (possibly per-t)

$C \leftarrow \sum_{t=i+1}^{j} y_t; \quad N \leftarrow \sum_{t=i+1}^{j} r_t;$

$\log A0 \leftarrow \sum_{t=i+1}^{j} \log \binom{y_t + r_t - 1}{y_t} + \log B(a_0 + N, b_0 + C) - \log B(a_0, b_0);$

$A0 \leftarrow \exp(\log A0);$

$a_B \leftarrow a_0 + N; \quad b_B \leftarrow b_0 + C;$

`mean_per_disp` $\leftarrow b_B/(a_B - 1)$ if $a_B > 1$;
`var_per_disp` $\leftarrow b_B(a_B + b_B - 1)/\{(a_B-1)^2(a_B-2)\}$ if $a_B > 2$;

Output: $A0$, posterior summaries for p, and observation-mean moments $r_* \times$ `mean_per_disp` when a prediction dispersion r_* is specified

4.2.5 Beta-distributed observations with known precision

Proposition 4.8 (Beta-block evidence as a one-dimensional integral)**.** *Assume $y_t \in (0,1)$ with latent mean μ constant within a block, known precisions $\phi_t > 0$, and prior $\mu \sim \mathrm{Beta}(a_0, b_0)$. Define*

$$\ell_{ij}(\mu) := \sum_{t=i+1}^{j} \log \mathrm{BetaPDF}(y_t \mid \phi_t \mu, \phi_t(1-\mu)),$$
$$\log \pi(\mu) := \log \mathrm{BetaPDF}(\mu \mid a_0, b_0).$$

Then the exact block evidence and moment numerators are the one-dimensional integrals

$$A_{ij}^{(r)} = \int_0^1 \mu^r \exp\{\ell_{ij}(\mu) + \log \pi(\mu)\}\, d\mu, \qquad r \in \{0, 1, 2, \dots\}.$$

A Q-node Gauss–Legendre rule on $(0,1)$ gives the deterministic approximation

$$\widehat{A}_{ij}^{(r)} = \sum_{q=1}^{Q} w_q^{\mathrm{GL}}\, \mu_q^r \exp\{\ell_{ij}(\mu_q) + \log \pi(\mu_q)\}.$$

Its convergence and error rate depend on the endpoint behavior and regularity of the transformed integrand. No monotonicity in Q and no routine-wide algebraic rate are asserted here.

Proof. The unknown block parameter is one-dimensional and supported on $(0,1)$. Multiplication of the Beta-observation likelihood by the Beta prior gives the displayed nonnegative integrand. Its integral is exactly the block evidence; multiplication by μ^r gives the moment numerator. The finite weighted sum is the Gauss–Legendre approximation after the standard affine map from $(-1,1)$ to $(0,1)$. □

Remark 4.9 (Quadrature error assessment). The block log-posterior $\ell_{ij}(\mu) + \log \pi(\mu)$ is not generally log-concave and may have non-smooth endpoint behavior after exponentiation. Implementations therefore compare nested node counts or an independently more accurate reference calculation, evaluate in log space, and inspect endpoint contributions. A certified error statement must be tied to the chosen transformation, node rule, and integrand regularity.

Implementation. Algorithm 5 implements this one-dimensional quadrature routine. It is useful for continuous proportions and other $(0,1)$-valued signals, and is a deterministic non-conjugate block integral rather than a conjugate closed form.

Algorithm 5: BetaObsBlock$(y, \phi, i, j, a_0, b_0, \{\mu_g, w_g\}_{g=1}^{G})$

Input: Block $(i, j]$; observations $y_{i+1:j} \in (0,1)$; known precisions $\phi_{i+1:j}$; Beta prior (a_0, b_0); quadrature nodes $\{\mu_g, w_g\}$

$S_0 \leftarrow 0$, $S_1 \leftarrow 0$, $S_2 \leftarrow 0$;

for $g = 1$ **to** G **do**

  $\log \ell_g \leftarrow \sum_{t=i+1}^{j} \log \mathrm{BetaPDF}\big(y_t \mid \phi_t \mu_g, \phi_t(1-\mu_g)\big)$;

  $\log \pi_g \leftarrow \log \mathrm{BetaPDF}(\mu_g \mid a_0, b_0)$;

  $v_g \leftarrow \exp(\log \ell_g + \log \pi_g)$;

  $S_0 \leftarrow S_0 + w_g v_g$; $S_1 \leftarrow S_1 + w_g v_g \mu_g$; $S_2 \leftarrow S_2 + w_g v_g \mu_g^2$;

$A0 \leftarrow S_0$; $\mu_{\mathsf{mean}} \leftarrow S_1/S_0$; $\mu_{\mathsf{var}} \leftarrow S_2/S_0 - (\mu_{\mathsf{mean}})^2$;

Output: $A0$, μ_{mean}, μ_{var}

4.2.6 Summary of block families

Table 4.1 collects the block families derived in this section in a single reference view. The conjugate rows expose closed-form ratios of normalizers, while the Beta-observation row supplies deterministic quadrature values. In both cases the DP layer of Section 5 only requires a compatible block-evidence array and is invariant to the family used to fill that array. Numerical-implementation guidance common to all families (log-space arithmetic, `logsumexp` normalization, prefix-sum ordering, and sanity checks on the DP outputs) is collected in the implementation checklist of Section 6.0.6; every numerical routine in this section and in Section 10 should be read against that checklist.

Family	Prior	Posterior	Block evidence $A_{ij}^{(0)}$	Mean $\mathbb{E}[m_\star(\theta)\|(i,j]]$
Gaussian$(\mu, \sigma^2/w)$	$\mathcal{N}(\nu, \rho^2)$	$\mathcal{N}(\nu_B, \rho_B^2)$	closed-form normal ratio	$(\rho^2 S + \sigma^2 \nu)/\kappa$
Poisson(λw)	$\mathrm{Gamma}(a_0, b_0)$	$\mathrm{Gamma}(a_0 + C, b_0 + W)$	$\propto b_0^{a_0}\Gamma(a_0 + C)/(b_0 + W)^{a_0 + C}$	$(a_0 + C)/(b_0 + W)$
Binomial(m, p)	$\mathrm{Beta}(a_0, b_0)$	$\mathrm{Beta}(a_0 + C, b_0 + M - C)$	$B(a_0 + C, b_0 + M - C)/B(a_0, b_0)$	$(a_0 + C)/(a_0 + b_0 + M)$
NegBin(r, p)	$\mathrm{Beta}(a_0, b_0)$	$\mathrm{Beta}(a_B, b_B)$	$B(a_B, b_B)/B(a_0, b_0)$	$r_* b_B/(a_B - 1)$
Beta-obs(μ, ϕ_t)	$\mathrm{Beta}(a_0, b_0)$	(1D quadrature)	Gauss–Legendre nodes	from quadrature

Table 4.1. Block families derived in Section 4.2. Base-measure factor $\exp\{H_{ij}\}$ is implicit in the conjugate rows. The first three rows are closed-form conjugate examples; the Negative-Binomial row distinguishes parameter-scale Beta moments from the observation-mean target; and the Beta-observation row is non-conjugate but remains compatible with the DP layer through one-dimensional quadrature. This is a derivation/reference table; no source CSV is associated with it.

4.2.7 Posterior-predictive formulas for common conjugate families (optional)

For groups that export segment-level conjugate posteriors, the segment posterior-predictive (11.1) reduces to standard closed forms: Gaussian (known variance) with $\mu \mid \text{train} \sim \mathcal{N}(\nu_B, \rho_B^2)$ and $y \mid \mu \sim \mathcal{N}(\mu, \sigma^2/w)$ gives $y \mid \text{train} \sim \mathcal{N}(\nu_B, \rho_B^2 + \sigma^2/w)$; Poisson–Gamma with $\lambda \mid \text{train} \sim \mathrm{Gamma}(a_B, b_B)$ and exposure w gives a negative-binomial predictive in y; Binomial–Beta with $p \mid \text{train} \sim \mathrm{Beta}(a_B, b_B)$ and trial count m gives a Beta-Binomial,

$$p(y \mid m, \text{train}) = \binom{m}{y} \frac{B(a_B + y, b_B + m - y)}{B(a_B, b_B)}.$$

For M new Gaussian points within one segment, the joint posterior-predictive has rank-one-plus-diagonal covariance and admits $\mathcal{O}(M)$ log-density evaluation via Sherman–Morrison.

Sum-product partition inference

CHAPTER PATH

Forward/backward identities, evidence, segment-count uncertainty, boundary marginals, and model-averaged Bayes curves.

For $k \in \{0, \dots, k_{\max}\}$ and $j \in \{0, \dots, n\}$, let L_{kj} denote the evidence of the prefix $y_{1:j}$ under a segmentation into k segments (after integrating out segment parameters), and let R_{ki} denote the evidence of the suffix $y_{i+1:n}$ under k segments. We adopt the conventions

$$L_{0,0} = 1, \quad L_{0,j} = 0 \ (j > 0), \qquad R_{0,n} = 1, \quad R_{0,i} = 0 \ (i < n).$$

5.0.1 Forward/backward recursions

Given single-block evidences $A^{(0)}_{ij}$, the forward and backward recursions are

$$L_{k+1,j} = \sum_{h=k}^{j-1} L_{k,h}\, A^{(0)}_{hj}, \qquad R_{k+1,i} = \sum_{h=i+1}^{n-k} A^{(0)}_{ih}\, R_{k,h}, \tag{5.1}$$

for $k \geq 0$, $1 \leq j \leq n$, and $0 \leq i \leq n-1$, where sums are restricted to valid indices.

Proposition 5.1 (Forward–backward duality and total-evidence identity)**.** *Under Assumption 3.1 and Assumption 3.6, with prior-adjusted block scores* $\widetilde{A}^{(0)}_{ij} = A^{(0)}_{ij}\gamma_x(i,j)$ (5.2)*, let* $\{\widetilde{L}_{k,j}\}$ *and* $\{\widetilde{R}_{k,i}\}$ *denote the prefix and suffix sum-product messages of* (5.1)*. Then for every admissible* $k \in \{1, \dots, k_{\max}\}$*:*

(i) Boundary conditions. $\widetilde{L}_{0,0} = 1$, $\widetilde{L}_{0,j} = 0$ *for* $j > 0$, $\widetilde{R}_{0,n} = 1$, *and* $\widetilde{R}_{0,i} = 0$ *for* $i < n$. *Furthermore* $\widetilde{L}_{k,j} = 0$ *whenever* $j < k$ *and* $\widetilde{R}_{k,i} = 0$ *whenever* $i > n - k$.

(ii) Total-evidence identity. $\widetilde{L}_{k,n} = \widetilde{R}_{k,0} = \sum_{t \in \mathcal{T}^+_{k,n}} \prod_{q=1}^{k} \widetilde{A}^{(0)}_{t_{q-1}t_q}$, *i.e. the forward-evidence terminal and the backward-evidence initial value both equal the same prior-adjusted partition sum.*

(iii) Boundary marginal product form. *For interior boundary position* $h \in \{p, \dots, n-(k-p)\}$ *and* $1 \leq p \leq k-1$,

$$p(t_p = h \mid y, k) = \frac{\widetilde{L}_{p,h}\, \widetilde{R}_{k-p,h}}{\widetilde{L}_{k,n}},$$

and the boundary-event marginal of Definition 3.4(a) satisfies $\mathrm{MB}_h(k) = \sum_{p=1}^{k-1} p(t_p = h \mid y, k)$.

Proof. Boundary conditions follow from definition; the support constraints $j \geq k$ and $i \leq n-k$ restate that a k-segment prefix requires at least k observations and a k-segment suffix requires at least k observations. For (ii), iterate (5.1) forward from $\widetilde{L}_{0,0}$: by induction on k, $\widetilde{L}_{k,j} = \sum_{t_0<\cdots<t_k=j} \prod_{q=1}^{k} \widetilde{A}^{(0)}_{t_{q-1}t_q}$, so $\widetilde{L}_{k,n}$ enumerates all k-segmentations of $\{1,\ldots,n\}$. The same induction backward from $\widetilde{R}_{0,n}$ gives $\widetilde{R}_{k,i} = \sum_{i=t_0<\cdots<t_k=n} \prod_{q=1}^{k} \widetilde{A}^{(0)}_{t_{q-1}t_q}$, and $\widetilde{R}_{k,0}$ enumerates the same set. Both equal the asserted partition sum. For (iii), split each segmentation at the pth boundary: a segmentation with $t_p = h$ is uniquely described by its first p blocks (counted by $\widetilde{L}_{p,h}$) and its last $k-p$ blocks (counted by $\widetilde{R}_{k-p,h}$). Their product, divided by $\widetilde{L}_{k,n}$, is the conditional probability that $t_p = h$. Summing over $p \in \{1,\ldots,k-1\}$ yields the boundary-event marginal because the boundaries are strictly ordered, so for fixed h at most one p satisfies $t_p = h$ in any given t. □

The identity $\widetilde{L}_{k,n} = \widetilde{R}_{k,0}$ is also one of the DP invariants used in the runtime correctness checklist of Section 6.0.6; Proposition 5.1 states the algebraic basis of that numerical check.

Implementation. Algorithm 6 presents the algebraic sum-product recursions corresponding to (5.1). It is a mathematical specification rather than the recommended floating-point implementation: Section 6.0.2 gives the implementable log-space version using `logsumexp`. The algebraic routine takes a triangular block-evidence array and returns the forward messages, backward messages, and segment-count posteriors needed for boundary and moment extraction.

Algorithm 6: Generic algebraic DP for EF segmentation (prefix/suffix form)

Input: Block evidences $\widetilde{A}^{(0)}_{ij}$ for $0 \leq i < j \leq n$; $k_{\max}$; C_k (default $C_k = \binom{n-1}{k-1}$)

Initialize $\widetilde{L}_{0,0} = 1$, $\widetilde{L}_{0,j>0} = 0$; $\widetilde{R}_{0,n} = 1$, $\widetilde{R}_{0,i<n} = 0$;
for $k = 0$ **to** $k_{\max} - 1$ **do**
  for $j = 1$ **to** n **do**
    $\widetilde{L}_{k+1,j} \leftarrow \sum_{h=k}^{j-1} \widetilde{L}_{k,h} \widetilde{A}^{(0)}_{hj}$;
  for $i = 0$ **to** $n-1$ **do**
    $\widetilde{R}_{k+1,i} \leftarrow \sum_{h=i+1}^{n-k} \widetilde{A}^{(0)}_{ih} \widetilde{R}_{k,h}$;
Compute $P(k \mid y) \propto p(k)\, \widetilde{L}_{kn}/C_k$ (cf. (5.4)) and choose $\widehat{k}$;
for $p = 1$ **to** $\widehat{k} - 1$ **do**
  For $h \in \{p, \ldots, n - (\widehat{k} - p)\}$ set $P(t_p = h \mid y, \widehat{k}) \leftarrow \widetilde{L}_{p,h} \widetilde{R}_{\widehat{k}-p,h} / \widetilde{L}_{\widehat{k}n}$;
  Record the marginal mode $\arg\max_h P(t_p = h \mid y, \widehat{k})$ as a diagnostic only;

Output: $\{\widetilde{L}_{kj}\}$, $\{\widetilde{R}_{ki}\}$, $P(k \mid y)$, $\widehat{k}$, and boundary-marginal diagnostics

5.0.2 Incorporating segment-factorized boundary priors

Suppose the boundary prior given k factorizes over segment lengths as in (3.8):

$$p(t \mid k) = \frac{1}{C_k} \prod_{q=1}^{k} g(\Delta_x(t_{q-1}, t_q)), \qquad C_k := \sum_{t:\, 0=t_0<\cdots<t_k=n} \prod_{q=1}^{k} g(\Delta_x(t_{q-1}, t_q)).$$

Then we can absorb g into block evidences by defining

$$\widetilde{A}^{(0)}_{ij} := A^{(0)}_{ij}\, \gamma_x(i,j), \qquad \widetilde{A}^{(r)}_{ij} := A^{(r)}_{ij}\, \gamma_x(i,j). \tag{5.2}$$

Running the DP (5.1) with $\widetilde{A}^{(0)}_{ij}$ computes $\widetilde{L}_{kn}$, the *unnormalized* evidence sum over all segmentations weighted by g. The normalizer C_k can be computed by the same DP with $A^{(0)}_{ij} \equiv 1$:

$$C_k = \sum_t \prod_{q=1}^{k} g(\Delta_x(t_{q-1}, t_q)) = L^{(g)}_{kn}, \quad \text{where} \quad L^{(g)}_{k+1,j} = \sum_{h=k}^{j-1} L^{(g)}_{k,h}\, \gamma_x(h, j),\ L^{(g)}_{0,0} = 1. \tag{5.3}$$

In the index-uniform case $g \equiv 1$, (5.3) reduces to $C_k = \binom{n-1}{k-1}$.

5.0.3 Posteriors over k, boundary marginals, and joint MAP segmentations

Let $p(k)$ be a prior on segment count. Under index-uniform $p(t \mid k)$ we have $C_k = \binom{n-1}{k-1}$ and

$$P(y \mid k) = \frac{L_{kn}}{C_k}, \qquad P(k \mid y) \propto p(k)\, \frac{L_{kn}}{C_k}.$$

For general segment-factorized $p(t \mid k)$, replace L by $\widetilde{L}$ and C_k by (5.3). In all cases we can write compactly

$$P(k \mid y) \ \propto\ p(k)\, \frac{\widetilde{L}_{kn}}{C_k}, \qquad k = 1, \dots, k_{\max}. \tag{5.4}$$

Let $\widehat{k} = \arg\max_k P(k \mid y)$ denote the MAP segment count.

Conditional on k, the marginal posterior for the pth boundary t_p is

$$P(t_p = h \mid y, k) = \frac{\widetilde{L}_{p,h}\, \widetilde{R}_{k-p,h}}{\widetilde{L}_{kn}}, \qquad h \in \{p, \dots, n - (k - p)\}, \tag{5.5}$$

where $(\widetilde{L}, \widetilde{R})$ are computed from $\widetilde{A}^{(0)}$ as in (5.1). The denominator is the unnormalized evidence under the length-weighted partition prior; the normalizer C_k cancels in (5.5) because the same C_k multiplies every boundary vector in the numerator and denominator. (When $g \equiv 1$, tildes can be dropped.)

> *Remark* 5.2 (Boundary-event marginal). A related object we use repeatedly is the *boundary-event marginal*
>
> $$P(b_i = 1 \mid y, k) := \sum_{p=1}^{k-1} P(t_p = i \mid y, k) = \sum_{p=1}^{k-1} \frac{\widetilde{L}_{p,i}\, \widetilde{R}_{k-p,i}}{\widetilde{L}_{kn}}, \qquad 1 \le i \le n-1, \tag{5.6}$$
>
> the posterior probability that interior index i coincides with *some* boundary, without conditioning on which boundary. The objects $P(t_p = h \mid y, k)$ and $P(b_i = 1 \mid y, k)$ are distinct: the former is a probability distribution over h for each fixed p (and sums to 1 over h), while the latter is a per-index Bernoulli probability summing to $k - 1$ over i. Calibration experiments in Section 13 target $P(b_i = 1 \mid y, k)$; a full derivation is given in Appendix 5.

A useful diagnostic is the *marginal boundary mode*

$$\widetilde{t}^{\,\mathrm{marg}}_p(k) \in \arg\max_h P(t_p = h \mid y, k).$$

However, maximizing the marginals independently need not yield an ordered boundary vector and, in general, does *not* recover the joint MAP segmentation.

Remark 5.3 (A small counterexample). Consider $n = 5$ and $k = 3$, so the two interior boundaries satisfy $1 \le t_1 < t_2 \le 4$. For example, let the joint posterior put masses $0.30, 0.28, 0.22, 0.20$ on the four ordered configurations $(1,4), (2,3), (2,4), (3,4)$, respectively. The marginal modes are then $t_1 = 2$ and $t_2 = 4$, so the independently selected marginal-mode vector is $(2,4)$, even though the joint MAP configuration is $(1,4)$. This is exactly the kind of discrepancy the max-sum DP resolves.

The joint MAP segmentation is the solution of

$$\widehat{t}^{\,\mathrm{MAP}}(k) \in \arg \max_{0=t_0<\cdots<t_k=n} \left\{ \sum_{q=1}^{k} \log \widetilde{A}^{(0)}_{t_{q-1} t_q} \right\}, \tag{5.7}$$

with the convention that any k-specific prior terms such as $\log p(k) - \log C_k$ are added only when selecting among different values of k. Equation (5.7) is solved exactly by a max-sum DP with backpointers. Inline, the max-sum recursion reads

$$M_{k+1,j} = \max_{h \in \{k,\ldots,j-1\}} \left\{ M_{k,h} + \log \widetilde{A}^{(0)}_{hj} \right\}, \qquad h^\star(k+1, j) \in \arg\max_{h} \left\{ M_{k,h} + \log \widetilde{A}^{(0)}_{hj} \right\}, \tag{5.8}$$

with initial conditions $M_{0,0} = 0$, $M_{0,j} = -\infty$ for $j > 0$, and backtracking from (k, n) as detailed in Section 6. In the sequel, whenever we report an *exported MAP segmentation*, we mean the joint optimizer $\widehat{t}^{\,\mathrm{MAP}} := \widehat{t}^{\,\mathrm{MAP}}(\widehat{k})$ rather than a vector of marginal boundary modes.

5.0.4 Segment-level and regression-curve posterior moments

For segment-wise posterior moments attached to a *fixed* segmentation $t = (t_0, \ldots, t_k)$, let segment $q \in \{1, \ldots, k\}$ correspond to block $(i, j] = (t_{q-1}, t_q]$. Then

$$\mu_q^{(r)}(t) = \frac{\widetilde{A}^{(r)}_{ij}}{\widetilde{A}^{(0)}_{ij}} = \frac{A^{(r)}_{ij}}{A^{(0)}_{ij}}, \qquad r = 1, 2, \ldots, \tag{5.9}$$

where the final equality holds because the prior factor $\gamma_x(i, j)$ cancels. In particular, for an exported joint MAP segmentation $\widehat{t}^{\,\mathrm{MAP}}$, the segment summaries are obtained by setting $t = \widehat{t}^{\,\mathrm{MAP}}$ in (5.9).

To compute the Bayes regression curve within a fixed k (typically $k = \widehat{k}$), define, for each block $(i, j]$ and moment order r,

$$F^{(r)}_{ij}(k) := \frac{1}{\widetilde{L}_{kn}} \sum_{m=1}^{k} \widetilde{L}_{m-1,i}\, \widetilde{A}^{(r)}_{ij}\, \widetilde{R}_{k-m,j}.$$

Then the rth posterior moment of the latent mean at index t given k is obtained by summing $F^{(r)}_{ij}(k)$ over all candidate blocks $(i, j]$ that contain the index t (i.e., all blocks with $i < t \le j$), so that each candidate segmentation contributes exactly once through the block that actually covers t:

$$\widehat{\mu}'^{(r)}_t := \mathbb{E}[(\mu'_t)^r \mid y, k] = \sum_{i<t\le j} F^{(r)}_{ij}(k), \qquad t = 1, \ldots, n. \tag{5.10}$$

In particular, $\widehat{\mu}'_t := \widehat{\mu}'^{(1)}_t$ is the Bayes regression curve and $\mathbb{V}[\mu'_t \mid y, k] = \widehat{\mu}'^{(2)}_t - (\widehat{\mu}'^{(1)}_t)^2$. If a fully k-marginalized curve is desired, compute the same conditional curve for each admissible k and average the resulting moments using $P(k \mid y)$ from (5.4). The empirical sections report curves conditional on the selected $\widehat{k}$ unless explicitly stated otherwise.

Proposition 5.4 (Block-covering decomposition for Bayes-curve moments)**.** *Fix k and an index $t \in \{1, \dots, n\}$. Under the support and factorization hypotheses of Theorem 5.5, the conditional posterior moment of the latent mean at t is obtained by summing the block contributions $F_{ij}^{(r)}(k)$ over exactly those candidate blocks $(i, j]$ with $i < t \le j$. No segmentation contributes more than once to the sum in (5.10).*

Proof. Every admissible ordered boundary vector partitions $\{1, \dots, n\}$ into disjoint blocks, hence contains a unique block $(i, j]$ satisfying $i < t \le j$. Conditioning on that covering block splits the remaining boundaries into an $(m-1)$-segmentation of the prefix $y_{1:i}$ and a $(k-m)$-segmentation of the suffix $y_{j+1:n}$ for some segment position m. The prefix, block, and suffix contributions multiply as $\widetilde{L}_{m-1,i}\, \widetilde{A}_{ij}^{(r)}\, \widetilde{R}_{k-m,j}$, and summing over m and over all covering blocks gives (5.10). Disjointness of the covering block proves that no segmentation is double counted. □

Theorem 5.5 (Correctness of the DP)**.** *Assume finite block evidences on all admissible blocks, a finite segment-count range, and a segment-factorized conditional partition prior satisfying Assumption 3.6 with the boundary-coordinate convention of Section 3.1. Under the EF–conjugate single-segment model of Section 4.1, or under any other block model whose exact evidence array is supplied, the recursions (5.1) compute exact marginal evidences for all prefixes/suffixes: $\widetilde{L}_{kj}$ and $\widetilde{R}_{ki}$ are exact evidence sums over all k-segmentations of the corresponding prefix/suffix. Moreover:*

1. *The posterior $P(k \mid y)$ given by (5.4) is exact.*
2. *The boundary marginal $P(t_p = h \mid y, k)$ given by (5.5) is exact.*
3. *The segment-wise moments (5.9) and regression-curve moments (5.10) are exact posterior moments conditional on k.*

The separate max-sum recursion and backtracking theorem for the joint MAP partition is stated and proved immediately after its statement in Theorem 6.1.

Proof. We argue for general $\widetilde{A}_{ij}^{(0)}$; $g \equiv 1$ is the untilded specialization.

Step 1: $\widetilde{L}_{kj}$ is the prefix evidence sum. For any k-segmentation $(t_0, \dots, t_k)$ of $y_{1:j}$ with $t_0 = 0$, $t_k = j$, integrating out segment parameters yields $p(y_{1:j} \mid t, k) = \prod_q \widetilde{A}_{t_{q-1}, t_q}^{(0)}$. Summing over admissible t gives $\widetilde{L}_{kj} = \sum_{t:\, t_k = j} \prod_{q=1}^{k} \widetilde{A}_{t_{q-1}, t_q}^{(0)}$.

Step 2: forward recursion. Conditioning a $(k+1)$-segmentation of $y_{1:j}$ on its last boundary $h \in \{k, \dots, j-1\}$ splits it into a k-segmentation of $y_{1:h}$ and the final block $(h, j]$; summing over h gives the forward recursion in (5.1). The backward recursion follows by the same argument applied to suffixes.

Step 3: posterior over k. $P(y \mid k) = \widetilde{L}_{kn} / C_k$ by definition of C_k; Bayes' rule yields (5.4).

Step 4: boundary posteriors. Conditioning on $t_p = h$ splits the sequence into a p-segmentation of $y_{1:h}$ and a $(k-p)$-segmentation of $y_{h+1:n}$, contributing $\widetilde{L}_{p,h}\, \widetilde{R}_{k-p,h}$; normalizing by $\widetilde{L}_{kn}$ yields (5.5).

Step 5: moments. Theorem 4.1 gives $A_{ij}^{(r)} = A_{ij}^{(0)}\,\mathbb{E}[m_\star(\theta)^r \mid (i,j]]$; the length factor $\gamma_x(i,j)$ cancels in the ratio, so (5.9) holds. Equation (5.10) follows by grouping segmentation contributions through the unique block $(i,j]$ that contains index t. □

Posterior-evidence sanity checks. After every DP run, verify four invariants: normalized segment-count weights sum to one; the full-sequence forward and backward evidences agree, $\widetilde{L}_{kn} = \widetilde{R}_{k0}$; boundary-event marginals satisfy $\sum_{i=1}^{n-1} P(b_i = 1 \mid y, k) = k-1$; and the max-sum backtracking score equals the stored terminal max-sum value. In log-space arithmetic these checks are evaluated with a declared tolerance, typically a small multiple of machine epsilon times the number of accumulated summands. A failure of the first two checks is treated as a normalization or admissibility mismatch; a failure of the boundary-sum check indicates an indexing or support error; and a failure of the backtracking check indicates inconsistent tie-breaking or stale backpointers.

Theorem 5.6 (Time and space complexity)**.** *Assume single-block evidences $A_{ij}^{(0)}$ are available for all $0 \le i < j \le n$. Computing all forward and backward messages via (5.1) requires $\mathcal{O}(k_{\max} n^2)$ arithmetic operations. The triangular block-evidence matrix requires $\mathcal{O}(n^2)$ memory, while all forward and backward layers require $\mathcal{O}(k_{\max} n)$ memory. Max-sum values and backpointers each require $\mathcal{O}(k_{\max} n)$ additional memory. Boundary-event probabilities for every k and candidate index require $\mathcal{O}(k_{\max} n)$ output storage. Ordered-boundary distributions for every triple (k,p,h) require $\mathcal{O}(k_{\max}^2 n)$ output storage, or $\mathcal{O}(kn)$ for one fixed segment count k. For R requested reporting moments, retaining all block moment numerators requires $\mathcal{O}(Rn^2)$ memory and the final Bayes curves require $\mathcal{O}(Rn)$; explicitly retaining every block-cover contribution over all k can require $\mathcal{O}(k_{\max} n^2)$ additional workspace but is not necessary when those contributions are accumulated sequentially.*

Proof. For each fixed k, the forward recursion computes $L_{k+1,j}$ for $j = 1, \dots, n$, and each $L_{k+1,j}$ sums at most $j-k$ terms. Thus the forward work per k is $\sum_{j=1}^{n} \mathcal{O}(j) = \mathcal{O}(n^2)$. The backward recursion is analogous. Repeating for $k = 0, \dots, k_{\max} - 1$ gives $\mathcal{O}(k_{\max} n^2)$ time. A message or backpointer array has one entry per (k,j) state. Boundary-event output has one value per (k,h), whereas the full ordered-boundary output has one value per (k,p,h) and therefore $\sum_{k \le k_{\max}} \mathcal{O}(kn) = \mathcal{O}(k_{\max}^2 n)$ entries. The remaining statements follow by counting the triangular block arrays, reporting-moment arrays, and optional block-cover workspace. □

Corollary 5.7 (Decision rules for full-vector and coordinatewise losses)**.** *For the loss $\ell_{01}(t,t') = \mathbb{1}\{t \ne t'\}$ on the full boundary vector, the Bayes estimator is the joint MAP segmentation $\widehat{t}^{\,\mathrm{MAP}}(k)$ from (5.7). For a separable coordinatewise loss and an unconstrained Cartesian action space, each coordinate can instead be chosen by minimizing its marginal posterior risk; under coordinatewise 0–1 loss this gives the marginal modes. If the reported action must belong to the strictly ordered partition space $\mathcal{T}_{k,n}^{+}$, coordinatewise marginal modes can be inadmissible, and the Bayes action must minimize posterior expected loss over that constrained set. It can differ from both the coordinatewise marginal modes and the joint MAP partition.*

Proof. The posterior risk of t' under full-vector 0–1 loss is $1 - p(t = t' \mid y, k)$, so any joint posterior mode minimizes it. Under a separable loss on an unconstrained Cartesian action space, the posterior risk is a sum of coordinatewise risks, each minimized using the corresponding marginal posterior. Restricting the action to strictly ordered boundary vectors couples the

coordinates, so independent coordinatewise minimization no longer solves the constrained decision problem. □

Remark 5.8 (Memory/time trade-offs for large n). Keeping only the current and previous k-layers reduces message memory to $\mathcal{O}(n)$, in addition to any stored block array, when only terminal fixed-k evidences are required. Posterior marginals and MAP paths then require retained layers or a documented recomputation pass. Checkpoint spacing can trade retained layers against recomputation, but its exact cost depends on which summaries are requested and is therefore an implementation choice rather than a universal bound. When the n^2 block-evidence matrix is prohibitive and every segment score is available in constant time from prefix summaries, scores can instead be evaluated on demand during the DP sweep. This uses less block storage but performs up to $\mathcal{O}(k_{\max} n^2)$ segment evaluations because the same segment can be revisited at different DP layers.

Remark 5.9 (Evidence across segment counts need not be monotone). The fixed-k evidence $P(y \mid k) = \widetilde{L}_{kn}/C_k$ is not monotone in general. Adding a boundary changes both the set of admissible partitions and the product of block evidences, while the normalized conditional prior contributes the normalizer C_k. Consequently, diagnose segment-count behavior through the normalized evidences $P(y \mid k)$ and posterior weights $P(k \mid y)$ rather than through an unnormalized pathwise "add one boundary" heuristic.

5.0.5 End-to-end BayesBreak inference routine

Collecting the ingredients from Sections 4.1 and 5—block-evidence precomputation, exact sum-product DP over partitions, and max-sum backtracking for exported segmentations—yields an end-to-end routine suitable for direct implementation. Algorithm 7 summarizes this workflow in a modular way: the only model-specific input is a function that evaluates (or approximates) block evidences $\log \mathcal{L}_{ij}$ and, when needed, block moment numerators.

Algorithm 7: BayesBreak$(y, x, w, \texttt{model}, \texttt{hyper}, k_{\max}, p(k), g)$

Input: Observations $y_{1:n}$ at locations $x_{1:n}$ (optional); observation-model weights/exposures $w_{1:n}$ (default determined by the chosen likelihood, often $w_i \equiv 1$);
model family $\texttt{model} \in \{\texttt{Gaussian}, \texttt{Poisson}, \texttt{Binomial}, \texttt{BetaObs}, \dots\}$;
hyperparameters $\texttt{hyper}$; maximum segment count $k_{\max}$; prior $p(k)$; length factor $g(\cdot)$ (default $g \equiv 1$).
Output: Posterior $P(k \mid y)$; MAP segment count $\widehat{k}$; boundary marginals $P(t_p \mid y, \widehat{k})$; exported joint MAP segmentation $\widehat{t}^{\,\mathrm{MAP}}$; segment-level moments; Bayes regression curve; (optional) evidence $P(y)$.

1. Precompute block evidences and moments;
for $0 \le i < j \le n$ **do**
  Compute block summaries (S_{ij}, W_{ij}, H_{ij}) using prefix sums (or running sums);
  Call a family-specific routine to obtain $A_{ij}^{(0)}, A_{ij}^{(1)}, A_{ij}^{(2)}$;
  Set $\widetilde{A}_{ij}^{(r)} \leftarrow A_{ij}^{(r)}\, \gamma_x(i,j)$ for $r \in \{0,1,2\}$ (cf. (5.2));
Compute C_k under the same admissibility and length-prior convention used in the score array; for $g \equiv 1$, use $C_k = \binom{n-1}{k-1}$, and otherwise use (5.3) (DP with $A_{ij}^{(0)} \equiv 1$);

2. Run the sum-product dynamic program;
Compute $\widetilde{L}_{kj}, \widetilde{R}_{ki}$ via (5.1) for $k = 0, \dots, k_{\max}$ using $\widetilde{A}_{ij}^{(0)}$;
Compute $P(k \mid y) \propto p(k)\, \widetilde{L}_{kn}/C_k$ via (5.4); set $\widehat{k} \leftarrow \arg\max_k P(k \mid y)$;

3. Boundary marginals;
for $p = 1$ **to** $\widehat{k} - 1$ **do**
  For all $h \in \{p, \dots, n - (\widehat{k} - p)\}$, set $P(t_p = h \mid y, \widehat{k}) \leftarrow \widetilde{L}_{p,h} \widetilde{R}_{\widehat{k}-p,h} / \widetilde{L}_{\widehat{k}n}$ (cf. (5.5));

4. Export a joint MAP segmentation;
Run the max-sum/backpointer recursion of Section 6 using $\log \widetilde{A}_{ij}^{(0)}$ and the chosen count $\widehat{k}$;
Backtrack to obtain $\widehat{t}^{\,\mathrm{MAP}} = (0 = \widehat{t}_0^{\,\mathrm{MAP}} < \cdots < \widehat{t}_{\widehat{k}}^{\,\mathrm{MAP}} = n)$;

5. Segment-level posterior moments on the exported segmentation;
for $q = 1$ **to** $\widehat{k}$ **do**
  Let $(i, j] \leftarrow (\widehat{t}_{q-1}^{\,\mathrm{MAP}}, \widehat{t}_q^{\,\mathrm{MAP}}]$;
  For $r \in \{1, 2\}$ set $\widehat{\mu}_q^{(r)} \leftarrow \widetilde{A}_{ij}^{(r)} / \widetilde{A}_{ij}^{(0)}$ (cf. (5.9));

6. Bayes regression curve within $\widehat{k}$;
Using $\widetilde{L}, \widetilde{R}, \widetilde{A}^{(r)}$ and (5.10), compute $\widehat{\mu}'_i$ and $\mathbb{V}[\mu'_i \mid y, \widehat{k}]$ for $i = 1, \dots, n$;

7. Optional: pooling/groups;
If multiple subjects or groups are present, pool block evidences (Theorem 8.2) and/or run the finite latent-group template procedure of Chapter 9, then repeat Steps 2–6.

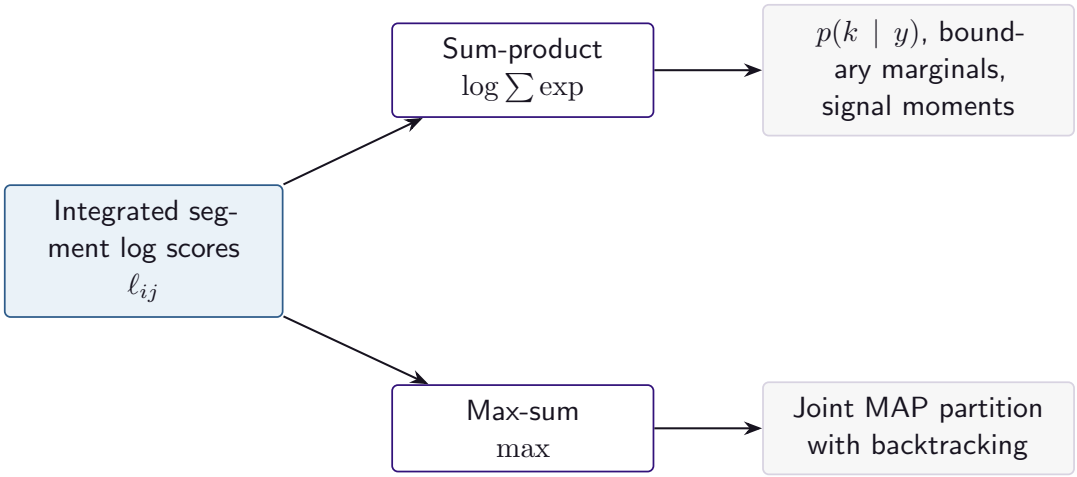


Figure 5.1. Same local scores, different semirings. Sum-product produces marginal posterior quantities; max-sum produces one joint maximizing path.

CHAPTER 6

Joint MAP recovery, algorithms, and numerical stability

CHAPTER PATH

Implementation-grade recursions, an in-place correctness proof, complexity, tie behavior, and log-space diagnostics.

ESTIMATOR SEPARATION

The vector of marginal boundary modes is not generally a valid joint optimizer and can even contain crossing or duplicate positions under naive assembly. BayesBreak exports the joint MAP path only from max-sum DP with backtracking. Sum-product outputs remain posterior summaries.

Section 5 derives the exact dynamic program (DP) that sits at the core of BayesBreak. The present section complements that development with implementation-ready pseudocode and a collection of numerical-stability considerations. The primary purpose is pragmatic: to make explicit what must be computed, stored, and normalized in order to reproduce the theoretical quantities (evidences, boundary posteriors, Bayes curves) without relying on undocumented conventions.

6.0.1 Block-evidence precomputation via prefix sums

For conjugate exponential-family blocks, Theorem 4.1 shows that the single block evidence and moments can be expressed as functions of aggregated sufficient statistics. This enables an *exact* $\mathcal{O}(n^2)$ precomputation of all candidate blocks $(i, j]$ with $0 \le i < j \le n$. Throughout this section we use the body convention: a block $(i, j]$ has length $j - i$ and contains the indices $i + 1, \dots, j$.

In practice, it is numerically preferable to store *log* block evidences,

$$\ell_{ij}^{(0)} \;:=\; \log \widetilde{A}_{ij}^{(0)}, \tag{6.1}$$

and similarly for positive moment numerators required for the Bayes regression curve. If a chosen moment numerator can be zero or negative, implementations use tagged signed-log storage rather than silently taking a logarithm: a signed quantity m_{ij} is represented as (s_{ij}, a_{ij}) with $s_{ij} \in \{-1, 0, 1\}$ and $a_{ij} = \log |m_{ij}|$ when $s_{ij} \neq 0$, while $s_{ij} = 0$ records exact zero. Each `BlockRoutine` must return a finite log evidence for admissible blocks, $-\infty$ for excluded

blocks, optional positive log-moment numerators, and optional signed-log moment records with this zero convention. Downstream DP code must reject untagged nonpositive moment numerators rather than treating them as log values. Algorithm 8 expresses the precomputation at an abstract level, with a single call to a `BlockRoutine` that implements the family-specific closed forms given in Section 4.2.

Algorithm 8: PrecomputeBlocks$(y, x, w,$ `BlockRoutine`$)$

Input: Ordered observations $\{(x_i, y_i, w_i)\}_{i=1}^{n}$ (with $w_i \equiv 1$ if unweighted); family-specific routine `BlockRoutine` implementing Theorem 4.1.

Output: Matrices of log-block evidences $\ell_{ij}^{(0)} = \log \widetilde{A}_{ij}^{(0)}$ on the index set $\{0 \le i < j \le n\}$ and (optional) moment numerators, stored as $\log \widetilde{A}_{ij}^{(r)}$ only when they are positive and otherwise in a signed-log or linear representation.

1. Prefix summaries;

Compute prefix cumulative sufficient statistics and weights required by `BlockRoutine`: store $W_j := \sum_{t=1}^{j} w_t$, $S_j := \sum_{t=1}^{j} w_t\, T(y_t)$, and any additional family-specific terms (e.g., $Q_j := \sum_{t=1}^{j} w_t y_t^2$ for Gaussian). Set $W_0 = S_0 = Q_0 = 0$.

2. Enumerate blocks;

for $i \leftarrow 0$ **to** $n-1$ **do**

  for $j \leftarrow i+1$ **to** n **do**

    Extract block summaries for $y_{i+1:j}$ via $W_{ij} = W_j - W_i$, $S_{ij} = S_j - S_i$, $Q_{ij} = Q_j - Q_i$ (constant time); `BlockRoutine` returns $\log A_{ij}^{(0)}$ and any moment numerators $\log A_{ij}^{(r)}$; Add any length factor $\log g(\Delta_x(i,j))$ and hazard constants as in (5.2); Store $\ell_{ij}^{(0)} \leftarrow \log \widetilde{A}_{ij}^{(0)}$ and, if needed, $\ell_{ij}^{(r)} \leftarrow \log \widetilde{A}_{ij}^{(r)}$.

Memory notes. Storing all $\ell_{ij}^{(0)}$ uses $\Theta(n^2)$ memory. When n is large but $k_{\max}$ is moderate, the memory footprint is reduced by streaming blocks in the inner DP loop (computing $\ell_{ij}^{(0)}$ on the fly from prefix statistics), at cost of recomputing some block summaries. The DP theory remains unchanged.

As sanity checks, verify that invalid blocks are represented by $-\infty$ log-scores, that $\log C_k$ is computed with the same admissibility rules as the DP, that forward and backward full-sequence evidences agree, and that boundary-event marginals sum to $k-1$ conditional on fixed k.

6.0.2 Log-space dynamic programming

The forward/backward DP in Section 5 sums products of block evidences. To avoid underflow/overflow, work in log space and use a stable `logsumexp` primitive. Algorithm 9 presents a log-space version of the recursions in (5.1)–(5.4).

Algorithm 9: DPLog$(\ell_{ij}^{(0)}, k_{\max}, \log p(k), \log C_k)$

Input: Log block evidences $\ell_{ij}^{(0)} = \log \widetilde{A}_{ij}^{(0)}$ on the index set $\{0 \le i < j \le n\}$ (block $(i, j]$); maximum segments $k_{\max}$; log prior on k, $\log p(k)$; log normalizers $\log C_k$ computed under the same admissible-block and length-prior convention as the score array. For the index-uniform fixed-k prior with $g \equiv 1$, $C_k = \binom{n-1}{k-1}$ unless the caller has intentionally absorbed a normalized prior into the block or count scores.

Output: Forward messages $\log \widetilde{L}_{k,j}$, backward messages $\log \widetilde{R}_{k,i}$, log evidence per k, and normalized posterior $p(k \mid y)$.

1. Initialize;

Set $\log \widetilde{L}_{0,0} \leftarrow 0$ and $\log \widetilde{L}_{0,j} \leftarrow -\infty$ for $j \ge 1$; set $\log \widetilde{R}_{0,n} \leftarrow 0$ and $\log \widetilde{R}_{0,i} \leftarrow -\infty$ for $i \le n-1$.

2. Forward recursion;

for $k \leftarrow 1$ **to** $k_{\max}$ **do**

  for $j \leftarrow k$ **to** n **do**

    $\log \widetilde{L}_{k,j} \leftarrow \texttt{logsumexp}_{h \in \{k-1,\dots,j-1\}} \big(\log \widetilde{L}_{k-1,h} + \ell_{h,j}^{(0)} \big)$;

3. Backward recursion;

for $k \leftarrow 1$ **to** $k_{\max}$ **do**

  for $i \leftarrow n-k$ **to** 0 *(decreasing)* **do**

    $\log \widetilde{R}_{k,i} \leftarrow \texttt{logsumexp}_{h \in \{i+1,\dots,n-k+1\}} \big(\ell_{i,h}^{(0)} + \log \widetilde{R}_{k-1,h} \big)$;

4. Evidence and posterior over k;

For each $k \in \{1, \dots, k_{\max}\}$ set $\log p(y \mid k) \leftarrow \log \widetilde{L}_{k,n} - \log C_k$; compute unnormalized log posterior scores $\log \pi_k \leftarrow \log p(k) + \log p(y \mid k)$; set $p(k \mid y) \leftarrow \exp\{\log \pi_k - \texttt{logsumexp}_{k'}(\log \pi_{k'})\}$.

The backward upper endpoint $n - k + 1$ is the last split that leaves $k - 1$ nonempty suffix blocks to cover indices $h + 1, \dots, n$. Empty summation ranges are interpreted as $-\infty$ in log space, so edge cases are handled by the same admissibility convention as the forward recursion.

Boundary marginals and Bayes curves. Once $\log \widetilde{L}$ and $\log \widetilde{R}$ are available, the boundary marginal in (5.5) can be computed stably by subtracting the common normalizer $\log \widetilde{L}_{k,n}$. Similarly, Bayes regression moments (Proposition 5.4; cf. (5.10)) can be computed by replacing $\ell_{ij}^{(0)}$ with compatible moment numerators and reusing the same log-space DP bookkeeping. Signed or nonpositive moment targets require the signed-log or scaled-linear convention noted in Section 6.0.1. A signed-log accumulator stores each target as a sign bit and a log-magnitude; sums of signed terms are evaluated by separately applying `logsumexp` to positive and negative parts and subtracting the two scaled results only at the final reporting step. This convention is required for any routine that returns moment numerators rather than marginal likelihoods. The backward-loop bound $h \le n - k + 1$ in Algorithm 9 enforces the nonempty-suffix convention: after choosing the next endpoint h for the block $(i, h]$, the remaining $k-1$ blocks must occupy at least one observation each in $(h, n]$. Thus the summation is empty, and the message is $-\infty$, whenever the remaining suffix cannot support the requested number of blocks.

6.0.3 MAP segmentation and backtracking

The DP used for evidences can be augmented to recover a MAP segmentation in $\mathcal{O}(k_{\max} n^2)$ time. During the forward recursion, one stores, for each (k, j), the maximizing split point

$$h^\star(k, j) \in \arg \max_{h \in \{k-1, \dots, j-1\}} \left\{ M_{k-1,h} + \ell^{(0)}_{h,j} \right\}, \tag{6.2}$$

where $M_{k,j}$ is the max-sum message defined in (5.8). Algorithm 10 computes $M_{k,j}$ and the backpointers; Algorithm 11 then backtracks from $(\widehat{k}, n)$ to recover the boundary vector $\widehat{b} = (\widehat{b}_0, \dots, \widehat{b}_{\widehat{k}})$.

Theorem 6.1 (Correctness of max-sum DP for joint MAP segmentation)**.** *For any choice of segment count $\widehat{k} \in \{1, \dots, k_{\max}\}$, Algorithms 10 and 11 jointly return the (or a) boundary vector $\widehat{b} = (\widehat{b}_0, \dots, \widehat{b}_{\widehat{k}})$ that attains*

$$\widehat{b} \in \arg \max_{t \in T_{\widehat{k},n}} \left\{ \log p(\widehat{k}) - \log C_{\widehat{k}} + \sum_{q=1}^{\widehat{k}} \ell^{(0)}_{t_{q-1}, t_q} \right\},$$

where $T_{\widehat{k},n}$ is the set of admissible boundary vectors of length $\widehat{k}$ on n observations under the segment-factorized partition prior of Section 5. In particular, $\widehat{b}$ is a joint MAP segmentation conditional on the chosen $\widehat{k}$, and any tie among maximizers is broken deterministically by the back-pointer convention encoded in $h^\star(k, j)$. The runtime is $\mathcal{O}(k_{\max} n^2)$ and the working space is $\mathcal{O}(k_{\max} n)$.

Proof. For $q = 1$, $V_{1,j} = \ell_{0j}$ is the best score among the sole one-block partition of $(0, j]$. Assume $V_{q-1,h}$ equals the best score over all admissible $(q-1)$-block partitions ending at h. Every admissible q-block partition ending at j has a unique penultimate endpoint h and score equal to a $(q-1)$-block prefix score plus ℓ_{hj}. Maximizing over h therefore gives $V_{q,j} = \max_h \{V_{q-1,h} + \ell_{hj}\}$. Induction proves optimality at (k, n). Storing one deterministically tie-broken maximizing predecessor at each state and following those predecessors backward reconstructs a path attaining $V_{k,n}$; hence the returned boundary vector is a joint MAP partition for fixed k. □

Algorithm 10: MAPForward$(\ell^{(0)}_{ij}, k_{\max})$

Input: Log block evidences $\ell^{(0)}_{ij}$ on $\{0 \le i < j \le n\}$; $k_{\max}$.
Output: Max-sum messages $M_{k,j}$ and backpointers $h^\star(k, j)$ for $k \le k_{\max}$.

Set $M_{0,0} \leftarrow 0$, $M_{0,j} \leftarrow -\infty$ for $j \ge 1$;
for $k \leftarrow 1$ **to** $k_{\max}$ **do**
  for $j \leftarrow k$ **to** n **do**
    $h^\star(k, j) \leftarrow \arg\max_{h \in \{k-1, \dots, j-1\}} \{M_{k-1,h} + \ell^{(0)}_{h,j}\}$;
    $M_{k,j} \leftarrow M_{k-1,h^\star(k,j)} + \ell^{(0)}_{h^\star(k,j),j}$;

Algorithm 11: MAPBacktrack$(h^\star, \widehat{k})$

Input: Backpointers $h^\star(k, j)$ as in (6.2); chosen segment count $\widehat{k}$.
Output: MAP boundary vector $\widehat{b}_0 = 0 < \widehat{b}_1 < \cdots < \widehat{b}_{\widehat{k}} = n$.

Set $\widehat{b}_{\widehat{k}} \leftarrow n$ and $q \leftarrow \widehat{k}$. **while** $q \geq 1$ **do**
  $\widehat{b}_{q-1} \leftarrow h^\star(q, \widehat{b}_q)$;
  $q \leftarrow q - 1$;
Return $\widehat{b}$.

6.0.4 Pooling across sequences and groups

For replicated or multi-subject data with shared boundaries (Section 8), Theorem 8.2 implies that subject-level block evidences multiply. In log space, pooling is therefore a simple sum:

$$\ell_{ij}^{(0,\text{pooled})} = \sum_{s=1}^{S} \ell_{ij}^{(0,s)}. \tag{6.3}$$

The same applies to known groups, with the sum taken over subjects in a group. The resulting pooled matrices can be passed unchanged to Algorithm 9.

Algorithm 12: Pooled-DP for shared-boundary multi-subject inference (log-space)

Input: Per-subject log-block-evidence arrays $\{\ell_{ij}^{(0,s)}\}_{s=1}^{S}$ for $0 \leq i < j \leq n$; local prior factors $\log c_x(i, j)$ and interior hazards $\log h_x(j)$; $k_{\max}$; group-label set $\{z_s\}_{s=1}^{S}$ if pooling by known group, else $z_s \equiv 1$ for global pooling.
Output: Per-group pooled posteriors $P_g(k \mid y)$, $P_g(t_p \mid y, k)$, MAP segmentations $\widehat{t}_g^{\,\text{MAP}}(\widehat{k}_g)$, and segment-conditional parameter posteriors.

for *each distinct group label* $g \in \{z_s\}$ **do**
  Let $\mathcal{S}_g = \{s : z_s = g\}$;
  $\ell_{ij,g}^{(0,\text{pool})} \leftarrow \sum_{s \in \mathcal{S}_g} \ell_{ij}^{(0,s)}$ for every admissible $(i, j]$;
  `// The local prior factor is absorbed once at the pooled level.`
  $\widetilde{\ell}_{ij,g}^{(0,\text{pool})} \leftarrow \ell_{ij,g}^{(0,\text{pool})} + \log \gamma_x(i, j)$;
  Run Algorithm 9 with score $\widetilde{\ell}_{ij,g}^{(0,\text{pool})}$ to obtain $P_g(k \mid y)$, $P_g(t_p \mid y, k)$, $\widehat{k}_g$;
  Run the max-sum routine (Algorithms 10–11) on the same score to obtain $\widehat{t}_g^{\,\text{MAP}}(\widehat{k}_g)$;
  Conditional on $\widehat{t}_g^{\,\text{MAP}}$, recover subject-specific segment-parameter posteriors from each $s \in \mathcal{S}_g$ block update;
return $\{P_g(k \mid y), P_g(t_p \mid y, k), \widehat{t}_g^{\,\text{MAP}}, \text{per-subject parameter posteriors}\}_g$;
`// Complexity:` $\Theta\big(\sum_g (|\mathcal{S}_g| n^2 + k_{\max} n^2)\big)$ `time and` $\Theta(n^2 + k_{\max} n)$ `working memory per group when subject sums are streamed.`
`// Numerical: pool in log space and use logsumexp; carry per-subject log-blocks to floating-point underflow safely; sum` $-\infty$ `contributions are absorbed at the pooled step.`

6.0.5 Complexity summary and stability

Component	Time	Memory	Numerical note
Conjugate block precomputation	$\Theta(n^2)$	$\Theta(n^2)$	prefix summaries; invalid blocks stored as $-\infty$
Exact sum-product messages	$\Theta(k_{\max}n^2)$	$\Theta(k_{\max}n)$ in addition to the block matrix	logsumexp normalization
Boundary and Bayes-curve extraction	$\Theta(k_{\max}n^2)$ for streamed fixed-k summaries	output dependent; up to $\Theta(k_{\max}^2 n)$ for every ordered-boundary table	common evidence normalizer
Max-sum recursion and backtracking	$\Theta(k_{\max}n^2)$	$\Theta(k_{\max}n)$ for values and backpointers	deterministic tie rule
Non-conjugate precomputation	$\Theta(n^2 T_{\text{block}})$	$\Theta(n^2)$ plus routine state	routine convergence/failure flags
Shared-boundary pooling	$\Theta(Sn^2 + k_{\max}n^2)$ after subject blocks	$\Theta(Sn^2)$ or streamed subject sums	sum subject log scores
Latent-group template ascent	per iteration $\Theta(GSn^2 + Gk_{\max}n^2)$	$\Theta(Sn^2 + Gk_{\max}n)$	log-responsibility normalization
Prediction scoring	$\Theta(Gm)$ after segment assignment; $\Theta(Gk_{\max}m^2)$ for resegmentation	problem dependent	log predictive scores

Table 6.1. Implementation-level complexity summary. This is a theoretical accounting table; no source CSV is associated with it.

Proposition 6.2 (BayesBreak time and space complexity)**.** *Suppose the chosen exponential-family block model admits constant-time block-evidence evaluation from cumulative sufficient statistics (Sections 4.1–4.2), and let n denote the number of observations and $k_{\max}$ the maximum segment count considered by the DP. Then:*

1. *Block-evidence precomputation (Algorithm 8) runs in $\Theta(n^2)$ time and stores $\Theta(n^2)$ values.*
2. *The forward and backward DP (Algorithm 9) runs in $\Theta(k_{\max}n^2)$ time and stores $\Theta(k_{\max}n)$ messages. Boundary-event summaries for every k require $\Theta(k_{\max}n)$ output storage; all ordered-boundary distributions for every (k, p, h) require $\Theta(k_{\max}^2 n)$ output storage. Explicit storage of every block-cover contribution can require $\Theta(k_{\max}n^2)$, but it is not required for streamed accumulation.*
3. *The max-sum recursion plus backtracking (Algorithms 10–11) runs in $\Theta(k_{\max}n^2)$ time and $\Theta(k_{\max}n)$ working space, including one predecessor per DP state.*
4. *For non-conjugate block models with per-block approximation cost T_{block} (e.g. Laplace, JJ, EP, PG mean field, 1D quadrature), block precomputation costs $\Theta(n^2 T_{\text{block}})$ and the rest of the DP is unchanged.*

Proof. There are $n(n+1)/2 = \Theta(n^2)$ admissible blocks. When cumulative sufficient statistics are available, each block score requires constant time and one stored scalar, proving (i). For every $k \le k_{\max}$ and endpoint j, the forward recurrence scans at most j predecessors. Hence its operation count is bounded above by $\sum_{k=1}^{k_{\max}} \sum_{j=k}^{n} j = O(k_{\max}n^2)$ and bounded below by a constant multiple of the same order when all blocks are admissible, giving $\Theta(k_{\max}n^2)$. Storing one message per (k, j) is $\Theta(k_{\max}n)$. Boundary-event output has one value per (k, h), while the complete ordered-boundary output has one value per (k, p, h) and therefore $\Theta(k_{\max}^2 n)$ entries. Block-cover terms can be accumulated without retaining every (k, i, j) contribution. The max-sum recurrence visits the same predecessor sets, stores one value and one predecessor per state, and backtracking takes only $O(k_{\max})$, proving (iii). Finally, if one local block evaluation costs T_{block}, evaluating all $\Theta(n^2)$ blocks costs $\Theta(n^2 T_{\text{block}})$; the already constructed array is then opaque to the unchanged partition recursions. This proves (iv). □

Two numerical practices are essential in realistic data sets: (i) always store evidences and DP messages in log space and normalize via `logsumexp`; (ii) standardize/center sufficient statistics when available (e.g., for Gaussian blocks) to reduce catastrophic cancellation in moment computations. These practices do not affect correctness, but they materially improve stability when n is in the hundreds or thousands.

6.0.6 A numerical-implementation checklist

The following checklist distills the common pitfalls we have encountered and translates them into implementation tests. Each item should be enforced either by a unit test, an assertion in the runtime path, or a recorded metadata check in the experiment record.

1. **Log-space storage test.** Store all block evidences, moment numerators, and DP messages as $\log$ values from the start, with explicit tags for signed quantities. Convert to linear space only for display.
2. **Stable-logsumexp test.** Implement `logsumexp` as $\mathrm{lse}(x) = m + \log \sum \exp(x_i - m)$ with $m = \max_i x_i$ [6]. Cache the maximum once and reuse for both numerator and denominator when normalizing.
3. **Prefix-array test.** Compute prefix sums of w_t, $w_t T(y_t)$, and (for Gaussian) $w_t y_t^2$ *before* enumerating blocks. Store them as double-precision arrays of length $n+1$ initialized at index 0. Block summaries then cost one subtraction per block.
4. **DP-invariant test.** After the forward recursion, verify that $\log \widetilde{L}_{0,0} = 0$ and $\log \widetilde{L}_{k,j} = -\infty$ for $j < k$; after the backward recursion, verify the analogous boundary conditions on $\log \widetilde{R}$. Verify that $\log \widetilde{L}_{k,n} = \log \widetilde{R}_{k,0}$ up to floating-point tolerance, that $\sum_k p(k \mid y) = 1$, that the fixed-k boundary marginal sums to $k-1$, and that the reported C_k matches the prior convention used to form the block scores.
5. **Moment-scaling test.** When computing Bayes regression curves via (5.10), subtract a block-specific shift (e.g., the block posterior-mean estimate) before exponentiating to reduce dynamic range, then add the shift back at the end.
6. **Degenerate-block test.** Guard against $W_{ij} = 0$ (missing-data blocks) by returning $\log A_{ij}^{(0)} = -\infty$ or a user-specified fallback. Guard against Q_{ij} numerically negative (from catastrophic cancellation) by clamping to zero.
7. **Reproducibility-metadata test.** Fix a seed for EM initializations and record it in the run metadata; record hyperparameters, $k_{\max}$, the count prior $p(k)$, length factor g, approximation method and settings, data-preprocessing hashes, and the exact version of every family-specific `BlockRoutine` used.
8. **Unit tests tied to theory.** Test the required block-routine outputs, the forward/backward identity, MAP backtracking, fixed-k boundary sums, signed-log moment reconstruction, and every reported complexity mode on small instances where exhaustive enumeration is possible.

CHAPTER 7

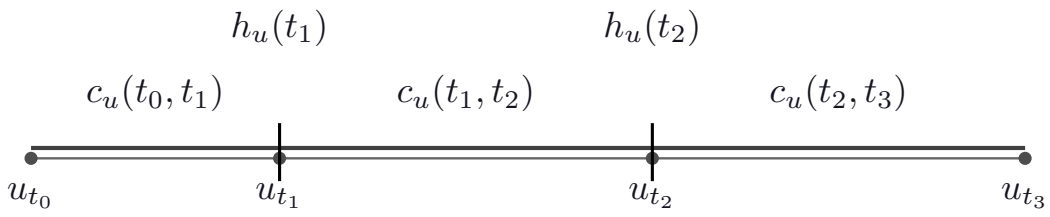
Ordered priors: segment cohesion and boundary hazard

CHAPTER PATH

Irregular designs enter through explicitly separated local factors, retaining exact dynamic programming without a false Poisson-process equivalence.

7.1 Two distinct geometric beliefs

A segment cohesion $c_x(i,j)$ scores the span of a complete block. An interior-boundary hazard $h_x(j)$ scores the candidate split at j. They answer different questions and cannot be justified by the same Poisson-process argument unless the model explicitly contains both factors.



$$p(t \mid k,u) \propto \prod_{q=1}^{k} c_u(t_{q-1},t_q) \prod_{q=1}^{k-1} h_u(t_q)$$

Figure 7.1. Segment cohesion labels complete intervals, whereas a boundary hazard labels interior endpoints only. Both remain local and compatible with dynamic programming.

Theorem 7.1 (Local prior factors preserve exact partition DP)**.** *Under Assumption 3.6, replace every block evidence by* $\widetilde{A}^{(r)}_{ij} = A^{(r)}_{ij}\gamma_x(i,j)$*, where* $\gamma_x(i,j) = c_x(i,j)h_x(j)^{\mathbf{1}\{j<n\}}$*. The forward and backward recursions then sum exactly the unnormalized joint weights* $p(y,t \mid k,x)$*, and division by* $C_k(x)$ *yields the proper fixed-*k *evidence.*

Proof. For any ordered partition t, multiplication of its local factors gives

$$\prod_{q=1}^{k} A^{(0)}_{t_{q-1},t_q}\gamma_x(t_{q-1},t_q) = \left\{\prod_{q} A^{(0)}_{t_{q-1},t_q}\right\}\left\{\prod_{q} c_x(t_{q-1},t_q)\right\}\left\{\prod_{q<k} h_x(t_q)\right\}.$$

The terminal indicator removes $h_x(n)$. This is precisely the likelihood times the unnormalized prior in Equation (3.8). The ordinary partition recursion enumerates every admissible path once, so it returns their sum. Dividing by the prior normalizer completes the claim. □

Proposition 7.2 (Fixed-count interval occupancy from a Poisson boundary process)**.** *Let potential interior boundaries arise from an inhomogeneous Poisson process of intensity $\lambda(u)$ over disjoint physical coordinate intervals $(u_{j-1}, u_j]$, and let $\Lambda_j = \int_{u_{j-1}}^{u_j} \lambda(v)\, dv$. If the process is reduced to the occupancy indicators of these intervals and one conditions on exactly m distinct occupied candidate intervals, then a subset B with $|B| = m$ has probability proportional to*

$$\prod_{j \in B} \frac{1 - e^{-\Lambda_j}}{e^{-\Lambda_j}} = \prod_{j \in B} \left(e^{\Lambda_j} - 1\right).$$

Thus the compatible fixed-count local boundary factor is $h_x(j) \propto e^{\Lambda_j} - 1$. For constant intensity and equal-width intervals this factor is constant, so the induced fixed-count boundary prior is index-uniform when $c_x \equiv 1$.

Proof. Counts in disjoint Poisson-process intervals are independent. Their occupancy indicators are therefore independent Bernoulli variables with $p_j = 1 - e^{-\Lambda_j}$. For a subset B of occupied intervals,

$$\Pr(B) = \prod_{j \in B} p_j \prod_{j \notin B} (1 - p_j) = \left\{ \prod_j (1 - p_j) \right\} \prod_{j \in B} \frac{p_j}{1 - p_j}.$$

After conditioning on $|B| = m$, the first factor is common to all subsets and the local odds are $p_j/(1 - p_j) = e^{\Lambda_j} - 1$. Equal integrated intensities give equal odds and hence a uniform prior over subsets of the fixed size. □

Remark 7.3 (What the proposition does not imply). The proposition weights candidate boundary intervals after conditioning on their number. Without that conditioning, both occupied and unoccupied Bernoulli factors remain in the probability. It does not imply that multiplying each segment likelihood by its physical length is a Poisson boundary-process prior. A length cohesion $c_x(i, j) = g(u_j - u_i)$ is a separate modeling choice.

Proposition 7.4 (Coarse-to-fine likelihood consistency for inherited partitions)**.** *Suppose a fine grid refines a coarse grid without changing observations or the physical contents of any inherited coarse block, and the block model's sufficient summaries are additive. Then the likelihood block evidence $A_{ij}^{(r)}$ for every inherited block is the same whether computed on the coarse representation or by aggregating the fine representation. Prior invariance additionally requires the chosen c_x and h_x to assign the same factors to inherited boundaries.*

Proof. Additivity makes S_{ij}, W_{ij}, and H_{ij} identical for an inherited physical block, so the block integral and moments are identical. The prior statement follows only if its local factors are also representation invariant; it is logically separate from likelihood equality. □

7.2 Recommended configurations

Use $c_x = h_x = 1$ for a truly index-uniform fixed-count prior. Use $c_x(i, j) = g(u_j - u_i)$ when the scientific belief concerns segment duration. Use $h_x(j)$ when external geometry or intensity changes the prior plausibility of a boundary at candidate location j. Combined models are valid, but sensitivity to both components must be reported.

CHAPTER 8

Exact shared-boundary pooling

CHAPTER PATH

Finite-sample block-evidence products, known groups, and a valid finite-candidate consistency theorem.

Plate model (shared boundaries, subject-specific parameters). With a shared segmentation $t = (t_0, \dots, t_k)$ and subject-specific conjugate priors, the generative model nests an inner *segment* plate inside an outer *subject* plate:

$$t \sim p(t \mid k), \qquad \theta_q^{(s)} \overset{\text{ind}}{\sim} p(\theta \mid \alpha_0^{(s)}, \beta_0^{(s)}), \quad y_i^{(s)} \mid \theta_{q(i)}^{(s)}, w_i^{(s)} \overset{\text{ind}}{\sim} p(y \mid \theta_{q(i)}^{(s)}, w_i^{(s)}), \tag{8.1}$$

where $q(i)$ is the segment index containing observation i under t.

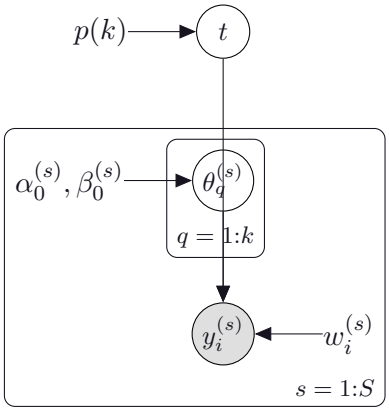


Figure 8.1. Plate model for multi-subject pooling under shared boundaries. The boundary vector t is shared across subjects; each subject has its own segment parameters $\theta_q^{(s)}$ and exposures $w_i^{(s)}$.

Suppose we observe S independent subjects, each with its own observations $\{(x_i^{(s)}, y_i^{(s)})\}$ and exposure weights $w_i^{(s)}$, but we wish to model them as sharing a common segmentation (boundary locations) while allowing subject-specific segment parameters. For simplicity of exposition, assume a common index grid $i = 1, \dots, n$ (e.g., after alignment/union of grids); subject-specific missing observations are handled by setting $w_i^{(s)} = 0$ at missing indices.

Let $(i, j]$ denote a block and write subject-specific block sums as $(S_{ij}^{(s)}, W_{ij}^{(s)}, H_{ij}^{(s)})$ with subject-specific prior hyperparameters $(\alpha_0^{(s)}, \beta_0^{(s)})$.

Assumption 8.1 (Common-grid shared-boundary support and conditional independence)**.** Subjects are aligned to a common ordered candidate-boundary grid, and every shared boundary vector $t = (t_0, \ldots, t_k)$ used by the pooled DP lies in the intersection of the subject-specific admissible supports after missingness/exposure conventions are applied. Conditional on this shared segmentation, subjects $s = 1, \ldots, S$ are independent, each subject has its own segment parameters $\theta_{1:k}^{(s)}$ with subject-specific conjugate priors, and subject-level block evidences are finite and strictly positive on the pooled support.

Theorem 8.2 (Exact pooling of block evidences)**.** *Under Assumption 8.1, the pooled block evidence for $(i, j]$ factorizes as*

$$A_{ij,\mathrm{pool}}^{(0)} = \prod_{s=1}^{S} \exp\{H_{ij}^{(s)}\} \, \frac{Z(\alpha_0^{(s)} + S_{ij}^{(s)}, \beta_0^{(s)} + W_{ij}^{(s)})}{Z(\alpha_0^{(s)}, \beta_0^{(s)})}.$$

Equivalently, $\log A_{ij,\mathrm{pool}}^{(0)} = \sum_s \log A_{ij,s}^{(0)}$. Running the DP recursions (5.1)–(5.10) with $A_{ij}^{(0)}$ replaced by $A_{ij,\mathrm{pool}}^{(0)}$ yields the exact posterior over shared boundaries. Conditional on any selected or averaged boundary configuration, the subject-specific segment-parameter posteriors are then recovered from the corresponding subject-level conjugate updates.

Proof. **Step 1: subject-wise block evidences.** For each subject s, Theorem 4.1 yields the subject-specific block evidence $A_{ij,s}^{(0)}$ computed from $(S_{ij}^{(s)}, W_{ij}^{(s)}, H_{ij}^{(s)})$.

Step 2: independence implies product pooling. Conditional on shared boundaries, subjects are independent and have independent segment parameters. Therefore, the joint likelihood (with segment parameters integrated out) for the block is the product of the subject-wise block evidences: $A_{ij,\mathrm{pool}}^{(0)} = \prod_s A_{ij,s}^{(0)}$, giving the stated form.

Step 3: DP correctness under pooled evidences. The DP in Section 5 depends only on the block evidence matrix. Replacing $A_{ij}^{(0)}$ by $A_{ij,\mathrm{pool}}^{(0)}$ therefore computes the exact shared-boundary posterior for the pooled model; parameter posteriors remain available conditionally on boundaries from the subject-specific block updates. □

8.1 Groups with known membership

When subject group labels $z_s \in \{1, \ldots, G\}$ are observed, BayesBreak applies the pooled DP of Theorem 8.2 independently within each group: for group g with subject index set $\mathcal{S}_g$, pool per-subject block evidences $\{A_{ij,s}^{(0)}\}_{s \in \mathcal{S}_g}$ into a group-specific pooled evidence $A_{ij,g}^{(0)}$, then run the DP recursions (5.1)–(5.10) with $A_{ij}^{(0)} = A_{ij,g}^{(0)}$. This yields group-specific $P_g(k \mid y)$, boundary posteriors $P_g(t_p \mid y, k)$, and segment/regression-curve moments at total cost G times the single-group cost. This is the supervised counterpart of the latent-template mixture in Section 9.3.

8.2 Finite-candidate concentration across independent subjects

Theorem 8.3 (Shared-boundary posterior consistency on a finite candidate set)**.** *Let $\mathcal{T}$ be a finite set of candidate segmentations and let $t^\star \in \mathcal{T}$ be the data-generating candidate. For subject s, let $M_s(t) > 0$ be the integrated score used by the shared-boundary model.*

Assume subjects are independent, the prior satisfies $p(t^\star) > 0$, and for every $t \neq t^\star$:
(i) $\mathbb{E}|\log M_s(t^\star) - \log M_s(t)| < \infty$;
(ii) a strong law holds for the centered log-score differences; and
(iii) $\liminf_{S\to\infty} S^{-1} \sum_{s=1}^{S} \mathbb{E}[\log M_s(t^\star) - \log M_s(t)] = \delta_t > 0$.
Then $p(t^\star \mid y^{(1:S)}) \to 1$ almost surely, and every joint MAP segmentation equals $t^\star$ eventually almost surely.

Proof. For any $t \neq t^\star$, the posterior ratio is

$$\frac{p(t \mid y^{(1:S)})}{p(t^\star \mid y^{(1:S)})} = \frac{p(t)}{p(t^\star)} \exp\left\{ -\sum_{s=1}^{S} \big[\log M_s(t^\star) - \log M_s(t)\big] \right\}.$$

By the strong-law assumption and the positive limiting average expectation gap, the normalized sum in brackets has positive liminf δ_t almost surely. Hence the ratio converges to zero exponentially on the S scale. Because $\mathcal{T}$ is finite, summing these ratios over all wrong candidates still converges to zero. Normalization gives posterior concentration at $t^\star$; the MAP conclusion follows because all wrong posterior ratios are eventually below one. □

Remark 8.4 (Why one informative subject is insufficient). A single fixed informative subject does not imply a positive *average* score gap as the number of subjects grows. The theorem requires information to accumulate through a positive limiting average gap; this repairs the earlier asymptotic statement without altering exact finite-sample pooling.

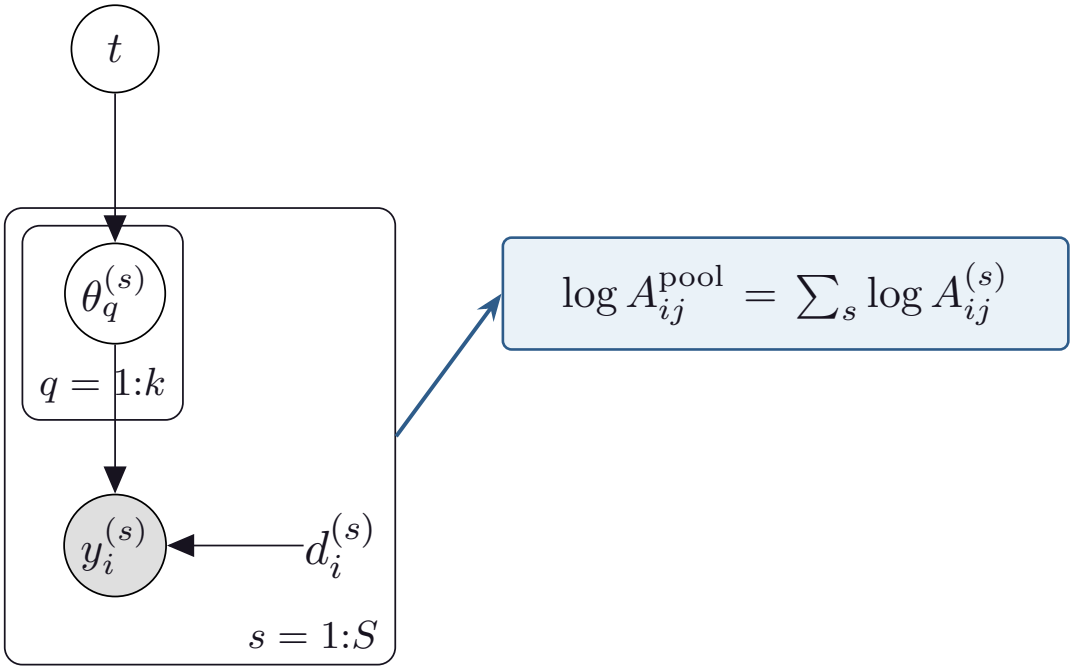


Figure 8.2. Conditional independence across subjects converts each pooled block score into a product of subject-level block evidences, or a sum in log space.

Grouped and latent-group models

CHAPTER PATH

Known-group shared-changepoint analysis and a finite latent-group criterion with Jensen minorization and exact template updates over a finite partition set.

9.1 Known groups

Suppose each sequence s has an observed group label $g(s) \in \{1, \ldots, G\}$. Within a group, sequences may share a common partition while retaining sequence-specific segment parameters. Under conditional independence, the group-specific marginal likelihood for a candidate segment is the product of the sequence-specific segment marginal likelihoods. The sum-product and max-sum recursions of Chapters 5 and 6 then apply separately to each group. This is a direct hierarchical extension of the shared-changepoint model in Chapter 8.

9.2 Finite latent-group criterion

When group labels are unknown, let $\tau_g = (k_g, t^{(g)})$ denote a deterministic candidate changepoint template for group g. The archived implementation uses the sequence–template score

$$S_s(\tau) = \frac{p(k)}{C_k} \prod_{q=1}^{k} A^{(0,s)}_{t_{q-1},t_q}\, \gamma_u(t_{q-1}, t_q), \tag{9.1}$$

with the convention that an impossible segment gives score zero. This subject-template score is not assumed to integrate to one over the sample space of sequence s. For weights $\pi_g \geq 0$ with $\sum_g \pi_g = 1$, define

$$\mathcal{F}(\pi, \tau) = \sum_{s=1}^{S} \log \left\{ \sum_{g=1}^{G} \pi_g S_s(\tau_g) \right\}. \tag{9.2}$$

Equation (9.2) is the optimization criterion studied in this chapter. The results below concern this finite criterion and do not require interpreting $S_s(\tau_g)$ as a normalized mixture component.

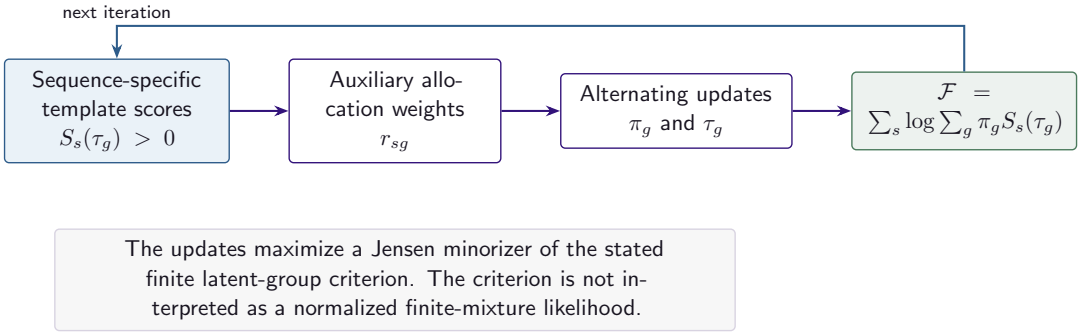


Figure 9.1. Alternating optimization of the finite latent-group criterion. Auxiliary allocation weights make a Jensen lower bound tight; group-template updates maximize the resulting weighted partition score over the finite candidate set.

9.3 Jensen minorization and alternating updates

For any nonnegative weights r_{sg} satisfying $\sum_g r_{sg} = 1$, the log-sum inequality gives

$$\mathcal{F}(\pi, \tau) \geq \mathcal{Q}(r, \pi, \tau) := \sum_{s,g} r_{sg} \left[\log \pi_g + \log S_s(\tau_g) - \log r_{sg}\right]. \tag{9.3}$$

Equality holds at

$$r_{sg} = \frac{\pi_g S_s(\tau_g)}{\sum_h \pi_h S_s(\tau_h)}. \tag{9.4}$$

For fixed r, maximization over the group weights gives $\pi_g = S^{-1} \sum_s r_{sg}$. Let $n_g = \sum_s r_{sg}$. Substituting Equation (9.1) into $\mathcal{Q}$ shows that, for a candidate segment count k, the group-g template contribution is

$$n_g\{\log p(k) - \log C_k\} + \sum_{q=1}^{k} \ell^{(g)}_{t_{q-1},t_q}, \qquad \ell^{(g)}_{ij} = \sum_{s=1}^{S} r_{sg} \log A^{(0,s)}_{ij} + n_g \log \gamma_u(i,j). \tag{9.5}$$

Thus the update is an exact max-sum recursion at fixed k, followed by comparison across k using the displayed count offset. The convention $0 \log 0 = 0$ applies when $r_{sg} = 0$, while a positive responsibility on a zero sequence–template score makes that candidate value $-\infty$.

> **Theorem 9.1** (Monotonicity of exact latent-group updates)**.** *Consider an iteration that (i) sets r according to Equation (9.4), (ii) maximizes $\mathcal{Q}$ over π, and (iii) exactly maximizes each responsibility-weighted template criterion using the local score and segment-count offset in Equation (9.5) over its declared finite candidate set. Then the iteration cannot decrease $\mathcal{F}$.*

Proof. At the current parameter values, Equation (9.4) makes the Jensen bound tight: $\mathcal{F}_{\text{old}} = \mathcal{Q}(r, \pi_{\text{old}}, \tau_{\text{old}})$. Exact maximization of the minorizer gives

$$\mathcal{Q}(r, \pi_{\text{new}}, \tau_{\text{new}}) \geq \mathcal{Q}(r, \pi_{\text{old}}, \tau_{\text{old}}).$$

The log-sum inequality gives $\mathcal{F}_{\text{new}} \geq \mathcal{Q}(r, \pi_{\text{new}}, \tau_{\text{new}})$. Combining the inequalities yields $\mathcal{F}_{\text{new}} \geq \mathcal{F}_{\text{old}}$. □

Algorithm 13: Alternating optimization for finite latent-group templates

Input: Sequence-specific segment-score arrays; number of groups G; admissible segment counts; number of restarts; convergence tolerance.

foreach *restart* **do**
- initialize π and templates $\tau_{1:G}$;
- evaluate the current value of $\mathcal{F}$;
- **repeat**
  - update r_{sg} by Equation (9.4);
  - update $\pi_g \leftarrow S^{-1} \sum_s r_{sg}$;
  - for each group g, set $n_g \leftarrow \sum_s r_{sg}$, build the local scores in Equation (9.5), solve the max-sum recursion at every allowed k, and select the offset-adjusted maximizer using $n_g\{\log p(k) - \log C_k\}$;
  - evaluate and append the new value of $\mathcal{F}$;
- **until** *the increase in $\mathcal{F}$ is at most the tolerance or the iteration limit is reached*;
- retain the restart with the largest final value in its recorded objective sequence;

Output: Group templates, group weights, auxiliary allocation weights, complete objective sequences, and restart ranking by the final objective value.

9.4 Non-uniqueness and implementation limitation

The criterion is invariant to permutation of the group labels. It may have duplicate or collapsed templates when G is larger than the effective number of groups, and multiple restarts do not guarantee a global maximum. These are optimization and identifiability limitations of the stated criterion.

The archived implementation contains an exit path that can return the objective value from the previous iteration rather than the last value in the objective sequence. In one of three diagnostic restarts, the difference was 0.003137360860080. The archived latent-group simulation remains a real computation, but the implementation verification must correct the returned objective and restart ranking before those values are treated as release-quality optimization results.

ESTABLISHED RESULT AND NON-CLAIM

For Equation (9.2), the auxiliary allocation update makes the Jensen minorizer tight, the group-weight update maximizes that minorizer, and the finite template step is solved exactly by max-sum recursion when Equation (9.5) applies. The monotonicity theorem is complete. No theorem is claimed for identifiability of a normalized finite mixture or consistency of the estimated number of groups.

CHAPTER 10

Nonconjugate block integration and numerical certification

CHAPTER PATH

Laplace, variational, expectation-propagation, Pólya–Gamma, and quadrature calculations for segment marginal likelihoods, with conditional propagation of segment-wise log-score error through the partition recursions.

EXACT GLOBAL ALGEBRA, APPROXIMATE LOCAL TARGET

Once an approximate block array is supplied, the partition recursions are evaluated exactly for that array. This is not the same as exact inference under the reference nonconjugate model. The gap is controlled only when a local certificate verifies Assumption 10.6 on all blocks reachable by the declared partition support.

Plate model (non-conjugate single segment with PG augmentation). The generative graph for a non-conjugate block $(i, j]$ is identical to Figure 4.1 with $\theta \sim \pi(\theta)$ replaced by a non-conjugate prior (e.g., Gaussian on a log or logit scale). For Pólya–Gamma-augmented binomial logistic blocks (Theorem 10.4), introducing the latent scale-mixture variables $\omega_t \sim \mathrm{PG}(m_t, 0)$ restores conditional conjugacy and is reflected in the plate as a second observed-indexed latent:

$$\theta \sim \mathcal{N}(\nu, \rho^2), \quad \omega_t \overset{\text{ind}}{\sim} \mathrm{PG}(m_t, 0), \quad y_t \mid \theta, \omega_t, m_t \overset{\text{ind}}{\sim} \mathrm{Binom}(m_t, \sigma(\theta)) \text{ with (10.5)}, \quad (10.1)$$

where σ is the logistic function.

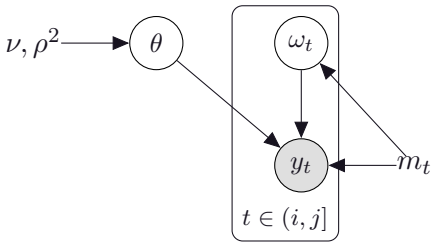


Figure 10.1. Plate model for a non-conjugate block with Pólya–Gamma augmentation. Conditional on ω_t, the block is quadratic in θ with a Gaussian posterior (Proposition 10.5); the same template applies to Laplace and variational routines with the ω_t node omitted.

In many applications the single-segment parameterization is not conjugate to the observation

model—for example, log-normal priors for positive means, heavy-tailed shrinkage priors, or logistic/negative-binomial links with structural offsets. This section develops a plug-in toolkit for computing approximate block evidences $A_{ij}^{(0)}$ and moments $A_{ij}^{(r)}$ when conjugacy is unavailable, while keeping the dynamic-programming layer of Section 5 unchanged. When approximate block evidences are used, the DP is exact for the approximate block-score model; additional error statements require explicit blockwise approximation control.

We focus on one-dimensional (or low-dimensional) canonical GLMs, where for a block $(i, j]$ we must approximate integrals of the form

$$A_{ij}^{(0)} = \int \Big[\prod_{t=i+1}^{j} p(y_t \mid \theta) \Big] p(\theta)\, d\theta, \qquad A_{ij}^{(r)} = \int \Big[\prod_{t=i+1}^{j} p(y_t \mid \theta) \Big] p(\theta)\, m_\star(\theta)^r\, d\theta,$$

where $m_\star(\theta) = m(\theta; d_\star)$ is the declared reporting functional at a fixed reference descriptor. We describe four complementary approaches: (i) first-order Laplace approximation; (ii) variational lower bounds (Jaakkola–Jordan [20] type for logistic regression and mean-field for more general GLMs); (iii) expectation propagation (EP); and (iv) Pólya–Gamma augmentation for logistic and negative-binomial blocks. Throughout we use the family-specific descriptor d_t and any separately declared power weight a_t and cumulative statistics (S_{ij}, W_{ij}, H_{ij}).

10.0.1 Model and notation for GLM blocks

Consider a one-parameter canonical GLM on a block $(i, j]$:

$$\log p(y_t \mid \theta) = w_t \big\{ y_t \theta - b(\theta) \big\} + \log h(y_t, w_t),$$

where b is the cumulant function and h collects base measures (including any w_t-dependent terms). The block log-likelihood is

$$\ell_{ij}(\theta) := \sum_{t=i+1}^{j} w_t \{ y_t \theta - b(\theta) \} + H_{ij}, \qquad H_{ij} := \sum_{t=i+1}^{j} \log h(y_t, w_t).$$

Let $\pi(\theta) = p(\theta)$ be an arbitrary prior (not necessarily conjugate), and write $\Psi_{ij}(\theta) := \ell_{ij}(\theta) + \log \pi(\theta)$ for the block log-posterior.

Derivatives are

$$\ell'_{ij}(\theta) = S_{ij} - W_{ij} b'(\theta), \qquad \ell''_{ij}(\theta) = -W_{ij} b''(\theta),$$

and

$$\Psi'_{ij}(\theta) = \ell'_{ij}(\theta) + (\log \pi)'(\theta), \qquad \Psi''_{ij}(\theta) = \ell''_{ij}(\theta) + (\log \pi)''(\theta).$$

10.0.2 Existence and uniqueness of block modes

Lemma 10.1 (Strict log-concavity and uniqueness). *Suppose b and $\log \pi$ are twice differentiable on a convex parameter domain, $b''(\theta) > 0$ for all θ (canonical GLM), the prior is log-concave $((\log \pi)''(\theta) \le 0)$, and $W_{ij} > 0$. Then Ψ_{ij} is strictly concave. If, in addition, Ψ_{ij} is proper and attains its supremum on the parameter domain (for example by coercive tails or compact-domain truncation), the maximizer $\hat{\theta}_{ij}$ exists and is unique.*

Proof. Since $W_{ij} > 0$ and $b''(\theta) > 0$, we have $-W_{ij} b''(\theta) < 0$. Adding $(\log \pi)''(\theta) \le 0$ yields $\Psi''_{ij}(\theta) < 0$ wherever the derivatives exist, so Ψ_{ij} is strictly concave. Strict concavity gives

uniqueness of any maximizer; existence is provided by the stated attainment assumption. □

10.0.3 Laplace approximation for single-block evidence

Theorem 10.2 (First-order Laplace approximation with a uniform remainder)**.** *Index a sequence of blocks by an effective information size $N = W_{ij} \to \infty$. Let $\widehat{\theta}_N$ be the unique interior maximizer of Ψ_N and $H_N = -\Psi_N''(\widehat{\theta}_N)$. Assume there are constants $c, C, \delta > 0$, independent of the block, such that: (i) $cN \le -\Psi_N''(\theta) \le CN$ throughout $|\theta - \widehat{\theta}_N| \le \delta$; (ii) Ψ_N is four times continuously differentiable on that neighborhood and $|\Psi_N^{(r)}(\theta)| \le CN$ for $r = 3, 4$; and (iii) the integral outside the neighborhood is at most the local Gaussian leading term times $O(N^{-1})$. Then*

$$A_{ij}^{(0)} = e^{\Psi_N(\widehat{\theta}_N)} \left(\frac{2\pi}{H_N}\right)^{1/2} \{1 + O(N^{-1})\}, \tag{10.2}$$

and hence

$$\log A_{ij}^{(0)} = \Psi_N(\widehat{\theta}_N) + \tfrac{1}{2}\log(2\pi) - \tfrac{1}{2}\log H_N + O(N^{-1}). \tag{10.3}$$

For a bounded twice continuously differentiable test function g whose first two derivatives are uniformly bounded in the same neighborhood,

$$\frac{M_{ij}^{[g]}}{A_{ij}^{(0)}} = g(\widehat{\theta}_N) + O(N^{-1}), \qquad M_{ij}^{[g]} = \int g(\theta) e^{\Psi_N(\theta)}\, d\theta. \tag{10.4}$$

Proof. Set $u = \sqrt{H_N}(\theta - \widehat{\theta}_N)$. Taylor's theorem through fourth order gives, on the local neighborhood,

$$\Psi_N(\theta) - \Psi_N(\widehat{\theta}_N) = -\tfrac{1}{2}u^2 + a_N u^3 N^{-1/2} + b_N(u) u^4 N^{-1},$$

where a_N and $b_N(u)$ are uniformly bounded by assumptions (i)–(ii). On a growing central region $|u| \le N^{1/12}$, expand the exponential. The term linear in u^3 integrates to zero against the symmetric Gaussian kernel; the remaining terms are bounded by an integrable Gaussian polynomial times N^{-1}. The part of the local neighborhood outside this central region is exponentially smaller by the quadratic term, and assumption (iii) controls the nonlocal tail. After the Jacobian $H_N^{-1/2}$, this proves the relative $1 + O(N^{-1})$ expansion. Applying the same argument to the numerator after the second-order Taylor expansion $g(\widehat{\theta}_N + u/\sqrt{H_N}) = g(\widehat{\theta}_N) + g'(\widehat{\theta}_N)u/\sqrt{H_N} + O(u^2/N)$ and again using Gaussian symmetry yields the ratio statement. □

Efficient Newton updates using block summaries. For canonical GLMs,

$$\Psi_{ij}'(\theta) = S_{ij} - W_{ij} b'(\theta) + (\log \pi)'(\theta), \quad \Psi_{ij}''(\theta) = -W_{ij} b''(\theta) + (\log \pi)''(\theta),$$

so Newton iteration is

$$\theta^{(m+1)} \leftarrow \theta^{(m)} - \frac{S_{ij} - W_{ij} b'(\theta^{(m)}) + (\log \pi)'(\theta^{(m)})}{-W_{ij} b''(\theta^{(m)}) + (\log \pi)''(\theta^{(m)})},$$

with cost independent of block length once S_{ij}, W_{ij} are available.

Implementation. Algorithm 14 implements the Laplace block-evidence approximation using a Newton or quasi-Newton solver for the block mode and the corresponding Hessian correction. The approximation is local but often highly accurate in moderate-to-large blocks. The remaining non-conjugate routines (Algorithms 15–18) follow the same template: a block-local optimizer feeds an approximate log-evidence into the DP.

Algorithm 14: GLMBlock_Laplace$(y, w, i, j, \mathtt{b}, \mathtt{prior})$

Input: $y_{i+1:j}$, weights $w_{i+1:j}$; cumulant $b(\theta)$; prior $\pi(\theta)$

Compute $S \leftarrow \sum_{t=i+1}^{j} w_t y_t$; $\quad W \leftarrow \sum_{t=i+1}^{j} w_t$; $\quad H \leftarrow \sum_{t=i+1}^{j} \log h(y_t, w_t)$;

Define $\Psi(\theta) = H + S\theta - Wb(\theta) + \log \pi(\theta)$ and its derivatives;

Initialize $\theta^{(0)}$ (e.g., solve $b'(\theta) = S/W$ or use prior mean);

for $m = 0, 1, \ldots$ *until convergence* **do**

  $g \leftarrow S - Wb'(\theta^{(m)}) + (\log \pi)'(\theta^{(m)})$;

  $h \leftarrow -Wb''(\theta^{(m)}) + (\log \pi)''(\theta^{(m)})$;

  if $|g|/|h| < \varepsilon_{newton}$ **then**

    break

  $\theta^{(m+1)} \leftarrow \theta^{(m)} - g/h$;

$\hat{\theta} \leftarrow \theta^{(m)}$; $\quad H^\star \leftarrow -\Psi''(\hat{\theta})$;

$\log \widehat{A}_{ij}^{(0)} \leftarrow \Psi(\hat{\theta}) + \frac{1}{2}\log(2\pi) - \frac{1}{2}\log H^\star$;

Output: $\widehat{A}_{ij}^{(0)}$ and (optional) moments via (10.4)

10.0.4 Variational lower bounds for GLM blocks

When a deterministic lower bound is desirable, introduce a variational distribution $q(\theta)$ and apply Jensen:

$$\log A_{ij}^{(0)} = \log \int q(\theta)\, \frac{e^{\Psi_{ij}(\theta)}}{q(\theta)}\, d\theta \;\geq\; \int q(\theta)\, \Psi_{ij}(\theta)\, d\theta + \mathcal{H}[q] \;:=\; \mathcal{L}_{ij}(q).$$

Jaakkola–Jordan bound for logistic blocks. For Bernoulli/logistic likelihoods, a standard quadratic bound yields a tractable Gaussian variational posterior. One convenient form (for sigmoid σ) is:

$$\log \sigma(z) \;\geq\; \log \sigma(\xi) + \frac{z - \xi}{2} - \lambda(\xi)(z^2 - \xi^2), \qquad \lambda(\xi) = \frac{\tanh(\xi/2)}{4\xi},$$

with variational parameter $\xi > 0$. Applying this bound to the appropriate signed linear predictors yields a quadratic surrogate in θ, so the optimal Gaussian $q(\theta)$ and ξ admit closed-form coordinate updates.

> **Proposition 10.3** (Monotone improvement of variational block bounds)**.** *Every exact coordinate-ascent update of the local variational parameters (for example ξ in a logistic block) or of a Gaussian $q(\theta)$ leaves the block evidence lower bound $\mathcal{L}_{ij}(q)$ non-decreasing. If the iterates converge to an interior fixed point and the objective is differentiable there, that fixed point is coordinatewise stationary. At every iteration, $\underline{A}_{ij}^{(0)} := \exp\{\mathcal{L}_{ij}(q)\}$ is a certified lower bound on $A_{ij}^{(0)}$.*

Proof. Each coordinate update is an exact maximization of $\mathcal{L}_{ij}(q)$ with respect to a subset of variables holding the others fixed, so the ELBO is non-decreasing. At a converged interior

fixed point, the first-order derivative along each updated coordinate block vanishes under the stated regularity, which is coordinatewise stationarity. Because $\mathcal{L}_{ij}$ is a Jensen lower bound on $\log A_{ij}^{(0)}$ at every iterate, exponentiating yields a valid evidence lower bound. □

Algorithm 15: LogisticBlock_JJ$(y, m, w, i, j, \texttt{prior})$ (variational lower bound)

Input: Binomial counts $y_t \sim \mathrm{Binom}(m_t, p)$, logit link; prior $\pi(\theta)$

Initialize ξ and a Gaussian $q(\theta) = \mathcal{N}(\mu_q, \sigma_q^2)$;
repeat
  Update the quadratic bound parameter $\xi \leftarrow \sqrt{\mathbb{E}_q[\theta^2]}$;
  Update Gaussian $q(\theta)$ by maximizing the block ELBO $\mathcal{L}_{ij}(q)$ (closed-form for quadratic surrogate);
until *converged*;
Return $\underline{A}_{ij}^{(0)} = \exp\{\mathcal{L}_{ij}(q)\}$ and moments from q.

10.0.5 Expectation propagation (EP) for GLM blocks

Expectation propagation (EP) [23] provides a deterministic approximation to the block posterior and block evidence by iteratively refining local site approximations and matching moments. In the formulation used here the prior is Gaussian, and each likelihood term $p(y_t \mid \theta)$ is approximated by a Gaussian site $\tilde{t}_t(\theta) = c_t \exp\{-\frac{1}{2} a_t \theta^2 + b_t \theta\}$. After convergence, the product of the Gaussian prior and sites is Gaussian and the approximate evidence is $\widehat{A}_{ij}^{(0,\mathrm{EP})} = \int \pi(\theta) \prod_{t=i+1}^{j} \tilde{t}_t(\theta)\, d\theta$. The constants c_t are part of the returned log evidence; dropping them changes the DP score. A non-Gaussian prior requires an additional site approximation or numerical integral and is not covered by this Gaussian-posterior description. EP is neither a bound nor guaranteed to converge.

Algorithm 16: GLMBlock_EP$(y, w, i, j, \texttt{prior})$

Input: Canonical GLM; Gaussian prior $\pi(\theta)$; tolerance and maximum iterations

Initialize site parameters $\{a_t, b_t, c_t\}_{t=i+1}^{j}$; set $\tilde{p}(\theta) \propto \pi(\theta) \prod_t \tilde{t}_t(\theta)$;
repeat
  for $t = i + 1$ **to** j **do**
    Form cavity $\tilde{p}^{\setminus t}(\theta) \propto \tilde{p}(\theta) / \tilde{t}_t(\theta)$ and verify positive cavity precision;
    Form tilted $p^{\mathrm{tilt}}(\theta) \propto \tilde{p}^{\setminus t}(\theta)\, p(y_t \mid \theta)$;
    Moment-match p^{tilt} and update the site parameters (a_t, b_t, c_t), damping if needed;
until *moments stable or maximum iterations reached*;
Return a convergence/failure flag, the approximate log evidence $\log \widehat{A}_{ij}^{(0,\mathrm{EP})} = \log \int \pi(\theta) \prod_{t=i+1}^{j} c_t \exp\{-a_t \theta^2/2 + b_t \theta\}\, d\theta$, and moments from $\tilde{p}$.

An implementation should export the convergence flag and the maximum site residual together with the block score. Downstream DP routines may then exclude failed EP blocks, retry them with stronger damping, or mark the whole EP approximation as uncertified for the stability analysis.

10.0.6 Pólya–Gamma augmentation for binomial-logistic blocks

Theorem 10.4 (Pólya–Gamma identity and conditional conjugacy)**.** *For binomial logistic blocks with counts m_t and successes y_t, let $w_t \geq 0$ denote either one or a declared likelihood-power/replication weight, define $b_t = w_t m_t$ and $\kappa_t = w_t(y_t - \frac{m_t}{2})$, and introduce latent $\omega_t \sim \mathrm{PG}(b_t, 0)$. Then the weighted logistic likelihood term admits the Gaussian scale-mixture representation*

$$\left\{ \frac{(e^\theta)^{y_t}}{(1+e^\theta)^{m_t}} \right\}^{w_t} = 2^{-b_t} e^{\kappa_t \theta} \int_0^\infty e^{-\omega_t \theta^2/2}\, p(\omega_t)\, d\omega_t. \tag{10.5}$$

The binomial coefficient, when present, is independent of θ and must be retained in the base-measure contribution to the block evidence. Conditionally on $\omega_{i+1:j}$, the remaining block likelihood is quadratic in θ:

$$\log p(y_{i+1:j} \mid \theta, \omega) = -\tfrac{1}{2}\theta^2 \sum_{t=i+1}^{j} \omega_t + \theta \sum_{t=i+1}^{j} \kappa_t + \text{const}. \tag{10.6}$$

With a Gaussian prior $\pi(\theta) = \mathcal{N}(\nu, \rho^2)$, the conditional posterior $p(\theta \mid y, \omega)$ is Gaussian.

Proof. **Step 1: the identity.** Equation (10.5) is the defining Pólya–Gamma augmentation of Polson et al. [27]: it expresses the binomial-logistic term as a Gaussian scale mixture in θ.

Step 2: condition on ω and collect quadratic terms. Given ω_t, each weighted term contributes $\kappa_t\theta - \frac{1}{2}\omega_t\theta^2$ up to constants, where the weight has already been absorbed into b_t and κ_t. Summing over t yields (10.6).

Step 3: combine with a Gaussian prior. Multiplying the quadratic-in-θ conditional likelihood by a Gaussian prior produces a Gaussian conditional posterior, because Gaussian priors are conjugate to Gaussian likelihoods. □

Proposition 10.5 (Mean-field variational approximation under Pólya–Gamma)**.** *Under the same setup as Theorem 10.4 and a Gaussian prior $\pi(\theta) = \mathcal{N}(\nu, \rho^2)$, the mean-field factorization $q(\theta, \omega) = q(\theta) \prod_{t=i+1}^{j} q(\omega_t)$ admits the following closed-form coordinate-ascent updates: (i) $q(\theta) = \mathcal{N}(\mu_q, \sigma_q^2)$ with $\sigma_q^{-2} = \rho^{-2} + \sum_t \mathbb{E}_q[\omega_t]$ and $\mu_q = \sigma_q^2(\rho^{-2}\nu + \sum_t \kappa_t)$; (ii) $q(\omega_t) = \mathrm{PG}(b_t, c_t)$ with $c_t^2 = \mathbb{E}_q[\theta^2] = \mu_q^2 + \sigma_q^2$, giving $\mathbb{E}_q[\omega_t] = \frac{b_t}{2c_t}\tanh(c_t/2)$. These updates are monotone in the block ELBO.*

Proof. For a mean-field factor, the coordinate optimum satisfies

$$\log q^*(\theta) = \mathbb{E}_{q(\omega)}[\log p(y, \theta, \omega)] + \text{const}.$$

Combining Equation (10.6) with the Gaussian prior gives a quadratic log density whose precision and precision-weighted mean are $\rho^{-2} + \sum_t \mathbb{E}_q[\omega_t]$ and $\rho^{-2}\nu + \sum_t \kappa_t$, respectively. This proves the stated Gaussian update. Similarly, $\log q^*(\omega_t) = \mathbb{E}_{q(\theta)}[\log p(y_t, \omega_t \mid \theta)] + \text{const}$ is the log density of an exponentially tilted $\mathrm{PG}(b_t, 0)$ variable with tilt $c_t = \sqrt{\mathbb{E}_q[\theta^2]}$, so $q^*(\omega_t) = \mathrm{PG}(b_t, c_t)$. Its standard mean is $b_t \tanh(c_t/2)/(2c_t)$, with the continuous limit $b_t/4$ at $c_t = 0$. Each update is the exact maximizer of the same block ELBO with the other factors fixed; therefore the ELBO is nondecreasing after every coordinate update. □

Algorithm 17: GLMBlock_PG_VB$(y, m, i, j, \nu, \rho^2)$ (Gaussian prior, PG mean-field)

Input: Binomial/logistic block with trial counts m_t; Gaussian prior $\mathcal{N}(\nu, \rho^2)$

Initialize $q(\omega_t)$ and $q(\theta) = \mathcal{N}(\mu, \sigma^2)$;

repeat

  Set $b_t \leftarrow w_t m_t$ and $\kappa_t \leftarrow w_t(y_t - m_t/2)$ if likelihood-power weights are used; otherwise $w_t = 1$;

  $\sigma^{-2} \leftarrow \rho^{-2} + \sum_{t=i+1}^{j} \mathbb{E}_q[\omega_t]$;

  $\mu \leftarrow \sigma^2 \left(\rho^{-2}\nu + \sum_{t=i+1}^{j} \kappa_t\right)$;

  foreach t **do**

    Update $q(\omega_t) = \mathrm{PG}(w_t m_t, c_t)$ using current $\mathbb{E}_q[\theta^2]$;

until *ELBO converged*;

Return a lower bound $\underline{A}_{ij}^{(0)}$ and approximate moments from $q(\theta)$.

Using approximate block evidences in BayesBreak. Once any block-evidence approximation is available (Laplace, JJ, EP, PG-VB, or another surrogate), BayesBreak uses it exactly as an input score array: build a triangular array of approximate block log-evidences and run the same DP recursions. The resulting posterior summaries are exact for that approximate score array, not automatically exact for the original non-conjugate model. Algorithm 18 provides a generic recipe for constructing this array in a model-agnostic way, emphasizing reuse of block-local optimizers and cached sufficient-statistic summaries.

Algorithm 18: Precompute block evidences for non-conjugate GLMs

Input: $\{(x_t, y_t, w_t)\}_{t=1}^{n}$, method $\in$ {Laplace, JJ, PG–VB, EP}

for $i = 0$ **to** $n - 1$ **do**

  Initialize running summaries (as needed);

  for $j = i + 1$ **to** n **do**

    Update block summaries for $(i, j]$;

    Call chosen block routine to compute $\widehat{A}_{ij}^{(0)}$ (and optional moments);

Run DP (Algorithm 6) with resulting block evidences.

10.0.7 Propagation of block approximation error through the DP

Assumption 10.6 (Uniform per-block log-evidence error)**.** Let $\widehat{A}_{ij}^{(0)}$ denote the approximate block evidence produced by the chosen non-conjugate block routine (one of Laplace, JJ, EP, PG mean field, or 1-D quadrature) on candidate block $(i, j]$, and set $\delta_{ij} := \log \widehat{A}_{ij}^{(0)} - \log A_{ij}^{(0)}$. There exists $\varepsilon > 0$ such that, for every candidate block $(i, j]$ with $0 \le i < j \le n$ reachable by some k-segmentation with $k \le k_{\max}$,

$$|\delta_{ij}| \le \varepsilon.$$

Assumption 10.6 is an additional numerical-analysis requirement, not a consequence of naming an approximation routine. It must be established independently for the chosen family, parameter domain, candidate-block support, transformation, initialization, convergence event, and tail treatment. Nested quadrature, a high-accuracy reference calculation, or a method-specific remainder theorem can provide such a certificate in a particular analysis. Convergence of an

optimizer, monotonicity of an ELBO, or an EP moment match alone does not provide the required two-sided uniform log-evidence bound.

Proposition 10.7 (Log-evidence and posterior-odds stability). *Under Assumption 10.6, for every DP state (k, j) with $k \le k_{\max}$,*

$$e^{-k\varepsilon} L_{kj} \le \widehat{L}_{kj} \le e^{k\varepsilon} L_{kj}, \qquad e^{-k\varepsilon} R_{kj} \le \widehat{R}_{kj} \le e^{k\varepsilon} R_{kj}.$$

Consequently:

1. *For any $k, k' \le k_{\max}$, the posterior odds for the segment count satisfy*

$$\left| \log \frac{\widehat{P}(k \mid y)}{\widehat{P}(k' \mid y)} - \log \frac{P(k \mid y)}{P(k' \mid y)} \right| \le (k + k')\varepsilon. \tag{10.7}$$

2. *For any fixed $k \le k_{\max}$ and any two candidate locations h, h' of boundary t_p,*

$$\left| \log \frac{\widehat{P}(t_p = h \mid y, k)}{\widehat{P}(t_p = h' \mid y, k)} - \log \frac{P(t_p = h \mid y, k)}{P(t_p = h' \mid y, k)} \right| \le 2k\varepsilon. \tag{10.8}$$

Thus uniform blockwise log-evidence control yields uniform control of global posterior odds. *Coarse absolute probability bounds follow by normalizing finite weights, as in Corollary 10.8; a margin condition is needed instead for preservation of a modal ranking.*

Proof. Any term contributing to L_{kj} is a product of exactly k block evidences. Hence the corresponding approximate term differs by a multiplicative factor in $[e^{-k\varepsilon}, e^{k\varepsilon}]$. Summing over all paths yields the same sandwich bound for $\widehat{L}_{kj}$, and the suffix bound is identical. Equation (10.7) follows immediately because $P(k \mid y)$ is proportional to $p(k)L_{kn}/C_k$ and the prior and normalization constants are unchanged. For boundary odds, note that $P(t_p = h \mid y, k) \propto L_{p,h} R_{k-p,h}$. The numerator for each candidate location involves k block factors in total, so each candidate-specific weight is perturbed by at most $e^{\pm k\varepsilon}$; comparing two candidates therefore yields the factor $e^{\pm 2k\varepsilon}$ and hence (10.8). □

Corollary 10.8 (Probability-error conversion for normalized finite weights). *Let $p_a = w_a / \sum_b w_b$ and $\widehat{p}_a = \widehat{w}_a / \sum_b \widehat{w}_b$ be two finite normalized distributions with $|\log \widehat{w}_a - \log w_a| \le \eta$ for every state a. Then $e^{-2\eta} \le \widehat{p}_a / p_a \le e^{2\eta}$ for every a, and $\|\widehat{p} - p\|_{\mathrm{TV}} \le \min\{1, e^{2\eta} - 1\}$. Consequently, Assumption 10.6 implies a fixed-k boundary-distribution total-variation bound with $\eta = k\varepsilon$, and a segment-count posterior bound with $\eta = k_{\max}\varepsilon$. These absolute probability bounds are coarser than the odds bounds but require no additional margin condition.*

Proof. The normalizing denominator $\sum_b p_b \exp(\delta_b)$ lies in $[e^{-\eta}, e^{\eta}]$ when $\delta_a = \log \widehat{w}_a - \log w_a$. Hence each probability ratio lies in $[e^{-2\eta}, e^{2\eta}]$. Summing the positive parts of $\widehat{p}_a - p_a$ gives the exponential bound, and total variation is always at most one. □

Corollary 10.9 (Preservation of boundary rankings under a margin condition). *Under the hypotheses of Proposition 10.7, if there is a* log-odds margin $\Delta > 0$ *separating the best candidate $h^\star$ from every alternative,*

$$\log \frac{P(t_p = h^\star \mid y, k)}{P(t_p = h \mid y, k)} \ge \Delta \quad \text{for all } h \ne h^\star,$$

then for every $\varepsilon < \Delta/(2k)$ the approximate posterior preserves the exact modal boundary location: $\arg\max_h \widehat{P}(t_p = h \mid y, k) = h^\star$.

Proof. By (10.8), the log-odds between $h^\star$ and any alternative h is perturbed by at most $2k\varepsilon < \Delta$, so the exact log-odds positivity is preserved. □

Proof obligation 10.10 (Routine-specific block-error certification). No routine-wide uniform rate is asserted for Laplace, Jaakkola–Jordan, expectation propagation, Pólya–Gamma mean field, or adaptive quadrature without assumptions tailored to the family, parameter domain, block support, initialization, convergence event, and tails. A valid run must instead provide a reachable-block certificate that separates optimization residual, truncation or tail error, quadrature error, and comparison-to-reference error. The conditional global theorem below consumes the maximum certified log-score error; it does not manufacture that certificate.

Remark 10.11 (A small numerical illustration of the stability bound). Consider $n = 500$, $k_{\max} = 10$, and a non-conjugate block routine whose verified uniform per-block log-evidence error over all reachable blocks is $\varepsilon = 0.02$. Proposition 10.7 then guarantees segment-count log-odds error at most $(5 + 10) \times 0.02 = 0.30$ when comparing $k = 5$ to $k = 10$, and boundary log-odds error at most $2 \cdot 10 \cdot 0.02 = 0.40$ under any fixed $k \le 10$. In terms of probability ratios, these correspond to factors of about $e^{0.30}$ and $e^{0.40}$. The numbers in this illustration are conditional on having already verified the uniform block-error bound; they are not a generic guarantee for any chosen approximation routine.

Remark 10.12 (When each non-conjugate routine struggles). The stability bound in Proposition 10.7 is tight only when the uniform blockwise error ε is actually small. Each non-conjugate routine has its own failure mode: Laplace (Theorem 10.2) requires posterior concentration near $\hat{\theta}_{ij}$, so it can be inaccurate on very short segments (W_{ij} small) or for heavy-tailed priors; mean-field variational Bayes (Proposition 10.5) can be overconfident and underestimate posterior variance, biasing moment numerators; EP can fail to converge for non-log-concave likelihoods; 1D quadrature is accurate for smooth 1D integrands but expensive in higher dimensions. For these reasons, monitor ε empirically per dataset rather than assuming it. A practical validation checklist is to compare a subset of blocks against quadrature or another trusted reference, report both maximum and high-quantile block errors, inspect errors along the exported MAP path, and assess sensitivity of $P(k \mid y)$ and boundary marginals.

Approximation-validation checklist. Before interpreting non-conjugate posterior summaries, compare a subset of block evidences against higher-accuracy quadrature or another trusted reference; report maximum and high-quantile block errors over reachable blocks; inspect errors on the MAP path; and assess sensitivity of $p(k \mid y)$, boundary marginals, and exported segmentations to the approximation method. The ranking corollary is meaningful only when the exact posterior margin is larger than the propagated approximation error.

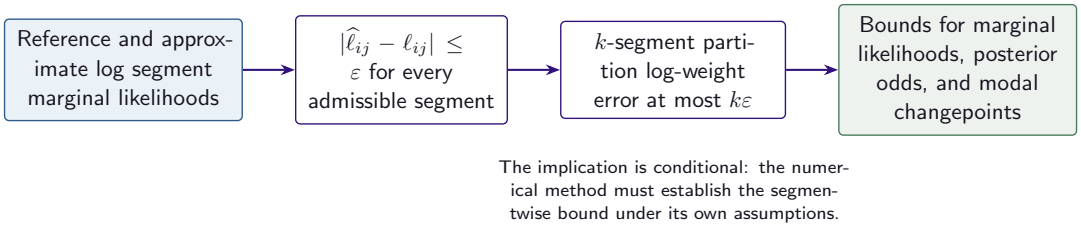


Figure 10.2. Conditional propagation from a reachable-block log-score tolerance to global evidence and posterior-odds bounds. The local numerical routine remains responsible for establishing the tolerance.

Interpretation of approximate local scores. The forward–backward and max–sum recursions are exact for the block-score array supplied to them. When the array is obtained by numerical approximation, this algorithmic exactness does not establish exactness with respect to the underlying sampling model. A Jaakkola–Jordan tangent gap averaged under a fitted Gaussian variational distribution is not an upper bound on the log-evidence error: the difference in log normalizing constants also depends on the changed posterior measure. Likewise, a truncated Laplace expansion is not a finite-sample error certificate unless its omitted remainder and tails have been bounded. Comparisons with a converged numerical-integration reference are necessary before interpreting either diagnostic as evidence of approximation accuracy.

Interpretation of nonconjugate method labels. The current software routes the options `jj` and `pg_vb` through the same Jaakkola–Jordan numerical routine. These labels therefore do not constitute two independent approximation methods. A distinct Pólya–Gamma variational implementation, if included later, requires a separately specified augmented model, objective, update equations, and numerical verification.

Prediction, coordinate support, and canonical metrics

CHAPTER PATH

Family-specific posterior prediction, explicit extrapolation, and evaluation rules that do not create false success from unmatched boundaries.

11.1 Posterior prediction

A prediction path must dispatch to the fitted observation family's posterior predictive. It may not fall through to a Gaussian default. The exported model also carries a coordinate support interval. Coordinates outside that interval raise an error by default; a named extrapolation policy may be used only when its behavior and provenance are recorded.

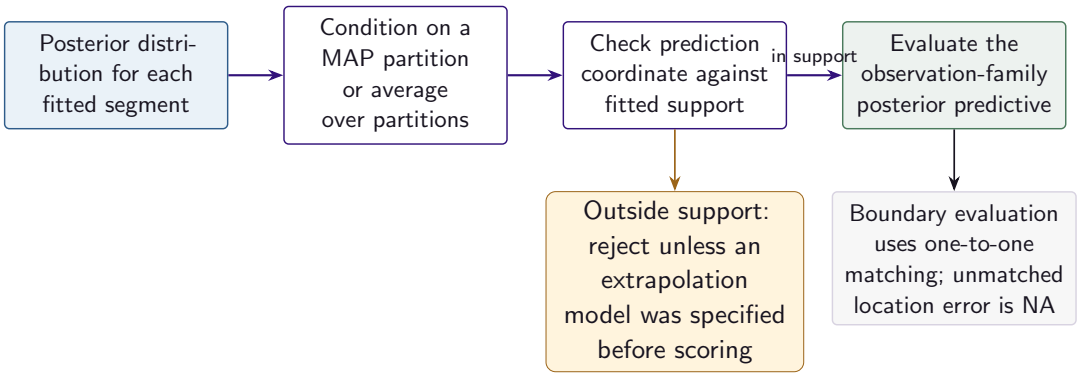


Figure 11.1. Posterior prediction and evaluation. The fitted observation family and coordinate support are checked before a proper score is evaluated.

Definition 11.1 (Canonical one-to-one boundary matching)**.** Given predicted boundaries $\widehat{\mathcal{B}}$, references $\mathcal{B}^\star$, and tolerance τ, construct the bipartite graph containing edges $(\widehat{b}, b^\star)$ with $|\widehat{b} - b^\star| \le \tau$. Select a matching with maximum cardinality; among all such matchings, select one with minimum total absolute distance, using a fixed deterministic tie rule. Precision, recall, and F_1 use the resulting matched count. No predicted or reference boundary is reused.

Definition 11.2 (Matched boundary-location error)**.** For the canonical matching $\mathcal{M}_\tau$, define $\mathrm{MAE}_\tau = |\mathcal{M}_\tau|^{-1} \sum_{(\widehat{b},b^\star)\in\mathcal{M}_\tau} |\widehat{b} - b^\star|$ when at least one match exists. If no match

exists, the quantity is *not defined* and is reported as NA, never as zero.

Definition 11.3 (Held-out predictive log score)**.** A held-out log score is valid only when the held-out observations are disjoint from training, the fitted family's posterior predictive is used, and every held-out coordinate is in support or is handled by a predeclared extrapolation policy. Failure of any condition makes the archived number inadmissible as a predictive score.

METHYLATION HELD-OUT COMPUTATION

The archived value -387.50040013308154 on $m = 381$ observations is real. It is not a valid Beta-observation held-out score because the implementation used a Gaussian predictive fallback and silently assigned coordinates beyond the training range to the final MAP block. It is retained as and excluded from predictive claims.

Having developed training-time posterior inference for partitions and segment parameters (Sections 5–10), we now turn to prediction for new (X, y) data. The prediction layer consumes posterior summaries (e.g., boundary posteriors, segment-parameter posteriors, and group evidences) and is therefore agnostic to whether block evidences were computed in closed form (Section 4.2) or via approximation (Section 10). We assume we have already trained (or otherwise obtained) a collection of G group-specific BayesBreak models $\{\mathcal{M}_g\}_{g=1}^{G}$, where each $\mathcal{M}_g$ comprises: (i) a fitted segmentation posterior (e.g. $P_g(k \mid y)$ and boundary posteriors), (ii) an exported MAP segmentation $\widehat{t}^{(g)}$ with segment-level posterior hyperparameters, and (iii) optionally a Bayesian curve representation (e.g. posterior mean/variance at design points).

Typical outputs are group log scores, posterior group probabilities, optional unit-level responsibilities, exported MAP signal values, Bayes-curve values, and optional pointwise uncertainty summaries. These prediction interfaces define the outputs consumed by the planned real-data prediction diagnostics in Section 14.

We pause to fix two terms that have been used informally above and that recur throughout the rest of this section.

Definition 11.4 (Exported segmentation)**.** An *exported segmentation* for a fitted group-specific model $\mathcal{M}_g$ is the tuple $\big(\widehat{k}_g,\ \widehat{t}^{(g)},\ \{\widehat{\theta}_q^{(g)}\}_{q=1}^{\widehat{k}_g}\big)$, where $\widehat{k}_g$ is a chosen segment count (either an estimator of k such as the posterior mode, or a deterministic choice fixed at training time), $\widehat{t}^{(g)} = (\widehat{t}_0^{(g)}, \ldots, \widehat{t}_{\widehat{k}_g}^{(g)})$ is the joint MAP boundary vector at $\widehat{k}_g$ produced by Algorithms 10–11, and each $\widehat{\theta}_q^{(g)}$ is a deterministic point summary of the segment-q posterior (typically the conjugate-family posterior mean or median of the expectation parameter). The exported segmentation is the signal-level analogue of a fitted model; it is what downstream prediction routines (P1 and P2) consume.

Definition 11.5 (Bayes curve)**.** The *Bayes curve* for a fitted group-specific model $\mathcal{M}_g$ evaluated at design point x is the posterior mean of the latent piecewise-constant signal under the segment-marginalized posterior:

$$\widehat{f}_g^{\,\mathrm{Bayes}}(x) := \mathbb{E}\big[\, f(x) \;\big|\; y,\, \mathcal{M}_g \,\big] = \sum_{k=1}^{k_{\max}} P(k \mid y, \mathcal{M}_g)\, \mathbb{E}\big[\, f(x) \;\big|\; y, k, \mathcal{M}_g \,\big]\,,$$

where the inner conditional expectation is taken over both the boundary vector t and the

segment parameters $\theta_{1:k}$, and where $f(x) = \theta_{q(x)}$ is the segment-q mean parameter for the segment $q(x)$ containing x under the boundary vector t.

Definition 11.6 (Segment-assignment map)**.** For an exported boundary vector $\widehat{t}^{(g)}$ on ordered coordinates $u_0^{(g)} < \cdots < u_{n_g}^{(g)}$, a segment-assignment map is a declared function $a_g(x) \in \{1, \ldots, \widehat{k}_g\}$ assigning each query coordinate x to a training segment. The default interval-containment rule is $a_g(x) = q$ when $u^{(g)}_{\widehat{t}^{(g)}_{q-1}} < x \leq u^{(g)}_{\widehat{t}^{(g)}_q}$, with endpoint conventions recorded by the implementation. Queries outside the training range are either rejected or assigned by an explicitly declared extrapolation rule; silent extrapolation is not part of the default interface.

The *exported segmentation* (Definition 11.4) and the *Bayes curve* (Definition 11.5) are distinct objects with different operational meanings: the exported segmentation is a fixed piecewise-constant function with a discrete change-point structure that is convenient for downstream tasks (e.g. region labeling, hypothesis testing, set-valued prediction); the Bayes curve is a continuous mixture that integrates segmentation uncertainty and is the appropriate target when smooth predictive quantities are needed.

Prediction supports two primary tasks:

(P1) Group membership likelihoods/posteriors. Given new inputs (X, y), compute for each group g an out-of-sample posterior-predictive likelihood $p(\text{new} \mid \mathcal{M}_g)$ (or its log), and convert these into posterior group membership probabilities $P(g \mid \text{new})$ under a prior π_g.

(P2) Signal prediction given group. Given query design points X and a specified (or inferred) group label g, return: (i) MAP piecewise-constant signal values $\widehat{f}_g^{\text{MAP}}(X)$ including the corresponding breakpoints and segment levels, and/or (ii) Bayesian curve values $\widehat{f}_g^{\text{Bayes}}(X)$ with pointwise uncertainty summaries when available.

Prediction output	Interpretation
ℓ_g	Group-specific log posterior-predictive score.
$P(g \mid \text{new})$	Group posterior after combining ℓ_g with prior weight π_g.
$\widehat{f}_g^{\text{MAP}}(X)$	Piecewise-constant signal under the exported MAP segmentation.
$\widehat{f}_g^{\text{Bayes}}(X)$	Bayes-curve posterior mean, optionally with pointwise uncertainty.
Unit responsibilities	Optional probabilities for set-valued units or grouped observations.

Table 11.1. Prediction-layer outputs exposed by fitted BayesBreak models. These interfaces support downstream scoring and calibration separately from the empirical result tables. This is an interface-definition table; no source CSV is associated with it.

Crucially, we distinguish *single point observations* (x_i, y_i) from *multivariate / set-valued observations* where each unit contains multiple internal pairs (x_{ur}, y_{ur}), i.e. X_u and y_u are vectors. We also separately address *vector-valued responses* $y_i \in \mathbb{R}^d$ observed at each design point.

11.1.1 Prediction inputs: pointwise vs multivariate/set-valued observations

Definition 11.7 (Prediction-input classes)**.** We distinguish three classes of new prediction inputs, jointly partitioning the observation-shape design space considered in this book.

(Case A) **Pointwise observations.** A new dataset

$$\mathcal{D}_{\mathrm{pt}}^{\mathrm{new}} = \{(x_i^{\mathrm{new}}, y_i^{\mathrm{new}}, w_i^{\mathrm{new}})\}_{i=1}^{m}$$

with $x_1^{\mathrm{new}} < \cdots < x_m^{\mathrm{new}}$, scalar (or low-fixed-dimensional) responses y_i^{new}, and optional exposure/precision weights w_i^{new} from the observation model. Each new observation lies on a single design point.

(Case B) **Multivariate/set-valued observations.** A new dataset

$$\mathcal{D}_{\mathrm{mv}}^{\mathrm{new}} = \left\{ \left([a_u, b_u], \{(x_{ur}, y_{ur}, w_{ur})\}_{r=1}^{R_u} \right) \right\}_{u=1}^{U},$$

where each *unit* u carries an interval support $[a_u, b_u]$ and a finite collection of internal observations indexed by $r = 1, \ldots, R_u$. Prediction is performed at the unit level by integrating or aggregating over each unit's internal observations.

(Case C) **Vector-valued response at each design point.** A new dataset

$$\{(x_i^{\mathrm{new}}, \mathbf{y}_i^{\mathrm{new}}, w_i^{\mathrm{new}})\}_{i=1}^{m}$$

where each $\mathbf{y}_i^{\mathrm{new}} \in \mathbb{R}^d$ is a vector response (or a mixed discrete/continuous component vector). Two sub-cases are admitted: (C-i) a factorized EF across dimensions, and (C-ii) a joint multivariate EF with a single joint log-partition function and conjugate or approximate block routine.

The three cases are mutually exclusive in the data type they consume and jointly cover the observation shapes considered by the BayesBreak prediction interface.

The remainder of this subsection treats the three cases in turn; the corresponding algorithms appear in Section 11.1.8.

11.1.2 Posterior-predictive scoring for a fixed segment under EF conjugacy

For group membership inference, we require out-of-sample likelihoods. The most computationally efficient and interpretable approach is to score new data against the *exported* group curve (MAP segmentation or Bayesian curve). We first derive segment-level posterior-predictive likelihoods under EF conjugacy; these become building blocks for both Case A and Case B.

Fix a group g and consider a segment (or block) B of its exported MAP segmentation. Let the posterior for the segment parameter θ after training be the conjugate posterior

$$p(\theta \mid \alpha_B^{(g)}, \beta_B^{(g)}) \;\propto\; \exp\{\eta(\theta)^\top \alpha_B^{(g)} - \beta_B^{(g)} A(\theta)\},$$

where $(\alpha_B^{(g)}, \beta_B^{(g)})$ are the training-updated hyperparameters for that segment. Now let $\mathcal{Y}_B^{\mathrm{new}}$ denote the collection of new observations whose design points fall into this segment.

Proposition 11.8 (EF posterior-predictive likelihood within a fixed segment)**.** *Assume the weighted EF likelihood and conjugate prior conventions of §4.1 and §3.1, and let* $(S_B^{new}, W_B^{new}, H_B^{new})$ *denote the EF block summaries of the new data restricted to segment* B*:*

$$S_B^{new} := \sum_{i \in B} w_i^{new} T(y_i^{new}), \qquad W_B^{new} := \sum_{i \in B} w_i^{new}, \qquad H_B^{new} := \sum_{i \in B} \log h(y_i^{new}, w_i^{new}).$$

Then the posterior-predictive likelihood of the new data in segment B *under group* g *is*

$$p(\mathcal{Y}_B^{new} \mid \mathcal{M}_g) = \exp\{H_B^{new}\} \frac{Z(\alpha_B^{(g)} + S_B^{new}, \beta_B^{(g)} + W_B^{new})}{Z(\alpha_B^{(g)}, \beta_B^{(g)})}. \tag{11.1}$$

Proof. The proof is identical to the single-block evidence derivation (Theorem 4.1) after replacing the prior hyperparameters (α_0, β_0) by the segment posterior hyperparameters $(\alpha_B^{(g)}, \beta_B^{(g)})$ learned from training data. □

Interpretation. Equation (11.1) integrates out the segment parameter θ under the *posterior* induced by training, yielding a strictly Bayesian out-of-sample likelihood for new observations assigned to that segment. It is therefore suitable for group scoring and group posterior computation.

11.1.3 Group likelihoods for pointwise observations (Case A)

MAP-segmentation-based scoring. Let $\widehat{t}^{(g)}$ denote the exported MAP boundaries for group g on the relevant domain. Assign each new design point x_i^{new} to its containing MAP segment $B = B(i; g)$ and compute segment-wise block summaries $(S_B^{\text{new}}, W_B^{\text{new}}, H_B^{\text{new}})$. The group log-likelihood is then:

$$\ell_g^{\text{MAP}}(\mathcal{D}_{\text{pt}}^{\text{new}}) := \log p(\mathcal{D}_{\text{pt}}^{\text{new}} \mid \mathcal{M}_g) = \sum_{B \in \widehat{t}^{(g)}} \log p(\mathcal{Y}_B^{\text{new}} \mid \mathcal{M}_g), \tag{11.2}$$

where each segment term uses (11.1). This yields $\mathcal{O}(m)$ evaluation time once segment membership is determined.

Bayesian-curve-based scoring (optional). If $\mathcal{M}_g$ exports a Bayesian curve representation that provides a local posterior over the mean parameter at each design point (or each small bin), then one can score each point under its local posterior predictive and sum over points. This can improve robustness near uncertain boundaries at additional computational cost.

Resegmentation scoring (optional, flexible). An alternative definition of group likelihood is to rerun the DP on the new data under group-specific hyperparameters (treating the new dataset as another draw from group g's prior). This yields an evidence $P(\mathcal{D}_{\text{pt}}^{\text{new}} \mid g)$ that does not condition on the trained MAP boundaries. This option is useful when groups differ primarily in hyperparameters rather than in the learned curve geometry, but is computationally $\mathcal{O}(k_{\max} m^2)$ per group.

11.1.4 Group likelihoods for multivariate/set-valued observations (Case B)

In Case B, each unit u contains multiple internal observations $\{(x_{ur}, y_{ur})\}_{r=1}^{R_u}$. Two output granularities are typically desired: (i) *unit-level* group likelihoods and group responsibilities, and (ii) a *global* group likelihood for the entire new dataset (product over units).

Unit-level predictive likelihood under a fixed group. Fix a group g. Partition the unit's internal design points by the MAP segments of group g. For each segment B, compute the block summaries $(S_{u,B}^{\text{new}}, W_{u,B}^{\text{new}}, H_{u,B}^{\text{new}})$ restricted to those internal points. Then define the unit likelihood as:

$$\ell_{ug} := \log p(\text{unit } u \mid \mathcal{M}_g) = \sum_{B \in \widehat{t}^{(g)}} \log p(\mathcal{Y}_{u,B}^{\text{new}} \mid \mathcal{M}_g), \tag{11.3}$$

where $\mathcal{Y}_{u,B}^{\text{new}}$ is the subset of unit u's internal observations that fall into segment B and each segment term uses (11.1) with summaries for that subset.

Dataset likelihood and independence assumption.

> **Assumption 11.9** (Prediction-unit conditional independence)**.** For Case B prediction, units are conditionally independent given the exported group model, and each unit contributes only through the segment-restricted predictive factors used in (11.3).

Under Assumption 11.9, the dataset likelihood under group g is:

$$\ell_g^{\text{MV}}(\mathcal{D}_{\text{mv}}^{\text{new}}) = \sum_{u=1}^{U} \ell_{ug}.$$

This is the direct multivariate generalization of (11.2) and corresponds to the earlier "fragment-level" intuition, but is fully application-agnostic.

11.1.5 Vector-valued responses (Case C): factorized and multivariate EF scoring

Suppose each observation is $\mathbf{y}_i = (y_{i1}, \ldots, y_{id})$.

Factorized EF across dimensions (recommended default). Assume conditional independence across dimensions given segment parameters:

$$p(\mathbf{y}_i \mid \boldsymbol{\theta}) = \prod_{\ell=1}^{d} p_\ell(y_{i\ell} \mid \theta_\ell), \qquad p(\boldsymbol{\theta}) = \prod_{\ell=1}^{d} p_\ell(\theta_\ell).$$

Then all block evidences and posterior-predictive terms factorize over ℓ, so log-likelihoods are sums over dimensions:

$$\ell_g(\mathbf{y}) = \sum_{\ell=1}^{d} \ell_{g,\ell}(y_{\cdot\ell}),$$

where each $\ell_{g,\ell}$ is computed by (11.2) or (11.3) for dimension ℓ. Computationally, this costs $\mathcal{O}(d)$ per observation (or per unit).

Multivariate EF (when available). If $(\mathbf{y}_i)$ admit a multivariate EF with conjugate prior on a joint parameter $\theta \in \mathbb{R}^p$, then the same ratio-of-normalizers structure applies with vector-valued

sufficient statistics. BayesBreak's DP layer remains unchanged; only the definition of S_{ij} and $Z(\alpha, \beta)$ changes.

11.1.6 From group likelihoods to posterior group membership probabilities

Single-dataset membership (one group per dataset). If the new dataset is assumed to come from a single (unknown) group g, define:

$$\ell_g := \log p(\text{new} \mid \mathcal{M}_g), \qquad P(g \mid \text{new}) = \frac{\pi_g \exp(\ell_g)}{\sum_{g'=1}^{G} \pi_{g'} \exp(\ell_{g'})}.$$

This yields a *group membership likelihood vector* (the ℓ_g) and a *group membership posterior vector* (the $P(g \mid \text{new})$), matching the requested prediction behavior in an application-agnostic way.

Unit-wise membership (one group per unit). If each multivariate unit u is assumed to originate from one group, then define unit-wise responsibilities:

$$r_{ug} := P(g \mid \text{unit } u) = \frac{\pi_g \exp(\ell_{ug})}{\sum_{g'=1}^{G} \pi_{g'} \exp(\ell_{ug'})}, \tag{11.4}$$

where ℓ_{ug} is given by (11.3). This is precisely the out-of-sample analog of the E-step responsibilities in the latent-group model (Section 9.3).

11.1.7 MAP signal evaluation given group membership

Given a group label g and query design points $X^\star = \{x_1^\star, \ldots, x_M^\star\}$, BayesBreak returns either: (i) *MAP piecewise-constant predictions* or (ii) *Bayesian curve predictions*.

MAP piecewise-constant predictions. Let $\widehat{t}^{(g)}$ be the exported MAP boundaries and let each MAP segment B provide the posterior mean of the mean parameter:

$$\widehat{\mu}_B^{(g)} := \mathbb{E}[m_\star(\theta) \mid \alpha_B^{(g)}, \beta_B^{(g)}].$$

Then the MAP signal at a query point $x^\star$ is:

$$\widehat{f}_g^{\text{MAP}}(x^\star) = \widehat{\mu}_{B(x^\star;g)}^{(g)},$$

where $B(x^\star; g)$ denotes the MAP segment containing $x^\star$.

Bayesian curve predictions. If the DP-derived Bayes regression curve is exported on a grid, return $\widehat{f}_g^{\text{Bayes}}(x^\star)$ by evaluating on the nearest grid point (or by piecewise-constant interpolation within each grid interval). When pointwise posterior variances are available, also return pointwise credible intervals; simultaneous bands require an additional construction and are not implied by the pointwise variances alone.

11.1.8 Algorithms for prediction

Algorithm 19: PredictGroup($\{\mathcal{M}_g\}_{g=1}^{G},\ \mathcal{D}^{\text{new}},\ \pi$)

Input: Trained group models $\{\mathcal{M}_g\}$; new data $\mathcal{D}^{\text{new}}$ (Case A or Case B); group prior weights π (default uniform)

Output: Log-likelihoods $\{\ell_g\}$ and posterior membership probabilities $\{P(g \mid \text{new})\}$; if Case B, also unit responsibilities $\{r_{ug}\}$

for $g = 1$ **to** G **do**
  if Case A (pointwise) then
    Assign each $(x_i^{\text{new}}, y_i^{\text{new}})$ to a MAP segment $B(i; g)$ under $\widehat{t}^{(g)}$;
    Accumulate segment summaries $(S_B^{\text{new}}, W_B^{\text{new}}, H_B^{\text{new}})$;
    Compute $\ell_g \leftarrow \sum_B \left[H_B^{\text{new}} + \log Z(\alpha_B^{(g)} + S_B^{\text{new}}, \beta_B^{(g)} + W_B^{\text{new}}) - \log Z(\alpha_B^{(g)}, \beta_B^{(g)})\right]$;
  else
    `// Case B (multivariate/set-valued units)`
    for $u = 1$ **to** U **do**
      Partition unit u internal pairs by MAP segments of group g; compute per-segment summaries;
      Compute ℓ_{ug} via (11.3) using (11.1);
    $\ell_g \leftarrow \sum_{u=1}^{U} \ell_{ug}$;
Compute $P(g \mid \text{new}) \propto \pi_g \exp(\ell_g)$ and normalize over g;
if Case B then
  Compute unit responsibilities $r_{ug} \propto \pi_g \exp(\ell_{ug})$ and normalize over g for each u (cf. (11.4));

Algorithm 20: PredictMAPSignal($\mathcal{M}_g,\ X^\star,\ \texttt{mode}$)

Input: A group model $\mathcal{M}_g$ with exported MAP segmentation $\widehat{t}^{(g)}$ and/or Bayes curve; query points $X^\star = \{x_1^\star, \ldots, x_M^\star\}$; mode $\in \{\texttt{MAP}, \texttt{Bayes}\}$

Output: Predicted signal values at $X^\star$ (and optionally uncertainty)

if *mode=MAP* **then**
  foreach $x^\star$ *in* $X^\star$ **do**
    Find containing MAP segment $B(x^\star; g)$ (binary search on breakpoint coordinates);
    Return $\widehat{f}_g^{\text{MAP}}(x^\star) = \widehat{\mu}_{B(x^\star;g)}^{(g)}$;
else
  Evaluate/interpolate the exported Bayes regression curve at $X^\star$; optionally return posterior variances;

11.1.9 Complexity

For Case A, prediction under a fixed group is $\mathcal{O}(m)$ after segment assignment ($\mathcal{O}(m \log k)$ worst case via binary search; often $\mathcal{O}(m)$ with a streaming pointer). For Case B, it is $\mathcal{O}(\sum_u R_u)$ per group, plus segment assignment for internal points. Vector-valued responses add a factor of d in the factorized EF construction. Importantly, prediction does *not* require rerunning the DP unless the optional resegmentation scoring mode is enabled. Membership posterior normalization is performed in log space by subtracting $\texttt{logsumexp}_g(\log \pi_g + \ell_g)$; unit responsibilities use the same normalization independently for each unit. Segment assignment

is a numerical convention, not an inferential identity: implementations must record whether binary search, streaming pointers, nearest-neighbor assignment, or a declared extrapolation rule was used for out-of-range query points.

11.1.10 Prediction diagnostics

Because the predictive layer reuses segment-level posteriors, the group posterior $P(g \mid \text{new})$ derived from (11.2) and the pointwise predictive summaries inherit their calibration quality from the segmentation. Use two diagnostics for real-data deployments:

- **Held-out log-likelihood (HLL) trace.** Split each test sequence into blocks and compute the per-block posterior-predictive log-likelihood under (11.1). The sum over blocks is the HLL for that sequence. Compare HLL trajectories across candidate segmentations (exported MAP vs. Bayes curve) and across groups; a well-calibrated model yields an HLL ranking consistent with the k-posterior ranking on held-out data.
- **PIT residuals.** For point-valued new observations with continuous predictive CDF, evaluate the probability-integral transform $u_i = F_g(y_i^{\text{new}} \mid \mathcal{M}_g)$. Under correct calibration the $\{u_i\}$ are Uniform$(0, 1)$; histograms or Kolmogorov–Smirnov summaries flag miscalibration (overconfident predictive widths show up as U-shaped histograms).

These diagnostics are cheap post-hoc checks. In this book they are planned outputs for the real-data analyses in Section 13; the planned per-case-study results tables identify the HLL and boundary-recovery fields that the completed analyses report.

CHAPTER 12

Software and reproducibility

CHAPTER PATH

Numerical responsibilities of the implementation, independent correctness tests, model and dataset identifiers, and the requirements for regenerating the reported quantities.

12.1 Implementation architecture

The package separates observation-family calculations from partition inference. A segment routine computes the log integrated likelihood, any required posterior moments and, for nonconjugate models, a clearly identified numerical approximation. The partition routine combines those quantities with the normalized prior over admissible boundaries. Two recursions serve different purposes: sum-product for integrated evidence and posterior marginals, and max-sum for the joint MAP partition. Replicate and group models construct pooled or responsibility-weighted segment scores before invoking the same recursion. Prediction must dispatch according to the fitted observation family, rather than falling back to an unrelated sampling law.

12.2 Independent checks of the exact recursions

For short sequences, enumerate the full admissible partition set. At each fixed segment count, compare the sum of prior-weighted block likelihood products against the forward and backward recursions. Verify the partition normalizer separately using unit block scores. Check that the ordered boundary and boundary-event marginals sum to one and $k - 1$, respectively. Compare posterior moments and global MAP decisions against direct enumeration, including tied maxima, zero-probability blocks, negative Gaussian segment means and minimum-length constraints. No self-consistency identity replaces an independent numerical oracle.

12.3 Observation-model semantics

Known Gaussian precisions, Poisson exposures and Binomial trial counts enter different likelihoods. Their sufficient-statistic and base-measure terms must be derived from the sampling model. In particular, raising a normalized observation density to an arbitrary positive power is not in general equivalent to changing its precision or exposure. If likelihood powers are used, posterior calculations must be described as generalized-Bayes inference rather than

as ordinary sampling-model evidence. Explicit masks distinguish missing observations from observations with invalid or zero-valued descriptors.

12.4 Numerical approximation diagnostics

A nonconjugate segment routine must return the approximation method, the value of its objective or evidence surrogate, and a convergence indicator. For each admissible test segment, compare log scores and posterior moments against independent adaptive quadrature when sufficiently low-dimensional. For a uniform log-score bound ε, global posterior-odds perturbation bounds are conditional on that bound holding on *every reachable segment*. Small empirical error on selected segments alone does not establish such a hypothesis. Labels such as Jaakkola–Jordan and Pólya–Gamma variational Bayes must not be reported as separate implementations when they share one computational path.

12.5 Data, output and verification records

A reproducible experiment identifies the original data source and observation model, the ordered coordinate axis, data and descriptor checksums, train/validation/test partitions, model hyperparameters, algorithm configuration, random seed and implementation version. Store both successful estimates and failures, including the statistical denominator for reported proportions and the numerical conditions behind timeouts. Synthetic fallback is disallowed in analyses declared to use an external empirical dataset.

12.6 Execution and publication controls

The controlled experiments E00–E10 in Appendix C are defined with their replication units, estimands, comparison methods and acceptance tests. Completed figures must be regenerated only from verified run records. Figures produced under an observation model later corrected in the implementation require refitting, not a change in caption or a numerical patch. The single-column paper and this book shall use the same accepted results. A successful LaTeX compilation is a typesetting check, not a scientific verification of the reported numerical data.

CHAPTER 13

Synthetic validation and executed diagnostics

CHAPTER PATH

Real recovery, calibration, portability, approximation, and scaling outputs interpreted at the exact scope supported by their archived protocols.

13.0.1 Single-sequence Gaussian illustration

We begin with a Gaussian sequence containing two true changepoints. Figure 13.1 shows two complementary outputs: the marginal posterior probability that each interior index is a boundary, and the exported joint MAP segmentation together with segment-level posterior intervals. The two dominant posterior spikes align with the true changepoints, while a smaller shoulder of posterior mass just before the first jump reflects local ambiguity caused by a short transitional run of noisy observations. That ambiguity is visible again in the bottom panel, where the exported fit inserts a short intermediate segment before settling into the large middle plateau.

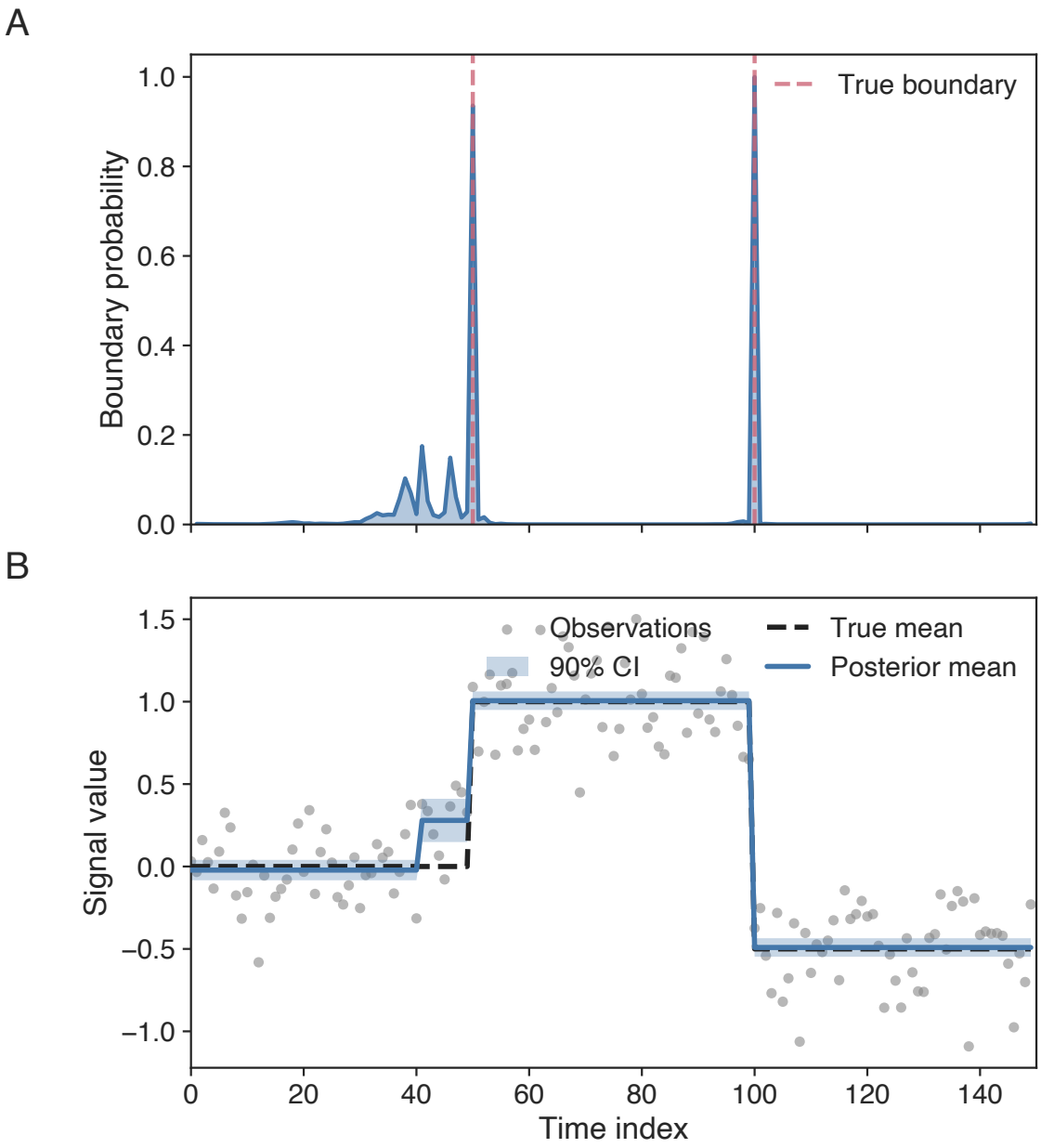


Figure 13.1. Single-sequence Gaussian example. Top: marginal posterior probability that each interior index $i \in \{1, \ldots, n-1\}$ is a changepoint, with dashed vertical lines marking the true changepoints. Bottom: observed data, true latent mean, exported joint MAP segmentation, and 90% segment-level posterior intervals.

Quantity	Value
Selected segment count $\widehat{k}$	4
Posterior mean $\mathbb{E}[k]$	4.742
MAP k	4
$\log p(y)$	-18.031

Table 13.1. Posterior summary for the single-sequence Gaussian example in Figure 13.1. The table separates the displayed exported segmentation from posterior summaries of the segment count.

Table 13.1 makes an important methodological point explicit: the displayed segmentation and

the posterior over the segment count are related but not identical objects. In the archived example, the exported joint MAP segmentation uses four segments, while the posterior mean $\mathbb{E}[k \mid y]$ is larger, indicating residual posterior mass on more highly segmented configurations. This is precisely why the book distinguishes sum-product posterior summaries from max-sum exported segmentations.

13.0.2 Likelihood-family showcase

BayesBreak is designed so that the global DP layer does not depend on the observation model once block evidences are available. Figure 13.2 visualizes this portability across three conjugate families (Gaussian, Poisson, Binomial) and one low-dimensional quadrature example for $(0, 1)$-valued responses. Across all four panels, the fitted piecewise-constant signal tracks the true latent parameter closely and preserves the correct qualitative segmentation pattern.

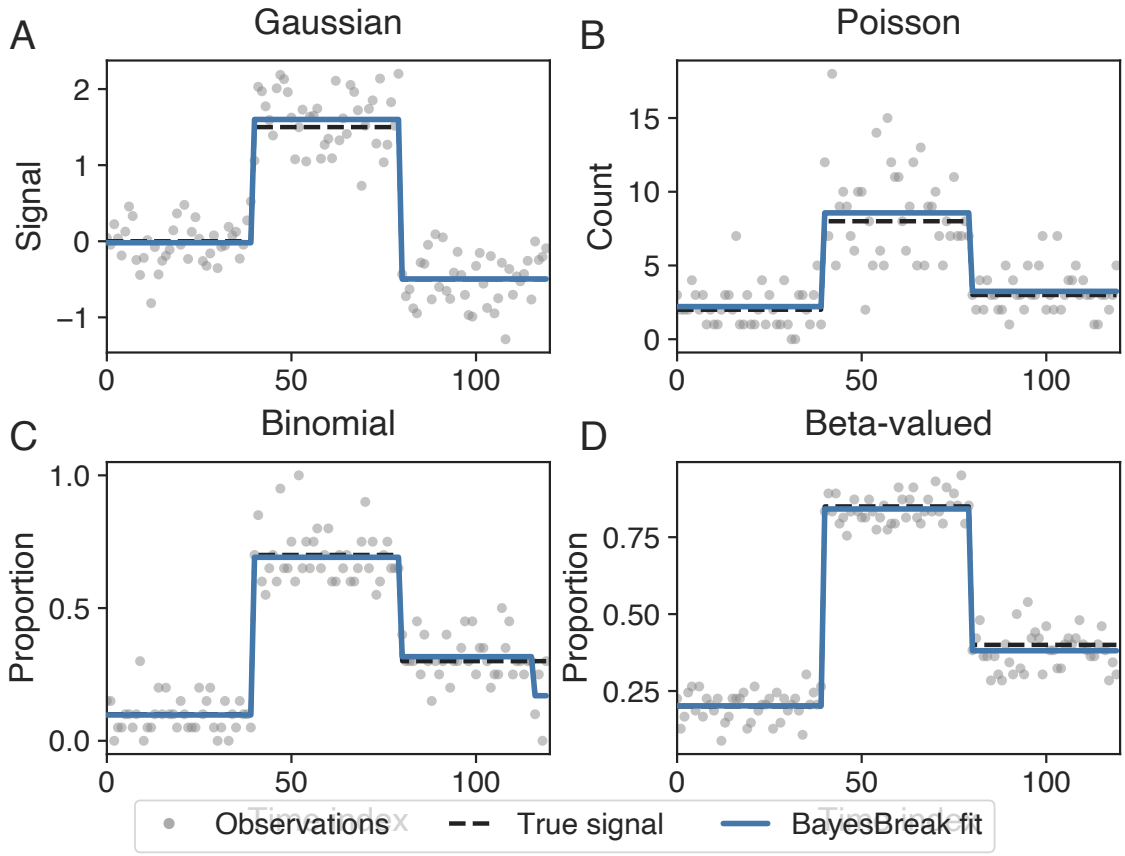


Figure 13.2. Likelihood-family showcase. The first three panels use closed-form conjugate block integrals (Gaussian, Poisson, and Binomial), while the final panel uses the one-dimensional Beta-response quadrature routine. In each case the fitted BayesBreak signal is close to the true piecewise-constant latent parameter.

13.0.3 Quantitative summary across families

Table 13.2 aggregates the small synthetic benchmark packaged with the archive. The main takeaway is not that one family universally dominates another, but that the same DP backend remains usable across very different block models. The Poisson setting is visibly the most challenging among the archived examples: it exhibits the weakest boundary F1 and the largest reconstruction error, which is consistent with the higher count variability shown in Figure 13.2. By contrast, the Beta-response quadrature example achieves the strongest boundary recovery

in this small benchmark.

Family	n	$k^\star$	$\hat{k}$	F1@τ	MAE	MSE	$-\log p(y)/n$
Gaussian	120	3	3	0.968	0.00	0.0023	0.191
Poisson	120	3	4	0.777	0.20	0.4501	2.162
Binomial	120	3	3	0.960	0.06	0.0004	2.108
Beta-valued	120	3	3	1.000	0.00	0.0001	2.620

Table 13.2. Available synthetic summary across the bundled family-specific examples. The final row corresponds to the Beta-response quadrature example rather than to a conjugate closed-form model.

Because the table reflects one archived benchmark configuration rather than a large Monte Carlo study, the numbers serve as a compact portability and relative-difficulty summary, not as a definitive cross-family ranking.

13.0.4 Calibration of boundary posterior probabilities

A central advantage of the Bayesian formulation is that it returns marginal posterior probabilities for boundary events. Figure 13.3 checks whether those probabilities are calibrated under repeated Gaussian simulations. The reliability curve tracks the diagonal closely across the full probability range, with an empirical calibration error reported in the figure inset of approximately $\mathrm{ECE} \approx 0.010$ and Brier score ≈ 0.011. The mid-probability bins remain the noisiest because boundary events are intrinsically rare per sequence, but the ordering of confidence levels is sensible: bins assigned larger posterior boundary probability correspond to larger empirical boundary frequency.

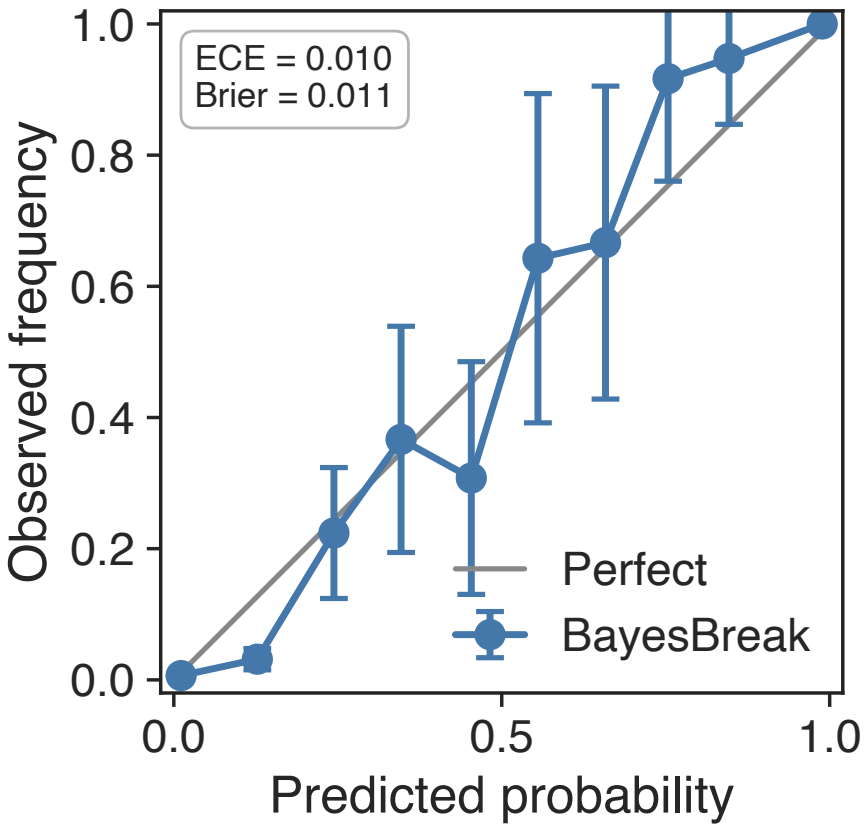


Figure 13.3. Calibration of marginal boundary posterior probabilities under synthetic Gaussian data. The dashed diagonal denotes perfect calibration; the orange curve is the empirical frequency within bins of predicted probability.

This experiment is valuable because segmentation methods are often judged only by recovered changepoints. For BayesBreak, well-behaved posterior probabilities are equally important: they control downstream thresholding, boundary ranking, and any decision rule that is sensitive to the strength of evidence rather than just to the best segmentation.

13.0.5 Latent-group pooling

Figure 13.4 evaluates the finite latent-group template model on $n_{\mathrm{seq}} = 24$ synthetic Gaussian sequences ($n = 80$, $\sigma = 1.0$) drawn from two groups with different changepoint patterns. The left panel shows the posterior responsibilities sorted by truth: the two groups are almost separable, with a small ambiguous band near the class boundary corresponding to a single mis-assignment (hard accuracy 96%). The right panel reports the group-specific boundary marginals induced by the fitted templates; both groups concentrate their posterior mass at the correct latent changepoints, with the cluster-1 marginal locating both boundaries near probability one and the cluster-0 marginal recovering its second boundary at $t \approx 53$ and a softer mode near $t \approx 26$ for the first.

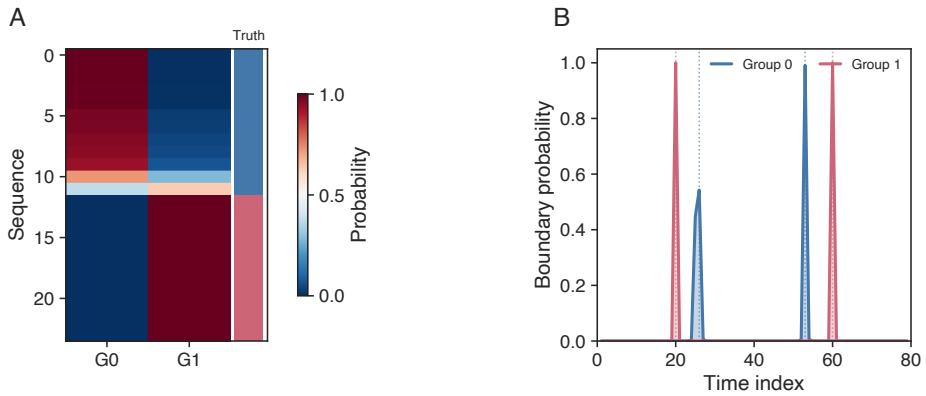


Figure 13.4. Latent-group pooling on synthetic Gaussian sequences ($n = 80$, $\sigma = 1.0$, $n_{\text{seq}} = 24$). Left: posterior responsibilities r_{sg} for each sequence and score-template group, sorted by truth; the right-hand stripe encodes the true label. Right: group-specific marginal boundary probabilities under the fitted latent templates, with dotted vertical lines marking the true per-group changepoints.

A six-panel companion diagnostic (uncropped figure `shared/figures/results/fig4_latent_groups.pdf`) unpacks the same fit further: per-group example sequences with the inferred piecewise mean, the boundary marginals, the responsibility heatmap, the group-averaged signal, and a per-sequence assignment-confidence diagnostic that plots r_{s,y_s} (responsibility on the true group) sorted along the x-axis, with the 0.5 decision threshold shaded and any mis-assigned sequences circled. Even when overall accuracy is high, this view exposes which sequences sit close to the decision boundary and is informative regardless of whether any hard mistakes are made.

The replicated stress analysis extends this single single illustrated setting to 400 seeded datasets over eight predeclared stress cells. The stated-design cell achieved mean hard accuracy 0.9742 (95% interval 0.9536–0.9948) and mean adjusted Rand index 0.9183. Every returned objective equaled the final trace value, all traces were monotone, and all 1,200 restarts were valid. Figure 13.5 also exposes the expected failure boundary: low separation and duplicate templates produce weak recovery or collapse, while higher noise and shorter sequences degrade accuracy. This supports the declared finite score-clustering criterion, not normalized-mixture identifiability or a global-optimum guarantee.

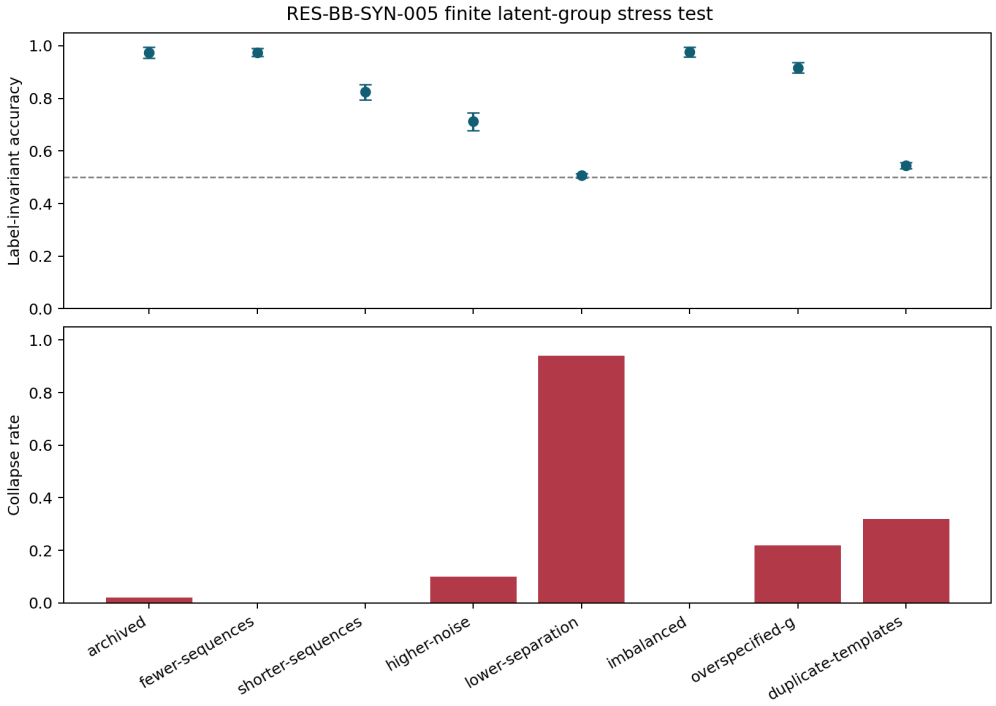


Figure 13.5. Finite-template stress study. Top: label-invariant hard accuracy with dataset-level 95% intervals. Bottom: collapse or duplicate-template rate across the eight declared cells.

13.0.6 Non-conjugate approximation trade-offs

For non-conjugate likelihoods, BayesBreak relies on approximate block evidences while reusing the same DP layer. Table 13.3 compares several such approximations in a bundled logistic-normal synthetic example, using numerical quadrature as a reference. The table shows the observed speed–accuracy trade-off in this archived example: Laplace is the fastest among the listed approximations, while exact quadrature is the slowest. At the same time, block-level approximation error is not perfectly monotone with segmentation quality, because the DP aggregates many local scores nonlinearly.

Method	$\max \lvert\Delta \log A^0\rvert$	$q_{95}\lvert\Delta \log A^0\rvert$	median $\lvert\Delta \log A^0\rvert$	time (s)	F1@τ	$\hat{k}$
quadrature	0.000	0.000	0.000	0.093	0.286	6
laplace	1.235	1.188	0.980	0.049	0.286	6
jj	1.265	1.216	1.002	0.071	0.500	3
ep	14.502	14.141	12.680	0.059	0.500	7
pg-vb	1.265	1.216	1.002	0.072	0.500	3

Table 13.3. Approximation trade-offs for a non-conjugate Bernoulli/logistic block model. Quadrature is used as the local reference block integral; the remaining methods trade block-level fidelity against computational cost.

Connection to Proposition 10.7. The large reported values of $\max \lvert\Delta \log A\rvert$ in Table 13.3 (several to tens of log units on the worst reachable block for some approximations) place this illustration outside the small-ε regime where the uniform-error bound in Proposition 10.7 is tight. The boundary F1 values are therefore descriptive diagnostics, not a confirmation of the small-ε regime. The table still usefully shows that block-level error, runtime, and segmentation

quality need not move monotonically together, but it is not a universal ranking of non-conjugate approximation methods.

13.0.7 Runtime scaling

Finally, Figure 13.6 and Table 13.4 report runtime behavior for the Gaussian implementation. Panel A sweeps the series length $n \in \{50, 100, 200, 400, 800\}$ at two fixed segment budgets $k_{\max} \in \{10, 20\}$; panel B sweeps the segment budget $k_{\max} \in \{5, 10, 20, 40, 80\}$ at fixed $n = 200$. Over the archived range, the empirical log-log slopes annotated on Figure 13.6 are ≈ 1.07 in n at $k_{\max} = 10$, ≈ 1.14 in n at $k_{\max} = 20$, and ≈ 0.88 in $k_{\max}$ at fixed $n = 200$. Both sweeps therefore sit well below the $\mathcal{O}(k_{\max} n^2)$ worst case predicted by the exact sum-product DP. The discrepancy reflects two practical effects: (i) cumulative sufficient-statistics caching makes each candidate block evaluation effectively constant time, so the DP cost reduces to traversing the $\mathcal{O}(k_{\max} n)$ reachable $(k, \text{end-point})$ pairs over the small archived range; and (ii) constant-factor overhead from prior evaluation and DP setup dominates at the smallest n, flattening the small-n end of the curve. The $\mathcal{O}(k_{\max} n^2)$ worst case still controls the asymptotic profile and is the relevant bound for the much larger n regimes flagged for the next-pass scaling study in Remark 5.8.

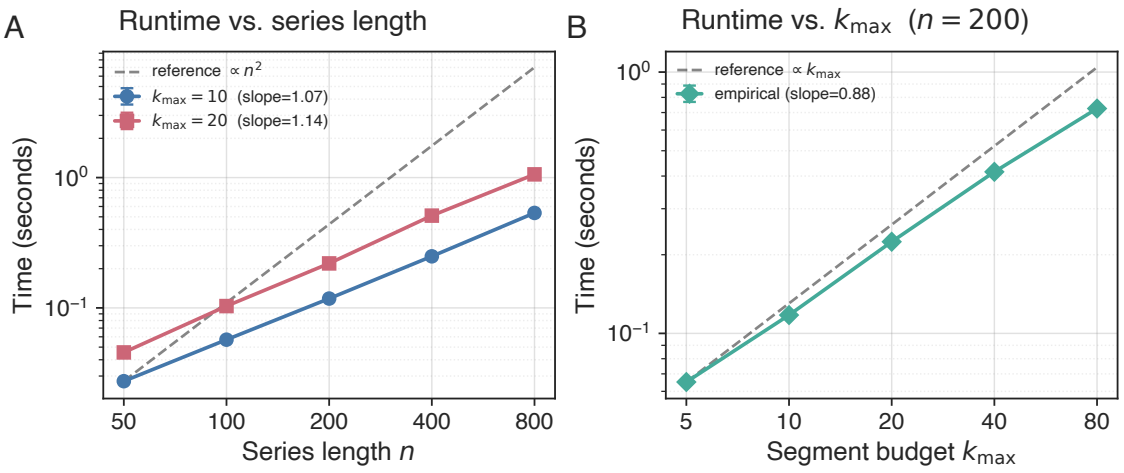


Figure 13.6. Runtime scaling for the Gaussian implementation. Panel A: log-log wall-clock time vs. series length $n \in \{50, 100, 200, 400, 800\}$ at two segment budgets $k_{\max} \in \{10, 20\}$; annotated slopes give the empirical $\log t$ vs. $\log n$ exponent for each curve, and a dashed reference line shows the theoretical $\mathcal{O}(n^2)$ worst case anchored at the smallest measured point. Panel B: log-log time vs. $k_{\max} \in \{5, 10, 20, 40, 80\}$ at fixed $n = 200$ with a dashed linear-in-$k_{\max}$ reference. Points are means over 5 repetitions; error bars denote one standard deviation.

n	$k_{\max}$	mean (s)	std (s)
50	20	0.0455	0.0008
100	20	0.1033	0.0011
200	20	0.2195	0.0011
400	20	0.5106	0.0307
800	20	1.0585	0.0345

Table 13.4. Numerical companion for the same archived Gaussian runtime configuration as Figure 13.6. The bundled table reports the $k_{\max} = 20$ configuration of panel A rather than every curve displayed in the figure; the $k_{\max}$ sweep of panel B is recorded in the figure's CSV sidecar.

13.1 Evidence interpretation

All populated values and panels in this chapter are real executions. Archived computations remain unchanged, while the wider stress study used a distinct run configuration and repeated seeds. Their role is method validation on the declared synthetic settings, not a comprehensive state-of-the-art claim. Boundary calibration reports ECE approximately 0.010 and Brier approximately 0.011. The archived latent-group template experiment reports 96% hard recovery at $\sigma = 1.0$ for 24 sequences; the corrected stress grid reports uncertainty and failure regimes separately. The runtime fits report log–log slopes about 1.07 at $k_{\max} = 10$ and 1.14 at $k_{\max} = 20$ for $n = 50, \ldots, 800$; these are local empirical summaries, not an asymptotic complexity result.

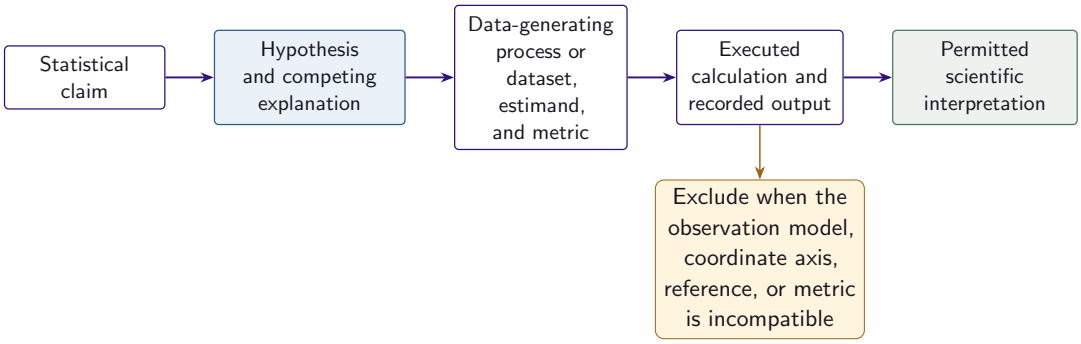


Figure 13.7. Relation among claims, experiments, and interpretation. An executed computation is excluded from a stated conclusion when its observation model, coordinate axis, reference quantity, or metric is incompatible with that conclusion.

13.2 Coverage still required

The archive does not yet provide fairness-controlled external comparators across every family, large-n scaling, or complete misspecification grids. These remain specified as protocols in Appendix C. The corrected latent stress cells are populated here; other planned cells remain explicitly unpopulated.

CHAPTER 14

Real-data studies and interpretation

CHAPTER PATH

Four real executed fits, their exact data and model semantics, comparator limits, and two immutable computations excluded from incompatible scientific interpretations.

14.1 Interpretation policy

The four case studies are real executed BayesBreak fits with archived data hashes and four-of-four DP diagnostic checks. Red vertical lines in the current figures are fitted MAP boundaries unless an independent annotation source is explicitly identified; they are not treated as geology, macro-event, copy-number, or atlas ground truth. Panel D in each generic case-study figure is an in-sample cumulative diagnostic, not held-out evaluation.

14.1.1 Well-log NMR response

The archived source describes the O Ruanaidh well-log NMR response with 4,050 original values and an executed downsampled sequence of $n = 507$. The Gaussian fit used $k_{\max} = 40$, returned $\widehat{k} = 23$, and recorded log evidence -4989.283473023698. A separately executed length-cohesion configuration returned $\widehat{k} = 25$ and log evidence -4997.147814570523; these values describe different priors and are not a universal ranking of irregular-design strategies.

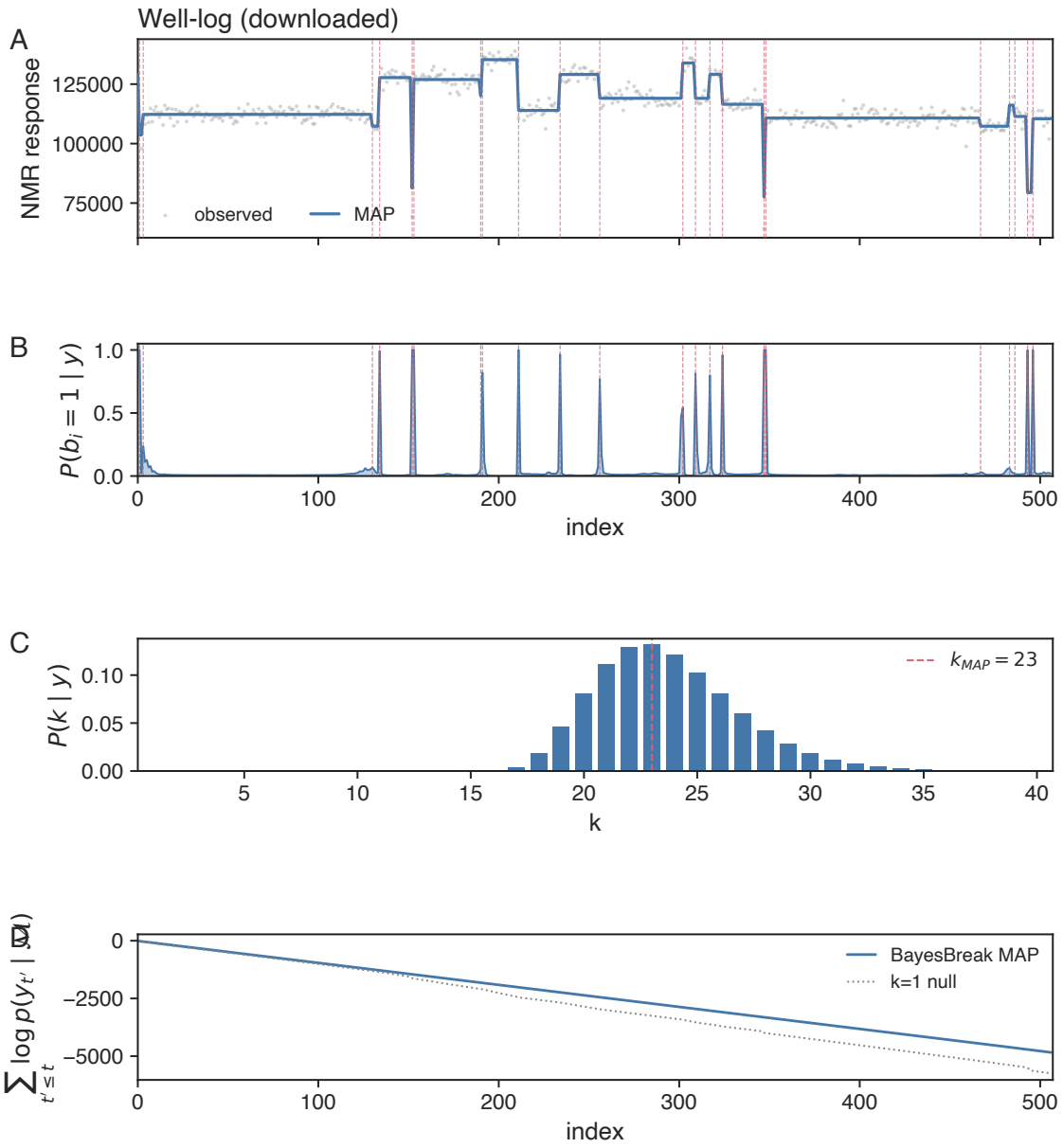


Figure 14.1. Executed well-log fit. The solid segmentation and red vertical markers are fitted BayesBreak summaries. Boundary probabilities and $P(k \mid y)$ are posterior outputs; the cumulative panel is in-sample.

Configuration	$\widehat{k}$	Log evidence	Boundary entropy (nats)
Index-uniform, $c = h = 1$	23	-4989.28	36.46
Archived length cohesion $c(i,j) \propto u_j - u_i$	25	-4997.15	45.46

Table 14.1. Real archived well-log configurations. External boundary accuracy is unavailable.

The comparator cache also contains agreement-to-BayesBreak diagnostics. At tolerance three, optimal partitioning and the Fearnhead-configured reference DP obtained agreement F_1 values 0.91 and 0.93, respectively; binary segmentation obtained 0.86, PELT 0.18, and the archived wild-binary run 0.05. These values measure resemblance to BayesBreak's MAP boundary set, not recovery of independently verified geology.

14.1.2 Array-CGH shared boundaries

The array-CGH archive contains 43 subjects on 2,215 probes. The shared-boundary Gaussian model used $k_{\max} = 15$, returned $\widehat{k} = 15$, and recorded pooled log evidence 76359.79958695151. The sum of 43 independently fitted subject evidences is 109617.69544504143; because the two quantities arise from different factorizations, their difference is descriptive and is not a Bayes factor for pooling.

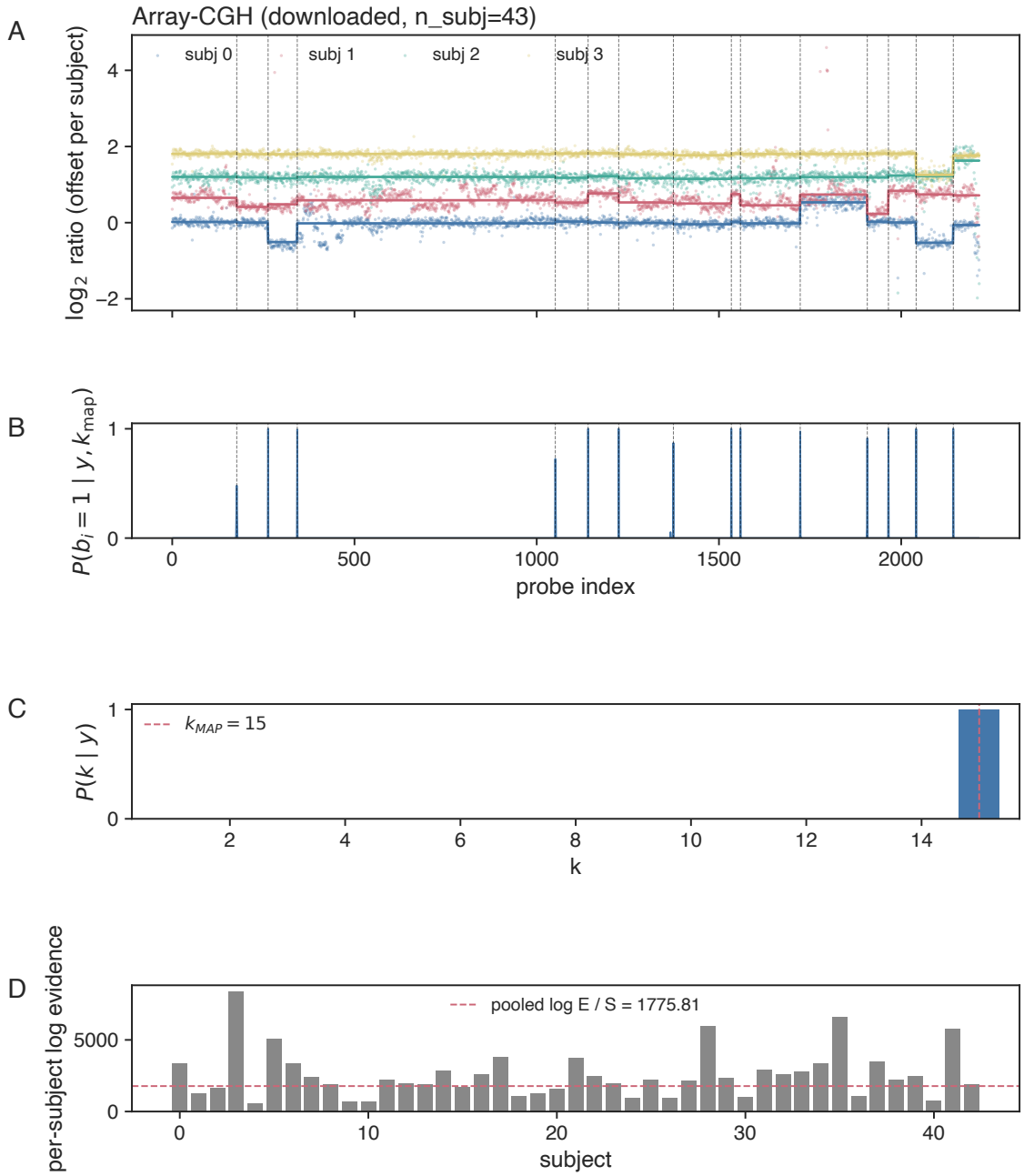


Figure 14.2. Executed shared-boundary array-CGH fit. The common MAP segmentation is fitted across the 43-subject matrix; no external boundary annotation is asserted.

oindent A previously computed comparison did not use a common statistical observation model and coordinate axis. Its scores are excluded from the reported evidence. The results below use the model and support described explicitly in their captions.

A common-probe-axis analysis is available. It uses the exact hashed raw $2{,}215 \times 43$ probe-

by-subject matrix and a common probe-index axis. Agreement $F_1@3$ with the model-derived BayesBreak MAP was 0.8000 for PELT, 0.9286 for dynamic programming, 0.7143 for binary segmentation, and 0.7857 for wild binary segmentation. The latter three used the 14-boundary target; the predeclared PELT grid did not attain that count and its closest 11-boundary candidate was retained without post-hoc tuning. Figure 14.3 is therefore a corrected agreement diagnostic, not external copy-number accuracy.

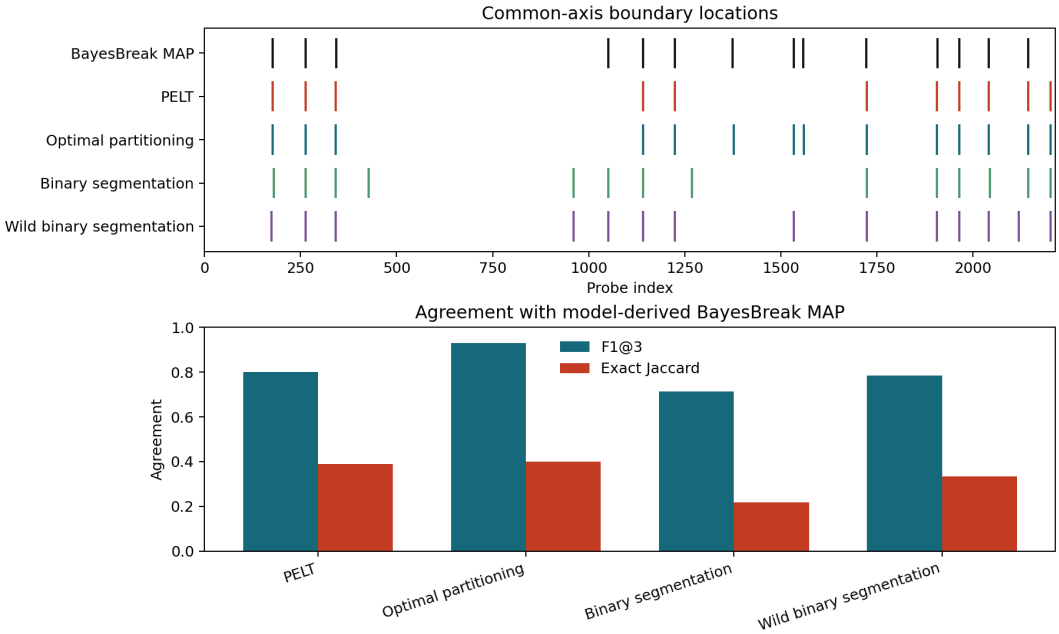


Figure 14.3. Probe-aligned array-CGH comparison. Top: boundaries on the common probe axis. Bottom: agreement with the model-derived BayesBreak MAP.

14.1.3 S&P 500 volatility regimes

The source covers January 1, 2015 through December 31, 2023, with approximately 2,263 raw daily observations and a stride-four executed series of $n = 566$. The primary Gaussian model on log-squared returns returned $\widehat{k} = 29$ and log evidence -1296.6537252802884. The secondary series is a Bernoulli indicator of crossing the empirical 95th-percentile threshold, not a Poisson count; it returned $\widehat{k} = 50$ and log evidence -114.41004640965144 under a different data model, so the raw evidences are not directly comparable.

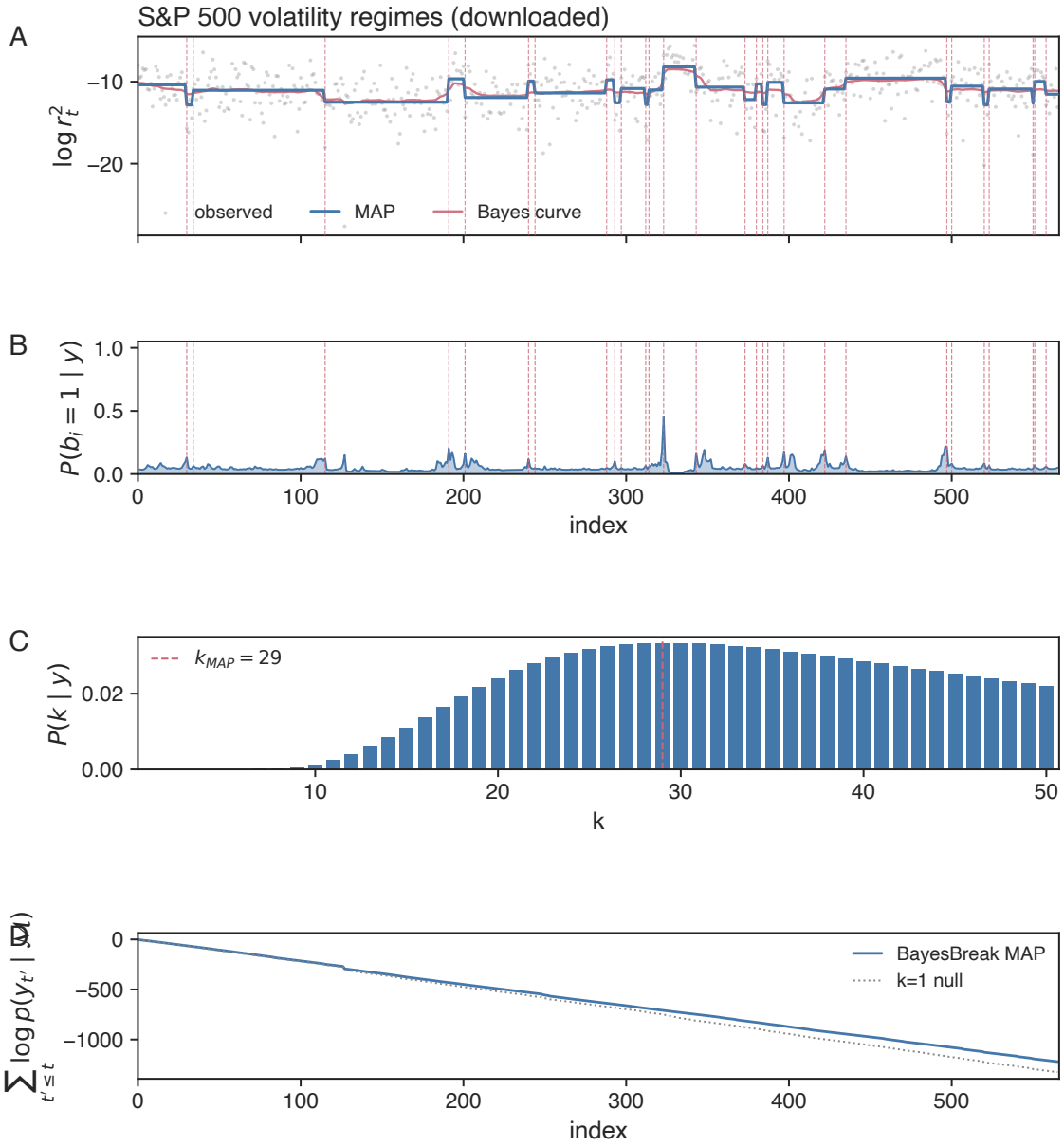


Figure 14.4. Executed S&P 500 volatility fit for 2015–2023 after the archived stride-four preprocessing. Red markers are fitted MAP boundaries, not independently annotated macro events.

Method or signal	$\widehat{k}$	Log evidence	Agreement $F_1@3$ vs BB Gaussian
BayesBreak Gaussian, $\log r_t^2$	29	-1296.65	reference
BayesBreak Bernoulli, threshold indicator	50	-114.41	different signal
Optimal partitioning, matched k	29	—	0.82
PELT	79	—	0.53
Binary segmentation, matched k	29	—	0.43
Wild binary segmentation, matched k	29	—	0.07
Fearnhead-configured reference DP	13	—	0.55

Table 14.2. Executed SPX outputs. Agreement columns use BayesBreak MAP boundaries as the reference and are not external accuracy.

14.1.4 Methylation on the methylKit chromosome-21 example

The executed data are the `methylKit test1.myCpG` chromosome-21 example with $n = 1904$ CpGs and per-CpG read coverage used as known precision. This is not the described Loyfer methylation-atlas analysis. The Beta-observation quadrature fit used $k_{\max} = 15$, returned $\widehat{k} = 15$, and recorded log evidence -9518.667508691196.

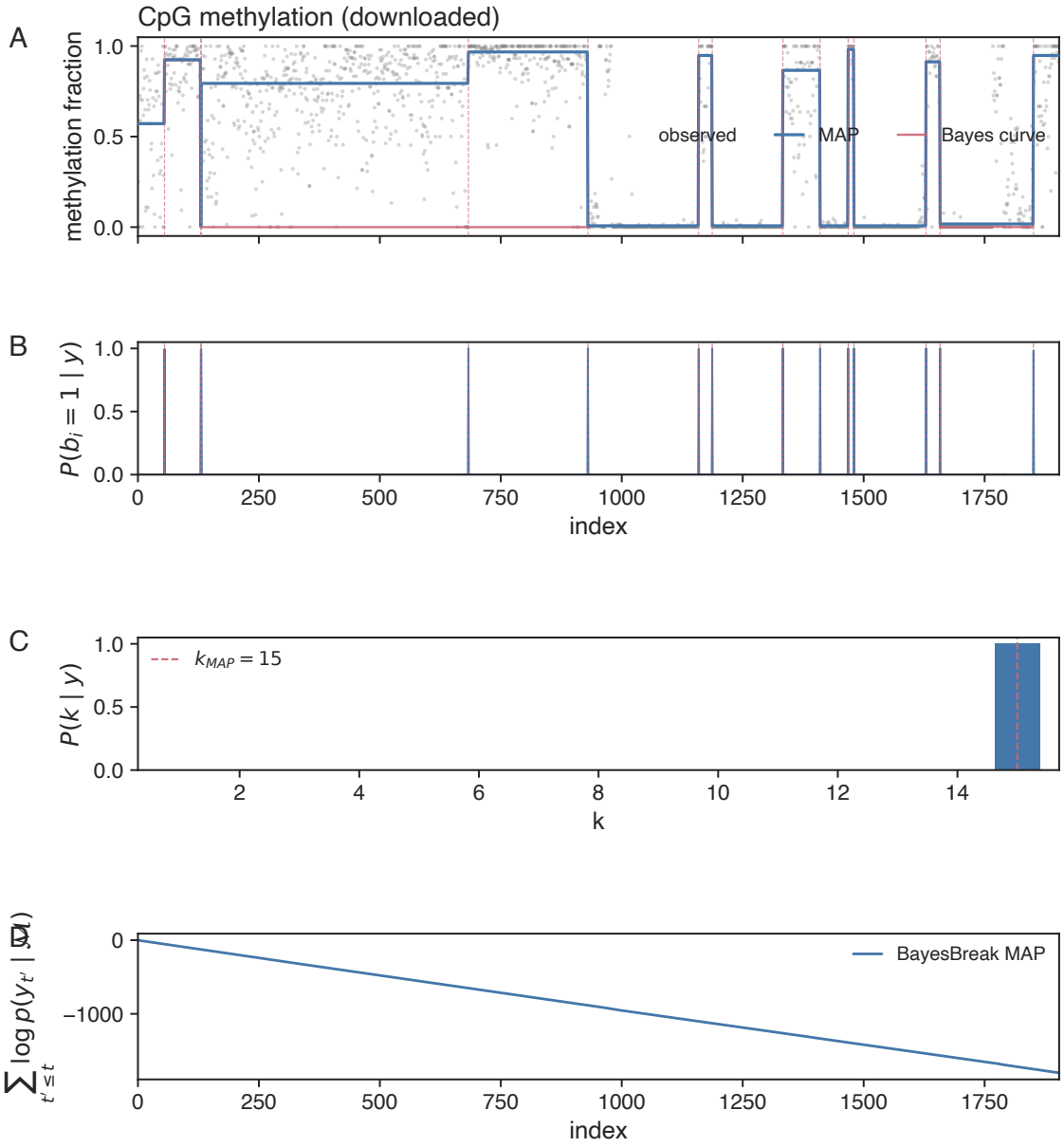


Figure 14.5. Executed methylKit chromosome-21 Beta-observation fit. Red markers are fitted MAP boundaries; no atlas transition annotation is present in the archived source.

Method	$\widehat{k}$	Runtime (s)	Agreement $F_1@3$ vs BB
BayesBreak Beta-observation	15	—	reference
Optimal partitioning, matched k	15	46.89	0.79
Binary segmentation, matched k	15	0.10	0.57
PELT	5	3.32	0.44
Wild binary segmentation, matched k	15	0.76	0.14

Table 14.3. Executed methylation comparator agreement. No external atlas accuracy is claimed.

oindent A previously computed comparison did not use a common statistical observation model and coordinate axis. Its scores are excluded from the reported evidence. The results below use the model and support described explicitly in their captions.

A family-correct held-out analysis uses the Beta-observation predictive distribution, positive held-out coverage as ϕ_{new}, and `extrapolation=error`. Ten disjoint stratified interior blocks held out 1,520 CpGs while retaining both fitted-support endpoints. Total log predictive score was -23605.6749, pooled mean score was -15.5300, and the split-mean 95% t interval was $[-23.1445, -7.9156]$. Mean boundary-stability $F_1@3$ against the model-derived full-data MAP was 0.8786. Figure 14.6 shows substantial regional variation. These chromosome blocks are not independent biological samples; no certified Beta-observation PIT or external atlas accuracy is claimed, and the score is not comparable with the excluded parent because both family and split changed.

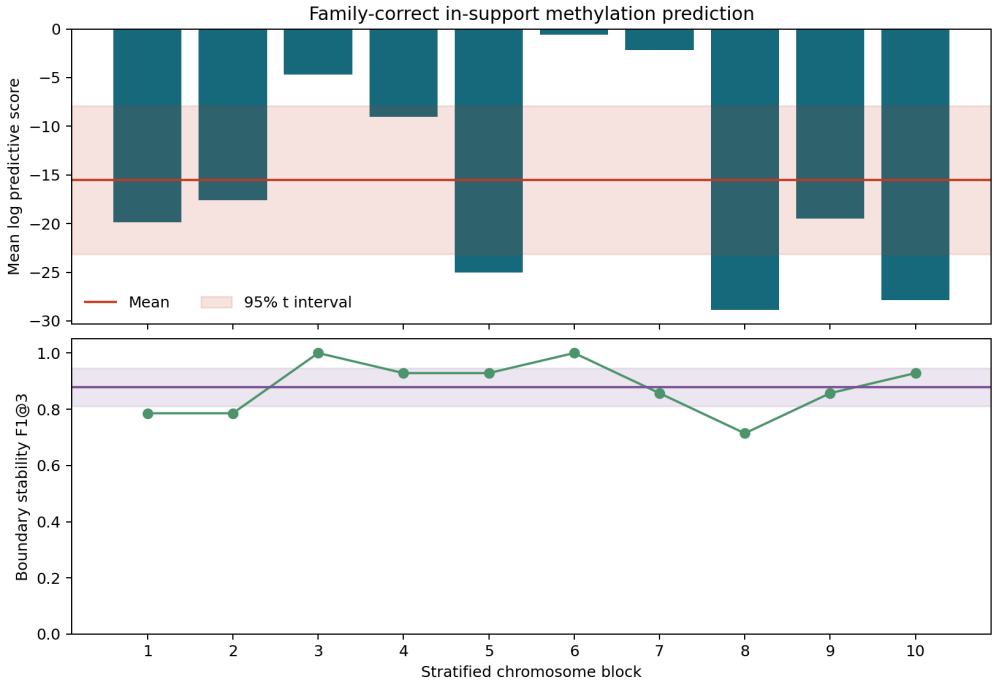


Figure 14.6. Beta-observation methylation posterior prediction. Top: mean log predictive score over ten in-support blocks. Bottom: boundary-stability $F_1@3$ against the model-derived full-data MAP.

14.2 What these case studies establish

Together, the fits and corrected children demonstrate that the implemented interface executed on Gaussian single-series, Gaussian shared-boundary, Gaussian volatility, Bernoulli threshold,

and Beta-observation settings, and that their archived DP invariants passed. The CGH child establishes common-axis agreement and the methylation child establishes family-correct in-support prediction for the declared blocks. They do not establish external boundary accuracy, causal correspondence to domain events, or superiority under independently tuned predictive comparisons. Those claims require the remaining predeclared protocol strata in Appendix C.

Limitations and failure modes

CHAPTER PATH

Conditions under which the factorized partition model, segment distribution, numerical approximation, hierarchical assumptions, prediction rule, or empirical comparison is not valid.

The exact results in this volume concern offline Bayesian segmentation on a finite ordered candidate set. They require finite segment marginal likelihoods and a prior whose contribution can be represented by segment or transition factors. Nonconjugate calculations require additional numerical assumptions. This chapter states the corresponding limitations.

15.1 Computational and model support

Sequence length and segment count. The dynamic program has worst-case time complexity $\Theta(k_{\max}n^2)$ and working memory at least $\Theta(k_{\max}n)$ for the principal posterior quantities (Proposition 6.2). It is an offline method. Large sequences may require a restricted changepoint set, pruning, or an approximate decomposition. Such modifications change the set of partitions or the summation unless their exactness is established separately; they do not automatically inherit Theorem 5.5.

Candidate changepoints. A changepoint outside the candidate set cannot receive posterior probability. This is a support restriction of the model, not a numerical error. Scientific locations that must remain possible should be included before fitting.

Nonlocal dependence among segments. The recursions assume that, after any necessary state variable is included, the probability of a partition can be written as products of local contributions. A prior depending on a global pattern of segment lengths, unrestricted interactions among nonadjacent segments, or labels with non-Markov dependence may violate this assumption. Some dependence structures can be handled by an enlarged state space, but the recurrence and complexity must then be derived for that model.

15.2 Observation-model misspecification and prior sensitivity

Misspecified segment distribution. Posterior probabilities are conditional on the chosen within-segment distribution, its descriptors, and its hyperparameters. Heavy-tailed observations

under a Gaussian segment model, zero inflation under a Poisson model, overdispersion under a Binomial model, or unmodeled serial dependence can distort segment marginal likelihoods and therefore the posterior over partitions. Posterior predictive diagnostics can reveal discrepancies, but the method does not automatically select the correct observation family.

Prior sensitivity. The posterior for k and the changepoint probabilities can depend materially on the segment-count prior, segment cohesion $c_u(i, j)$, and boundary hazard $h_u(j)$. This dependence can be stronger on an irregular grid because different priors assign different probabilities to long physical gaps. An analysis should report sensitivity to scientifically plausible prior choices rather than treat a single prior as neutral.

15.3 Numerical approximation

Nonconjugate segment marginal likelihoods. Proposition 10.7 assumes a verified bound on the largest log marginal-likelihood error over every admissible segment. The bound increases with the number of segments, so a fixed segmentwise error can have a larger effect for large k. If the numerical method does not establish this condition, the dynamic program returns the exact posterior for the approximate score array but no error bound relative to the intended posterior.

In the archived comparison, the maximum absolute discrepancy for expectation propagation was approximately 14.5 relative to the quadrature reference on the examined segment set. This is not a small-error setting. The associated boundary score is descriptive and does not verify the stability theorem. A scientifically interpretable nonconjugate analysis should compare numerical methods on the same admissible segments, examine convergence and tail behavior, and use a more accurate calculation when the discrepancy is too large for the desired posterior comparison.

15.4 Hierarchical assumptions

Common changepoints across sequences. The shared-changepoint model assumes aligned candidate coordinates and conditional independence across sequences given a common partition and sequence-specific segment parameters. Unmodeled cross-sequence dependence or genuine sequence-specific changepoints can make the common partition misleading. The finite-candidate consistency theorem also requires a positive average expected log-score gap between the true partition and every alternative; one informative sequence does not satisfy that condition as the number of sequences increases.

Latent-group non-uniqueness. The finite latent-group criterion is invariant under permutation of the group labels and may have duplicate or collapsed templates. Multiple restarts do not prove that a global maximum has been found. Because the sequence-template scores are not assumed to be normalized component densities, finite-mixture identifiability theorems do not apply. The archived implementation must also be corrected so that restart ranking uses the final value in each objective sequence.

Posterior summaries versus joint optimization. Marginal changepoint probabilities and the joint MAP partition are different summaries. Selecting the marginal mode for each ordered

changepoint coordinate need not yield a valid or jointly most probable partition. Applications should report the posterior quantities relevant to the decision rather than interpreting all returned summaries as interchangeable.

15.5 Prediction and evaluation

Coordinate support and extrapolation. Posterior prediction must use the fitted observation family. A coordinate outside the fitted range requires a pre-specified extrapolation model or rejection. Silent endpoint assignment changes the predictive model and can yield a finite but invalid score, as demonstrated by the archived methylation calculation excluded from predictive conclusions.

Changepoint metrics. Tolerance-based boundary F_1 depends on the tolerance and matching rule. When BayesBreak MAP boundaries are used as the reference, a comparator score measures agreement with BayesBreak, not accuracy against independent truth. A location error is undefined when no pair is matched and is reported as NA. The four real-data analyses do not contain independent changepoint annotations, so their fitted red lines cannot be interpreted as verified geological, genomic, or financial events.

Incompatible comparator coordinates. The array-CGH comparator flattened a 43×2215 fitted matrix to 95,245 values while the reference changepoints remained on the 2,215-probe axis. Those objects do not define a common matching problem. The executed values remain recorded, but they are excluded from comparator conclusions.

15.6 Choice of analysis

(1) For a single sequence with a conjugate segment model, use the exact segment integration and sum-product or max-sum recursions of Chapters 4–6.

(2) For a nonconjugate segment model, use a stated numerical method and assess its error on the admissible segment set. Use Proposition 10.7 only when Assumption 10.6 has been established.

(3) For aligned sequences expected to share changepoints, use the common-partition model of Chapter 8; examine sensitivity to sequence dependence and violations of common structure.

(4) For observed groups, fit the shared-changepoint model within each group. For unobserved groups, use the finite latent-group criterion of Chapter 9 and interpret its allocations and templates according to that criterion, without importing normalized-mixture claims.

(5) For prediction, use the posterior predictive distribution of the fitted observation family and state whether conditioning is on a selected partition or averaged over partition uncertainty.

15.7 Empirical coverage

The synthetic studies and four real-data fits demonstrate the calculations under their archived settings. They do not provide a complete fairness-controlled comparison with all relevant

frequentist and Bayesian methods, broad repeated uncertainty intervals, comprehensive misspecification studies, or large-n scaling. Those studies remain necessary before making broad comparative or robustness claims.

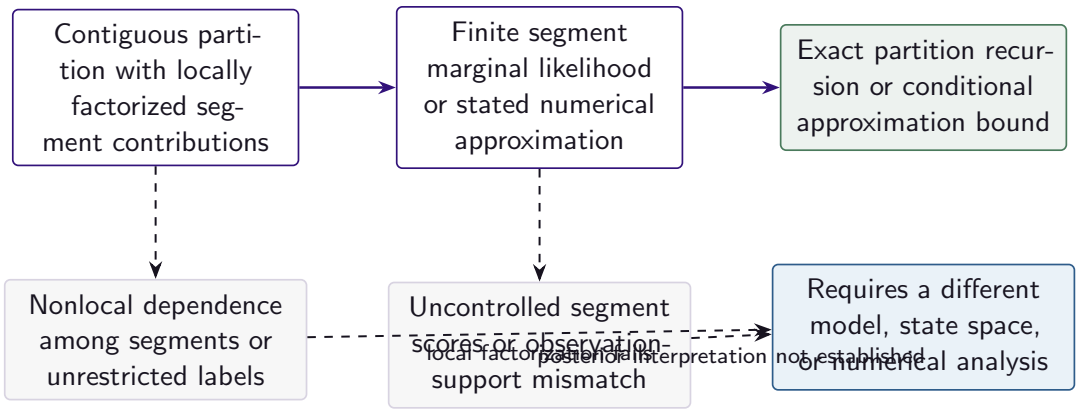


Figure 15.1. Scope of the exact partition calculation. Unsupported candidate sets, nonlocal dependence, misspecified segment distributions, uncontrolled numerical scores, undefined extrapolation, and incompatible comparator axes require a revised model or analysis.

CHAPTER 16

Reproducibility, remaining validation, and conclusion

CHAPTER PATH

A final account of the established statistical results, the executed empirical results, the unresolved implementation work, and the scope of the generalized method.

16.1 Established statistical results

BayesBreak is a finite Bayesian calculation over ordered partitions. For regular exponential-family segment models with proper conjugate priors, exact segment marginal likelihoods and posterior moments are obtained from sufficient statistics. Under a factorized partition prior, sum-product dynamic programming yields fixed-count marginal likelihoods, $p(k \mid y)$, changepoint marginals, changepoint-event probabilities, and Bayes curves. A separate max-sum recursion yields the joint MAP partition. For aligned conditionally independent sequences with a common partition, sequence-specific segment marginal likelihoods multiply. The finite latent-group criterion has monotone alternating updates under exact maximization of its Jensen minorizer. For nonconjugate segment models, the posterior-odds bounds are conditional on a uniform error bound for every admissible segment score.

16.2 Interpretation of the executed results

The archive contains executed synthetic recovery, boundary-calibration, observation-model, latent-group, nonconjugate-approximation, and finite-range timing studies, together with four real-data fits. The real-data analyses are descriptive because independent changepoint labels are not archived. Comparator scores that use the BayesBreak joint-MAP changepoints as the reference measure agreement rather than external accuracy. Unsupported computations are excluded from comparator and predictive conclusions.

16.3 Implementation and validation still required

The implementation must align known Gaussian precision, Poisson exposure and predictive observation-family semantics with the likelihoods given in the mathematical development. Prospective validation must cover independent exactness oracles, approximation convergence,

matched-coordinate baselines, posterior calibration and held-out predictive scores. The array-CGH weighted fit requires regeneration after this correction; segment-count saturation in CGH and methylation requires additional sensitivity analysis. The required experiment designs are collected in Appendix C, with exact machine-readable configurations in the repository's current experimental specification.

16.4 Conclusion

The main contribution of BayesBreak is a generalized hierarchical Bayesian segmentation method that combines exponential-family segment integration with exact posterior computation over ordered partitions and extends that construction to irregular designs, multiple aligned sequences, known groups, and finite latent groups. The generality lies in the observation-model independence of the partition recursion once mathematically valid segment marginal likelihoods are supplied. The hierarchical structure lies in explicit probability models for shared or group-specific changepoints and in the separately stated latent-group optimization criterion.

This formulation is grounded in conjugate analysis, product-partition models, Bayesian multiple-changepoint recursions, hierarchical modeling, and dynamic programming. Its limitations are equally mathematical: exactness depends on finite segment integrals and local factorization; shared-changepoint pooling depends on alignment and conditional independence; nonconjugate results depend on numerical error control; and empirical conclusions are limited by the available references, replications, and comparator definitions. Stating those conditions precisely preserves the breadth of the generalized method without replacing statistical claims by branding language.

CHAPTER A

Complete annotated literature review

CHAPTER PATH

Annotated sources that establish the theoretical and algorithmic setting of BayesBreak and the limits of directly transferring earlier results.

The references below are annotated by their direct relevance to the model, inference method or evaluation problem. The cited work is distinguished from the contributions and empirical claims of BayesBreak.

A.1 Recommended reading sequence

A reader focused on the BayesBreak statistical formulation should begin with product partitions and exact Bayesian recursion (Barry–Hartigan, Fearnhead, Hutter), then read design-aware partitioning (Scargle, PPMx, generalized PPM), multiple-sequence work (BASIC and JRPM), and finally the block-local approximation sources. The clustering extension should be read beside Corradin et al.; the finite latent-group criterion in BayesBreak is intentionally narrower than a normalized latent-mixture model.

A.2 Bayesian segmentation and exact-recursion lineage

`adamsmackay2007bocpd`**: Adams and MacKay [1].** Adams and MacKay formulate online Bayesian changepoint detection through a recursive posterior over run length. BayesBreak uses this work to delimit its scope: BayesBreak is an offline contiguous-partition method that returns a posterior over segment count, fixed-count boundary marginals, Bayes curves, and a separate joint MAP partition, rather than an online filtering distribution. The source therefore informs the online/offline distinction and potential future streaming comparisons, but it is not treated as evidence for BayesBreak's offline exact-DP claims.

`auger1989segment`**: Auger and Lawrence [2].** Auger and Lawrence develop dynamic-programming algorithms for identifying segment neighborhoods under broad segment-fit criteria, including least squares and maximum likelihood. The paper is part of the classical algorithmic lineage for Bellman-style segmentation and shows that an optimal-partition max-sum recurrence is not itself a BayesBreak contribution. BayesBreak differs by using integrated segment likelihoods in Bayesian recursions for normalization and posterior marginals, while retaining a separate max-sum recursion for the joint MAP partition. The paper is used as algorithmic background rather than as a source for Bayesian posterior calculations.

barry1992ppm**: Barry and Hartigan [3].** Barry and Hartigan introduce product partition models for changepoint problems, including the cohesion factorization that underlies segment-local prior contributions. This is direct prior art for BayesBreak's factorized partition prior. BayesBreak narrows the setting to ordered contiguous partitions and makes the local prior factorization explicit by separating segment cohesion $c_x(i, j)$ from an interior-boundary hazard $h_x(j)$, both of which preserve dynamic-programming factorization. The source prevents any claim that cohesion priors originate here.

barry1993bayesCP**: Barry and Hartigan [4].** Barry and Hartigan develop a Bayesian analysis for changepoint problems using the product-partition framework. BayesBreak cites this paper for the Bayesian changepoint and cohesion lineage, including posterior uncertainty about partitions. The exact BayesBreak delta is not generic Bayesian changepoint inference; it is the generalized exponential-family segment formulation, the collection of posterior quantities computed by the sum-product recursion, the hierarchical extensions, and the separately computed joint MAP partition. The comparison also motivates calibrated novelty language.

carlin1992hierBayesCP**: Carlin et al. [8].** Carlin, Gelfand, and Smith present hierarchical Bayesian changepoint analysis and show how uncertainty and pooling can be represented beyond a single flat partition model. BayesBreak uses this paper to locate its shared and grouped extensions within a broader hierarchical Bayesian lineage. The current exact BayesBreak results apply only where subject-level contributions factor through a common boundary vector or known grouping; richer hierarchical dependence generally requires additional computation. The source therefore informs both motivation and scope limits.

denison1998bayesian**: Denison et al. [11].** Denison, Mallick, and Smith develop automatic Bayesian curve fitting with a joint model for the number and positions of knots in piecewise-polynomial representations and reversible-jump Markov chain Monte Carlo for posterior computation. BayesBreak uses the paper as a contrasting Bayesian model-space and computational lineage rather than as an exact changepoint dynamic-programming reference. The comparison makes BayesBreak's scope precise: its exact recursions require a finite ordered candidate set and segment-factorized marginal likelihoods, and do not subsume arbitrary trans-dimensional curve models.

fearnhead2006exact**: Fearnhead [14].** Fearnhead gives exact and efficient Bayesian inference for a class of multiple-changepoint models with unknown changepoint count, conditionally independent segments, product-partition structure, and quadratic exact computation. This is the closest foundational work for BayesBreak's sum-product recursion and materially constrains novelty. BayesBreak's contribution is the generalized exponential-family segment formulation, posterior moments and changepoint summaries computed by common recursions, the separate max-sum calculation, irregular-design and multiple-sequence extensions, and the finite latent-group criterion.

green1995rjMCMC**: Green [18].** Green introduces reversible-jump Markov chain Monte Carlo for Bayesian model determination across spaces of differing dimension. Changepoint models can use this machinery to sample segment count and boundaries without the finite factorization

required by exact dynamic programming. BayesBreak cites the work as the principal trans-dimensional Bayesian alternative and uses it to state its own restriction: exactness comes from finite ordered support and local factorization, not from universal Bayesian computation.

`hutter2006bpcr`**: Hutter [19].** Hutter derives exact Bayesian regression for piecewise-constant functions, including unknown segment count and boundaries, evidence, break probabilities, error estimates, and a Bayesian regression curve. This is direct closest work for several BayesBreak headline outputs. The revised book therefore does not claim that evidence, boundary probabilities, or a Bayes curve originate with BayesBreak. The remaining contribution is the generalized use of exposure-aware segment sufficient statistics, shared and grouped extensions, separate posterior and joint-MAP calculations, and one implementation of these statistical quantities.

`punskaya2002bayesian`**: Punskaya et al. [28].** Punskaya and coauthors develop Bayesian curve fitting by MCMC with signal-segmentation applications. The paper supplies a sampling-based Bayesian alternative that can accommodate models outside BayesBreak's exact factorized regime. BayesBreak uses it to motivate posterior uncertainty while explaining why its own exact recursions require segment-local integrated scores. It is not used to justify the archived BayesBreak numerical values.

`yao1988biometrika`**: Yao [38].** Yao studies estimation of the number of changepoints in an independent normal sequence using Schwarz's criterion and establishes weak consistency of the proposed estimator. This paper represents information-criterion selection of segment count. BayesBreak uses it as a contrast to assigning a prior to the number of segments, integrating over that number, and reporting its posterior distribution together with posterior changepoint summaries and a separate joint MAP partition. These procedures address related but nonidentical inferential targets.

A.3 Conjugacy, approximate inference, and numerical computation

`bernardo1994bayesian`**: Bernardo and Smith [5].** Bernardo and Smith provide the general Bayesian foundations used to interpret marginal likelihoods, posterior model probabilities, posterior predictive quantities, and decision-relevant uncertainty. In BayesBreak, the book supports the distinction between evidence, posterior summaries, and exported point decisions. It does not supply a changepoint algorithm or validate any archived numerical result. Its role is foundational vocabulary and interpretation, especially where the manuscript separates a posterior over partitions from a joint MAP action.

`diaconis1979conjugate`**: Diaconis and Ylvisaker [12].** Diaconis and Ylvisaker characterize conjugate priors for exponential families. Their normalizer convention supplies the algebra behind BayesBreak's integrated block evidence: cumulative statistic contributions and cumulative log-partition multipliers update the conjugate hyperparameters, and the evidence is a ratio of normalizers times the powered base measure. BayesBreak extends the presentation with an exposure-aware descriptor d_t and a separately declared power/reliability weight a_t so Poisson exposures and binomial trial counts are not conflated with likelihood tempering.

`tierney1986laplace`**: Tierney and Kadane [36].** Tierney and Kadane develop Laplace approximations for posterior moments and marginal posterior densities under regularity conditions that include a suitably regular, unimodal likelihood-times-prior integrand. BayesBreak cites this paper as a foundation for Laplace evaluation of a nonconjugate segment marginal likelihood. The source does not imply a uniform error bound over all candidate segments, nor does it establish a BayesBreak-specific convergence rate. Any use in BayesBreak therefore requires the segment objective, mode, curvature, and approximation error to be assessed over the admissible segment set; the partition-level error statements apply only after such a segment-level bound has been established.

`jaakkola2000logisticvb`**: Jaakkola and Jordan [20].** Jaakkola and Jordan develop a variational bound for Bayesian logistic models. BayesBreak uses the construction only as an established segment-local approximation option for nonconjugate logistic marginal likelihoods or moments. The approximation is not a BayesBreak contribution, and the source does not supply a uniform segment-error rate for the present changepoint setting. Application of the BayesBreak posterior perturbation result therefore requires a separate, model-specific uniform bound over all reachable segments under explicit parameter, initialization, convergence, and tail conditions.

`minka2001ep`**: Minka [23].** Minka develops expectation propagation as an approximate Bayesian inference method based on iterative moment matching. BayesBreak treats EP as one possible block-local evidence routine for a nonconjugate family. The revised method makes no routine-wide convergence or accuracy claim: EP must return diagnostics and, where global perturbation claims are desired, a certified error against a trustworthy reference on every reachable block. The source contributes the approximation mechanism, not evidence for BayesBreak-specific rates.

`polson2013pg`**: Polson et al. [27].** Polson, Scott, and Windle introduce Pólya–Gamma data augmentation for Bayesian logistic models, yielding conditional Gaussian structure. BayesBreak uses this identity as an established route for block-local logistic computation and diagnostics. The source does not imply that a mean-field or approximate BayesBreak block score has a uniform rate over all reachable blocks. Such a rate would require additional assumptions and proof; in this book the global theorem is explicitly conditional on a supplied local error certificate.

`dempster1977em`**: Dempster et al. [10].** Dempster, Laird, and Rubin establish the expectation–maximization algorithm for maximum-likelihood estimation with incomplete data. The BayesBreak latent-template routine uses a responsibility-weighted alternating optimization, but its subject–template quantities are prior-weighted partition scores rather than normalized component densities. The book therefore cites EM as optimization background and describes the routine through its displayed finite objective and Jensen minorization; it does not claim an observed-data finite-mixture likelihood or import identifiability conclusions from mixture theory.

`blanchard2021lse`**: Blanchard et al. [6].** Blanchard, Higham, and Higham analyze stable computation of log-sum-exp and softmax functions. BayesBreak uses these numerical primitives throughout log-domain forward/backward recursions, pooled evidence aggregation, template

responsibilities, and normalization checks. The source informs implementation and numerical QA rather than the statistical model. It is particularly relevant where underflow would otherwise turn very small but nonzero partition mass into an apparent hard exclusion.

`teicher1963identifiability`**: Teicher [34].** Teicher establishes classical identifiability results for finite mixtures under conditions on normalized component distributions. The reference is retained to mark a boundary of applicability. BayesBreak's current subject–template scores are positive partition scores rather than normalized subject-level component densities, so finite-mixture identifiability does not follow from Teicher's result. The book limits the established latent-template theory to monotonicity of the stated objective and records label permutations, duplicate templates, and collapsed solutions as failure modes.

A.4 Optimization and frequentist comparator lineage

`frick2014smuce`**: Frick et al. [16].** Frick, Munk, and Sieling develop multiscale changepoint inference with simultaneous error control and confidence statements. SMUCE is a relevant baseline because it emphasizes inferential reliability rather than only point recovery. BayesBreak uses it to design task-matched comparisons for boundary recovery and uncertainty, while recognizing that its posterior probabilities and Bayes curves are not directly equivalent to multiscale confidence sets. Any benchmark must therefore specify the inferential target and calibration criterion.

`fryzlewicz2014wbs`**: Fryzlewicz [17].** Fryzlewicz introduces wild binary segmentation for efficient multiple-changepoint detection, especially when many changes may occur. It is a computationally attractive point-estimation baseline in the BayesBreak experiments. The source informs scaling and recovery comparisons, but it does not return the same posterior objects as BayesBreak. Archived F1 values computed against BayesBreak MAP boundaries are consequently labeled agreement diagnostics, not independent accuracy or evidence of superiority.

`jackson2005optpart`**: Jackson et al. [21].** Jackson and coauthors give an algorithm for optimal partitioning of interval data, reinforcing the dynamic-programming lineage for globally optimal segmentations. BayesBreak uses this paper to distinguish an optimal partition computed by max-sum recursion from posterior quantities computed by sum-product recursion. Both routes consume the same local scores, but they answer different questions; marginal boundary modes cannot be substituted for the joint MAP partition.

`killick2012pelt`**: Killick et al. [22].** Killick, Fearnhead, and Eckley introduce PELT, a pruned optimization method with linear expected computational cost under stated conditions. PELT is a central efficiency comparator for BayesBreak's worst-case $O(k_{\max} n^2)$ exact posterior engine. The comparison is intentionally asymmetric in output richness: PELT targets a penalized optimal segmentation, whereas BayesBreak also returns evidence, segment-count uncertainty, boundary marginals, and model-averaged curves. Fair evaluation must match objectives and tuning budgets.

`olshen2004cbs`**: Olshen et al. [25].** Olshen and coauthors introduce circular binary segmentation for array-based DNA copy-number analysis. CBS is a domain-standard comparator for the array-CGH case study. It operates on point-estimate segmentation rather than BayesBreak's posterior partition distribution. The archived CGH comparator computation is excluded from scientific comparison because a 43×2215 fitted array was flattened to 95,245 values while the BayesBreak boundaries remained on the 2,215-probe axis; that execution cannot support a scientific comparison.

`rigaill2010pruned`**: Rigaill [30].** Rigaill develops pruned dynamic programming for recovering best segmentations over a range of segment counts. The work belongs to the optimization/pruning lineage and informs possible efficiency comparisons. BayesBreak's current core deliberately preserves the unpruned exact posterior recursion, so the empirical slopes near 1.07 and 1.14 are reported only as finite-range observations and never as a change to the $O(k_{\max} n^2)$ worst-case complexity.

`rudin1992rof`**: Rudin et al. [31].** Rudin, Osher, and Fatemi introduce total-variation regularization for edge-preserving reconstruction. The work is a mathematical ancestor of piecewise-constant penalization, not a Bayesian changepoint posterior method. BayesBreak uses this lineage to contextualize fused and total-variation baselines and to distinguish a penalized point estimator from a posterior distribution over partitions.

`tibshirani2005fused`**: Tibshirani et al. [35].** Tibshirani and coauthors formulate the fused lasso, combining sparsity and adjacent-difference penalties to estimate piecewise-constant structure. It is a natural point-estimation baseline for offline segmentation. BayesBreak differs through probabilistic integration over segment parameters and posterior inference over changepoint configurations; a fused-lasso tuning path is not equivalent to a prior on the number of segments. The source informs baseline selection and misspecification analysis.

`truong2020review`**: Truong et al. [37].** Truong, Oudre, and Vayatis provide a selective review and taxonomy of offline changepoint methods, including costs, search methods, and evaluation considerations. BayesBreak uses the review to organize baseline families and to prevent category errors between detection, segmentation, online monitoring, and posterior inference. It is a secondary synthesis rather than evidence for a BayesBreak theorem or result.

A.5 Design-aware and generalized partition priors

`scargle2013bayesianblocks`**: Scargle et al. [33].** Scargle and coauthors develop Bayesian Blocks for optimal representations under arbitrary sampling modes, gaps, exposures, and several data modes. This is important prior art for irregularly sampled segmentation. BayesBreak therefore narrows its claim: likelihood exposure, segment cohesion, and boundary hazard are distinct modeling components, and ordered cohesion-plus-hazard factors can be consumed by the same DP. The source is closest work for irregular-design handling, not support for the earlier conflated Poisson-gap derivation.

`muller2011ppmx`**: Müller et al. [24].** Müller, Quintana, and Rosner introduce PPMx, adding covariate similarity to product-partition cohesion. This is direct lineage for data- or

design-dependent partition priors. BayesBreak's contribution is a restricted ordered-contiguous factorization that makes segment cohesion and boundary hazard explicit and preserves exact normalization by dynamic programming. The source prevents claims that covariate-dependent or design-aware cohesion is new in general.

`park2010gppm`**: Park and Dunson [26].** Park and Dunson develop generalized product partition models, broadening the ways local partition factors can encode prior structure. BayesBreak uses the work to situate its ordered prior specification: only factors that can be assigned locally to admissible contiguous blocks and interior endpoints can enter the unchanged finite DP. The book's result is therefore a transparent specialization and implementation rule, not a general invention of product-partition priors.

A.6 Multiple sequences, dependence, and shared boundaries

`bleakley2011groupfused`**: Bleakley and Vert [7].** Bleakley and Vert propose the group fused lasso for detecting changepoints shared across multiple one-dimensional signals by penalizing multidimensional total variation. It is a task-relevant frequentist comparator for common structural changes across sequences. BayesBreak differs by integrating sequence-specific segment parameters and, in its strict common-boundary model, multiplying sequence-level segment marginal likelihoods before Bayesian dynamic programming. The methods return different inferential objects, so comparisons must use comparable tuning budgets and distinguish point-estimate agreement from posterior uncertainty.

`fearnhead2011dependence`**: Fearnhead and Liu [15].** Fearnhead and Liu develop efficient Bayesian multiple-changepoint analysis when parameters can depend across adjacent segments. BayesBreak's protected exact regime instead assumes that, after integration and declared prior factors, a partition score decomposes into local block contributions. This source marks a clear boundary: dependence across segments is scientifically useful but is not covered by the current exact block API without state augmentation or a different recursion.

`ruggieri2014bcpvs`**: Ruggieri and Lawrence [32].** Ruggieri and Lawrence combine Bayesian changepoint inference with variable selection and apply the method to a paleoclimate proxy record. The work demonstrates that scientifically meaningful segmentation often couples regime changes with additional structural choices. BayesBreak cites it as an application and model-extension lineage while keeping variable selection outside the protected core. It informs future-work boundaries rather than a current contribution claim.

`fan2017basic`**: Fan and Mackey [13].** Fan and Mackey's BASIC model addresses simultaneous changepoints across many sequences and includes copy-number and S&P volatility applications. It is the closest multisequence comparator and overlaps the domains of two BayesBreak case studies. BayesBreak's narrower exact result assumes one common boundary vector and conditionally independent subject contributions whose integrated block evidences multiply. The revised book presents this as a restricted exact pooling regime, not the first Bayesian multisequence changepoint method.

`quinlan2024jrpm`**: Quinlan et al. [29].** Quinlan, Page, and Castro develop joint random partition models that permit dependence among multiple partitions. This is richer than BayesBreak's exact common-boundary specialization. The source is used to state the tradeoff precisely: BayesBreak gains a simple exact DP by imposing a shared partition (or known groups), while JRPM targets flexible dependence across series. Broad claims about general multivariate partition inference are therefore excluded.

A.7 Clustering by structural changes

`corradin2026commonstructural`**: Corradin et al. [9].** Corradin, Danese, KhudaBukhsh, and Ongaro cluster time-dependent observations according to common structural-change locations using a latent random-order representation and Bayesian computation. This is the closest verified work for the broad latent-group narrative. BayesBreak retains only a narrower computational object: a finite collection of deterministic changepoint templates optimized under the explicitly stated criterion in Chapter 9, using auxiliary allocation weights and exact responsibility-weighted max-sum updates. The two models should not be conflated, and no claim of first clustering by shared changepoints remains.

Computational implementation and validation

CHAPTER PATH

Algorithm interfaces and numerical acceptance criteria for exact and approximate segment calculations, posterior inference, grouped models and predictions.

B.1 Block routines and their outputs

For each admissible $(i, j]$, a block routine returns $\log A_{ij}^{(0)}$ and, when required, posterior segment moments or sufficient statistics for posterior predictive calculations. The function must be explicit about its family, known descriptors and prior hyperparameters. Where $A_{ij}^{(0)}$ is computed numerically, record the approximation, objective and convergence state without asserting unverified finite error bounds. Validate the Gaussian, Poisson, Binomial and negative-Binomial integrated likelihoods by independent one-dimensional quadrature with both unit and nonunit descriptors.

B.2 Partition normalizer and posterior calculations

For a fixed k, the admissible partitions have prior probabilities proportional to products of segment and interior-boundary factors. Compute their normalizing constant by the same factorized recursion with observation likelihoods set to one. Given the normalized prior, compute $p(y \mid k)$, $p(k \mid y)$, ordered changepoint marginals and boundary-event marginals by log-space forward and backward messages. Recover the joint MAP by max-sum DP and backtracking. For small n, exhaustive enumeration must agree on all normalized quantities.

B.3 Group structure and predictive inference

Shared-boundary pooling multiplies independent subject-specific segment integrals conditional on the common partition, which means adding log-block scores across subjects. Finite-template clustering updates assignments and templates for its declared score-clustering/MM objective, not an unstated generative mixture likelihood. Predictive scores must use the same observation family, known descriptors and coordinate support as the fitted model. Score held-out blocks only under prespecified training-only conditioning and split policies.

B.4 Invariants to verify

1. Agreement between independent direct integration and analytic segment scores.
2. Agreement between brute-force partitions and log-space forward, backward and max-sum DP.
3. Consistent treatment of impossible boundaries, empty/masked blocks and supported coordinates.
4. Monotone improvement of the declared finite-template optimization objective.
5. Independent identity and convergence checks for nonconjugate numerical methods.
6. Exact correspondence between observation axes, external annotations and comparison metrics.
7. Identical figure, table and manuscript metrics for a common, verified run manifest.

The corresponding current experiment definitions and execution criteria are maintained in the repository's single coding-agent skill. Publication claims require both a correct mathematical argument and matching verification, not merely a passing integration test.

Prospective evaluation protocols

CHAPTER PATH

Predefined statistical tests for posterior accuracy, calibration, irregular spacing, pooling, finite latent groups, numerical integration, prediction and computational cost.

The complete executable design matrices, seed construction, primary outcomes, confidence interval methods, benchmark tuning and output filenames reside in one current experiment specification. The experiments E00–E10 are *planned*. No table or figure in this appendix presents a result from those experiments. Earlier executed examples in Chapters 13 and 14 provide context under their original models, not substitute confirmatory outcomes.

C.1 Common statistical design

All simulation comparisons use the same simulated sequence for every method in each replicate. The independent generated dataset is the unit of replication. Posterior boundary calibration is summarized by proper scoring rules, reliability curves with bin counts and dataset-level confidence intervals. Boundary matching uses a maximum-cardinality, minimum-displacement one-to-one assignment within the declared tolerance. Distinguish the posterior over segment count from a conditional-on-k boundary distribution. For real-data comparisons, coordinate axes, train/validation/test splits and the reference label type must be verified before scoring. Model selection and tuning do not use the test observations.

Table C.1. Planned experiments; no outcomes are supplied in this table.

ID	Research question	Core planned comparison or verification
E00	Exactness of inference	Exhaustive enumeration and direct segment integration for short sequences under several proper priors.
E01	Recovery and calibration	Replicated Gaussian sequences with varied jump sizes, segment counts and posterior uncertainty targets.
E02	Family generality	Gaussian, Poisson, Binomial, negative-Binomial and Beta-valued block models under appropriate descriptors.

ID	Research question	Core planned comparison or verification
E03	Irregular design	Compare index-uniform, length-cohesion, gap-hazard and combined priors under controlled gaps.
E04	Shared boundaries	Evaluate pooled and unpooled segmentation as subject count and boundary heterogeneity vary.
E05	Finite-template clustering	Assess label-invariant clustering accuracy, objective monotonicity, restarts and template collapse.
E06	Nonconjugate integration	Compare independent adaptive quadrature with identifiable Laplace, variational and EP approximations.
E07	Model misspecification	Test heavy tails, count dispersion, missingness and dependence under declared alternative models.
E08	Posterior prediction	Compare family-correct out-of-sample scores under disjoint held-out regions and fit uncertainty.
E09	Applied comparisons	Use common coordinates, independently tuned comparators and genuinely external annotations where available.
E10	Runtime and resource cost	Report stage-separated timings and peak memory as n and $k_{\max}$ vary.

C.2 Primary reporting conventions

For every experiment, report the complete design grid, number of planned and executed cells, random-seed schedule, data and configuration hashes, failures and exclusion criteria. Do not pool performance scores across incompatible response scales or observational likelihoods. Numerical approximations must report both local evidence error and the effect on global segmentation posteriors. Model-derived boundary agreement and externally labeled boundary accuracy are different estimands and belong in separate tables. Only completed and independently checked outcomes may replace the planned result slots in the single-column paper.

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